

**Virtual University of Pakistan**  
**Statistics and Probability**  
**STA301**

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An Overview of the Science of Statistics in Today's World (including Latest

## LECTURE NO. 1

### MCQs WHAT IS STATISTICS?

- That science which enables us to draw conclusions about various phenomena on the basis of real data collected on sample-basis
- A tool for data-based research
- Also known as Quantitative Analysis
- A lot of application in a wide variety of disciplines Agriculture, Anthropology, Astronomy, Biology, Economic, Engineering, Environment, Geology, Genetics, Medicine, Physics, Psychology, Sociology, Zoology .... Virtually every single subject from Anthropology to Zoology .... A to Z!
- Any scientific enquiry in which you would like to base your conclusions and decisions on real-life data, you need to employ statistical techniques!
- Now a day, in the developed countries of the world, there is an active movement for of Statistical Literacy.

### THE NATURE OF THIS DISCIPLINE

#### DESCRIPTIVE STATISTICS



#### PROBABILITY



#### INFERENCEAL STATISTICS

### MEANINGS OF 'STATISTICS'

### MCQs

The word "Statistics" which comes from the Latin words *status*, meaning a political state, originally meant information useful to the state, for example, information about the sizes of population and armed forces. But this word has now acquired different meanings.

- In the first place, the word statistics refers to "numerical facts systematically arranged". In this sense, the word statistics is always used in plural. We have, for instance, statistics of prices, statistics of road accidents, statistics of crimes, statistics of births, statistics of educational institutions, etc. In all these examples, the word statistics denotes a set of numerical data in the respective fields. This is the meaning the man in the street gives to the word *Statistics* and most people usually use the word *data* instead.
- In the second place, the word statistics is defined as a discipline that includes procedures and techniques used to collect process and analyze numerical data to make inferences and to research decisions in the face of

uncertainty. It should of course be borne in mind that uncertainty does not imply ignorance but it refers to the incompleteness and the instability of data available. In this sense, the word statistics is used in the singular. As it embodies more or less all stages of the general process of learning, sometimes called *scientific method*, statistics is characterized as a science. Thus the word *statistics* used in the plural refers to a set of numerical information and in the singular, denotes the science of basing decision on numerical data. It should be noted that statistics as a subject is mathematical in character.

- *Thirdly*, the word statistics are numerical quantities calculated from sample observations; a single quantity that has been so collected is called a *statistic*. The mean of a sample for instance is a statistic. The word statistics is plural when used in this sense.

### **CHARACTERISTICS OF THE SCIENCE OF STATISTICS**

Statistics is a discipline in its own right. It would therefore be desirable to know the characteristic features of statistics in order to appreciate and understand its general nature. Some of its important characteristics are given below:

- Statistics deals with the behaviour of aggregates or large groups of data. It has nothing to do with what is happening to a particular individual or object of the aggregate.
- Statistics deals with aggregates of observations of the same kind rather than isolated figures.
- Statistics deals with variability that obscures underlying patterns. No two objects in this universe are exactly alike. If they were, there would have been no statistical problem.
- Statistics deals with uncertainties as every process of getting observations whether controlled or uncontrolled, involves deficiencies or chance variation. That is why we have to talk in terms of probability.
- Statistics deals with those characteristics or aspects of things which can be described numerically either by counts or by measurements.
- Statistics deals with those aggregates which are subject to a number of random causes, *e.g.* the heights of persons are subject to a number of causes such as race, ancestry, age, diet, habits, climate and so forth.
- Statistical laws are valid *on the average* or in the long run. There is no guarantee that a certain law will hold in all cases. Statistical inference is therefore made in the face of uncertainty.
- Statistical results might be misleading the incorrect if sufficient care in collecting, processing and interpreting the data is not exercised or if the statistical data are handled by a person who is not well versed in the subject matter of statistics.

### **THE WAY IN WHICH STATISTICS WORKS**

As it is such an important area of knowledge, it is definitely useful to have a fairly good idea about the way in which it works, and this is exactly the purpose of this introductory course.

The following points indicate some of the main functions of this science:

- Statistics assists in summarizing the larger set of data in a form that is easily understandable.
- Statistics assists in the efficient design of laboratory and field experiments as well as surveys.
- Statistics assists in a sound and effective planning in any field of inquiry.
- Statistics assists in drawing general conclusions and in making predictions of how much of a thing will happen under given conditions.

### **IMPORTANCE OF STATISTICS IN VARIOUS FIELDS**

As stated earlier, Statistics is a discipline that has finds application in the most diverse fields of activity. It is perhaps a subject that should be used by everybody. Statistical techniques being powerful tools for analyzing numerical data are used in almost every branch of learning. In all areas, statistical techniques are being increasingly used, and are developing very rapidly.

- A modern administrator whether in public or private sector leans on statistical data to provide a factual basis for decision.
- A politician uses statistics advantageously to lend support and credence to his arguments while elucidating the problems he handles.
- A businessman, an industrial and a research worker all employ statistical methods in their work. Banks, Insurance companies and Government all have their statistics departments.
- A social scientist uses statistical methods in various areas of socio-economic life a nation. It is sometimes said that “a social scientist without an adequate understanding of statistics, is often like the blind man groping in a dark room for a black cat that is not there”.

### THE MEANING OF DATA

The word “data” appears in many contexts and frequently is used in ordinary conversation. Although the word carries something of an aura of scientific mystique, its meaning is quite simple and mundane. It is Latin for “those that are given” (the singular form is “datum”). Data may therefore be thought of as the *results of observation*.

Mcqs in 2023

### EXAMPLES OF DATA

- Data are collected in many aspects of everyday life.
- Statements given to a police officer or physician or psychologist during an interview are data.
- So are the correct and incorrect answers given by a student on a final examination.
- Almost any athletic event produces data.
- The time required by a runner to complete a marathon,
- The number of errors committed by a baseball team in nine innings of play.
- And, of course, data are obtained in the course of scientific inquiry:
- the positions of artifacts and fossils in an archaeological site,
- The number of interactions between two members of an animal colony during a period of observation,
- The spectral composition of light emitted by a star.

### OBSERVATIONS AND VARIABLES

In statistics, an *observation* often means any sort of numerical recording of information, whether it is a physical measurement such as height or weight; a classification such as heads or tails, or an answer to a question such as yes or no.

### VARIABLES

A characteristic that varies with an individual or an object is called a *variable*. For example, *age is a variable* as it varies from person to person. A variable can assume a number of values. The given set of all possible values from which the variable takes on a value is called its Domain. If for a given problem, *the domain of a variable contains only one value, then the variable is referred to as a constant*.

### QUANTITATIVE AND QUALITATIVE VARIABLES

MCQs

Variables may be classified into *quantitative and qualitative* according to the form of the characteristic of interest. A variable is called a *quantitative variable* when a characteristic can be expressed numerically such as age, weight, income or number of children. On the other hand, if the characteristic is non-numerical such as education, sex, eye-colour, quality, intelligence, poverty, satisfaction, etc. the variable is referred to as a *qualitative variable*. A qualitative characteristic is also called an *attribute*. An individual or an object with such a characteristic can be counted or enumerated after having been assigned to one of the several mutually exclusive classes or categories.



**DISCRETE AND CONTINUOUS VARIABLES** *MCQs* Mcqs in 2023

A quantitative variable may be classified as **discrete or continuous**. A *discrete* variable is one that can take only a discrete set of integers or whole numbers, which is the values, are taken by **jumps or breaks**. A discrete variable represents *count* data such as the **number of persons in a family**, the **number of rooms in a house**, the **number of deaths in an accident**, the **income of an individual**, etc.

A variable is called a *continuous* variable if it can take on any value-fractional or integral—within a given interval, i.e. its domain is an interval with **all possible values without gaps**. A continuous variable represents measurement data such as the **age of a person**, the **height of a plant**, the **weight of a commodity**, the **temperature at a place**, etc.

A variable whether countable or measurable, is generally denoted by some symbol such as **X or Y** and  $X_i$  or  $X_j$  represents the  $i$ th or  $j$ th value of the variable. The subscript  $i$  or  $j$  is replaced by a number such as **1,2,3**, ... when referred to a particular value.

**MEASUREMENT SCALES** *Important* short question in 2022

By *measurement*, we usually mean the assigning of number to observations or objects and scaling is a process of measuring. **The four scales of measurements are briefly mentioned below:**

**NOMINAL SCALE** Nominal scale is a naming scale, where variables are simply “named” or labeled, with no specific order.

The classification or grouping of the observations into mutually exclusive qualitative categories or classes is said to constitute a *nominal scale*. For example, **students are classified as male and female**. Number 1 and 2 may also be used to identify these two categories. Similarly, rainfall may be classified as heavy moderate and light. We may use number 1, 2 and 3 to denote the three classes of rainfall. The numbers when they are used only to identify the categories of the given scale carry no numerical significance and there is no particular order for the grouping.

**Gender, marital status, religion, race, hair color, country**

**ORDINAL OR RANKING SCALE** The ordinal scale defines data that is placed in a specific order. While each value is ranked

It includes the characteristic of a nominal scale and in addition has the property of **ordering or ranking of measurements**. For example, the **performance of students (or players) is rated as excellent, good fair or poor, etc.** Number 1, 2, 3, 4 etc. are also used to indicate ranks. The only relation that holds between any pair of categories is that of “greater than” (or more preferred).

**INTERVAL SCALE** The interval scale contains properties of nominal and ordered data, but the difference between data points can be quantified.

A measurement scale possessing a constant interval size (distance) but not a true zero point, is called an *interval scale*. Temperature measured on either the Celsius or the Fahrenheit scale is an outstanding example of interval scale because the same difference exists between 20° C (68° F) and 30° C (86° F) as between 5° C (41° F) and 15° C (59° F). It cannot be said that a temperature of 40 degrees is twice as hot as a temperature of 20 degree, i.e. the ratio 40/20 has no meaning. The arithmetic operation of addition, subtraction, etc. is meaningful.

**40 degrees is not 20 degrees**

**RATIO SCALE**

It is a special kind of an interval scale where the scale of measurement has a true zero point as its origin. The ratio scale is used to measure **weight, volume, distance, money, etc.** The key to differentiating interval and ratio scale is that the **zero point is meaningful for ratio scale**.

**ERRORS OF MEASUREMENT**

Experience has shown that a continuous variable can never be measured with perfect fineness because of certain habits and practices, methods of measurements, instruments used, etc. the measurements are thus always recorded correct to the nearest units and hence are of limited accuracy. The *actual* or *true* values are, however, assumed to exist. For example, if a student's weight is recorded as 60 kg (correct to the nearest kilogram), his true weight in fact lies between **59.5 kg and 60.5 kg**, whereas a weight recorded as 60.00 kg means the true weight is known to lie between 59.995 and 60.005 kg. Thus there is a difference, however small it may be between the measured value and the true value. This sort of departure from the true value is technically known as the **error of measurement**. In other words, if the observed value and the true value of a variable are denoted by  $x$  and  $x + \varepsilon$  respectively, then the difference  $(x + \varepsilon) - x$ , i.e.  $\varepsilon$  is the error. This error involves the unit of measurement of  $x$  and is therefore called an *absolute error*. An absolute error

divided by the true value is called the *relative error*. Thus the **relative error** =  $\frac{\varepsilon}{x + \varepsilon}$ , which when multiplied by 100,

is *percentage error*. These errors are independent of the units of measurement of  $x$ . It ought to be noted that an error has both magnitude and direction and that the **word error in statistics does not mean mistake which is a chance inaccuracy**.

**BIASED AND RANDOM ERRORS**

An error is said to be *biased* when the observed value is consistently and constantly higher or lower than the true value.

Biased errors arise from the personal limitations of the observer, the imperfection in the instruments used or some other conditions which control the measurements. These errors are not revealed by repeating the measurements. They are cumulative in nature, that is, the greater the number of measurements, the greater would be the magnitude of error. They are thus more troublesome. These errors are also called *cumulative* or *systematic errors*.

An error, on the other hand, is said to be unbiased when the deviations, i.e. the excesses and defects, from the true value tend to occur equally often. *Unbiased errors are revealed when measurements are repeated and they tend to cancel out in the long run.* These errors are therefore *compensating* and are also known as *random errors* or *accidental errors*.

## LECTURE NO. 2

Steps involved in a Statistical Research-Project

- Collection of Data:
  - Primary Data
  - Secondary Data
- Sampling:
  - Concept of Sampling
  - Non-Random Versus Random Sampling
  - Simple Random Sampling
  - Other Types of Random Sampling

### STEPS INVOLVED IN ANY STATISTICAL RESEARCH

- Topic and significance of the study
- Objective of your study
- Methodology for data-collection
  - Source of your data
  - Sampling methodology
  - Instrument for collecting data

As far as the objectives of your research are concerned, they should be stated in such a way that you are absolutely clear about the goal of your study --- EXACTLY WHAT IT IS THAT YOU ARE TRYING TO FIND OUT?

As far as the *methodology for DATA-COLLECTION* is concerned, you need to consider:

- Source of your data (the statistical population)
- Sampling Methodology
- Instrument for collecting data

### COLLECTION OF DATA

The most important part of statistical work is perhaps the collection of data. Statistical data are collected either by a COMPLETE enumeration of the whole field, called **CENSUS**, which in many cases would be too costly and too time consuming as it requires large number of enumerators and supervisory staff, or by a PARTIAL enumeration associated with a SAMPLE which saves much time and money.

### PRIMARY AND SECONDARY DATA

MCQs

Data that have been originally collected (raw data) and have not undergone any sort of statistical treatment, are called **PRIMARY data**. Data that have undergone any sort of treatment by statistical methods at least ONCE, i.e. the data that have been collected, classified, tabulated or presented in some form for a certain purpose, are called **SECONDARY data**.

### COLLECTION OF PRIMARY DATA

One or more of the following methods are employed to collect primary data:

- Direct Personal Investigation
- Indirect Investigation
- Collection through Questionnaires
- Collection through Enumerators
- Collection through Local Sources

### DIRECT PERSONAL INVESTIGATION

In this method, an investigator collects the information personally from the individuals concerned. Since he interviews the informants himself, the information collected is generally considered quite accurate and complete. This method may prove **very costly and time-consuming** when the area to be covered is vast. However, it **is useful for laboratory experiments or localized inquiries**. Errors are likely to enter the results due to personal bias of the investigator.

### INDIRECT INVESTIGATION

Sometimes the direct sources do not exist or the informants hesitate to respond for some reason or other. **In such a case, third parties or witnesses having information are interviewed**. Moreover, due allowance is to be made for the personal bias. This method is useful when the information desired is complex or there is reluctance or indifference on the part of the informants. It can be adopted for extensive inquiries.

### COLLECTION THROUGH QUESTIONNAIRES

A questionnaire is an inquiry form comprising of a number of pertinent questions with space for entering information asked. The questionnaires are usually sent by mail, and the informants are requested to return the questionnaires to the investigator after doing the needful within a certain period. This method is cheap, fairly expeditious and good for extensive inquiries. But the difficulty is that the majority of the respondents (i.e. persons who are required to answer the questions) do not care to fill the questionnaires in, and to return them to the investigators. Sometimes, the questionnaires are returned incomplete and full of errors. Students, in spite of these drawbacks, this method is considered as the **STANDARD method for routine business and administrative inquiries.**

It is important to note that the questions should be few, brief, very simple, and easy for all respondents answer, clearly worded and not offensive to certain respondents.

### COLLECTION THROUGH ENUMERATORS

Under this method, the information is gathered by employing trained enumerators who assist the informants in making the entries in the schedules or questionnaires correctly. This method gives the most reliable information if the enumerator is well-trained, experienced and tactful. Students, it is considered the BEST method when a large-scale governmental inquiry is to be conducted. This method can generally not be adopted by a private individual or institution as its cost would be prohibitive to them.

### COLLECTION THROUGH LOCAL SOURCES

In this method, there is no formal collection of data but the agents or local correspondents are directed to collect and send the required information, using their own judgment as to the best way of obtaining it. This method is cheap and expeditious, but gives only the estimates.

### COLLECTION OF SECONDARY DATA

The secondary data may be obtained from the following sources:

- **Official**, e.g. the publications of the Statistical Division, Ministry of Finance, the Federal and Provincial Bureaus of Statistics, Ministries of Food, Agriculture, Industry, Labour, etc.
- **Semi-Official**, e.g., State Bank of Pakistan, Railway Board, Central Cotton Committee, Boards of Economic Inquiry, District Councils, Municipalities, etc.
- **Publications of Trade Associations**, Chambers of Commerce, etc
- **Technical and Trade Journals and Newspapers**
- **Research Organizations** such as universities, and other institutions

Let us now consider the POPULATION from which we will be collecting our data. In this context, the first important question is: Why do we have to resort to Sampling?

The answer is that: If we have available to us every value of the variable under study, then that would be an ideal and a perfect situation. But, the problem is that this ideal situation is very rarely available --- very rarely do we have access to the entire population.

The census is an exercise in which an attempt is made to cover the entire population. But, as you might know, even the most developed countries of the world cannot afford to conduct such a huge exercise on an annual basis!

More often than not, we have to conduct our research study on a sample basis. In fact, the goal of the science of Statistics is to draw conclusions about large populations on the basis of information contained in small samples.

## MCQs 'POPULATION'

A statistical population is the collection of every member of a group possessing the same basic and defined characteristic, but varying in amount or quality from one member to another.

### EXAMPLES

- **Finite population:**
  - IQ's of all children in a school.
- **Infinite population:**
  - Barometric pressure: (There are an indefinitely large number of points on the surface of the earth).
  - A flight of migrating ducks in Canada

(Many finite pops are so large that they can be treated as effectively infinite). The examples that we have just considered are those of existent populations.

A hypothetical population can be defined as the aggregate of all the conceivable ways in which a specified event can happen.

For Example:

- 1) All the possible outcomes from the throw of a die – however long we throw the die and record the results, we could always continue to do so for a still longer period in a theoretical concept – one which has no existence in reality.
- 2) The No. of ways in which a football team of 11 players can be selected from the 16 possible members named by the Club Manager.

We also need to differentiate between the sampled population and the target population. Sampled population is that from which a sample is chosen whereas the population about which information is sought is called the target population thus our population will consist of the total no. of students in all the colleges in the Punjab.

Suppose on account of shortage of resources or of time, we are able to conduct such a survey on only 5 colleges scattered throughout the province. In this case, the students of all the colleges will constitute the target population whereas the students of those 5 colleges from which the sample of students will be selected will constitute the sampled population. The above discussion regarding the population, you must have realized how important it is to have a very well-defined population.

The next question is: How will we draw a sample from our population?

The answer is that: **In order to draw a random sample from a finite population, the first thing that we need is the complete list of all the elements in our population.**

**This list is technically called the FRAME.**

### SAMPLING FRAME

A sampling frame is a complete list of all the elements in the population. For example:

- **The complete list of the BCS students of Virtual University of Pakistan on February 15, 2003**
- Speaking of the sampling frame, it must be kept in mind that, as far as possible, our frame should be free from various types of defects:
  - does not contain inaccurate elements
  - is not incomplete
  - is free from duplication, and
  - Is not out of date.

Next, let's talk about the sample that we are going to draw from this population.

As you all know, a **sample is only a part of a statistical population**, and hence it can represent the population to only to some extent. Of course, it is intuitively logical that the larger the sample, the more likely it is to represent the population. Obviously, the limiting case is that: when the sample size tends to the population size, the sample will tend to be identical to the population. But, of course, in general, the sample is much smaller than the population.

The point is that, in general, statistical sampling seeks to determine how accurate a description of the population the sample and its properties will provide. We may have to compromise on accuracy, but there are certain such advantages of sampling because of which it has an extremely important place in data-based research studies.

### ADVANTAGES OF SAMPLING

#### **1. Savings in time and money.**

- Although cost per unit in a sample is greater than in a complete investigation, the total cost will be less (because the sample will be so much smaller than the statistical population from which it has been drawn).
- A sample survey can be completed faster than a full investigation so that variations from sample unit to sample unit over time will largely be eliminated.
- Also, the results can be processed and analyzed with increased speed and precision because there are fewer of them.

#### **2. More detailed information may be obtained from each sample unit.**

#### **3. Possibility of follow-up:**

(After detailed checking, queries and omissions can be followed up --- a procedure which might prove impossible in a complete survey).

#### **4. Sampling is the only feasible possibility where tests to destruction are undertaken or where the population is effectively infinite.**

The next two important concepts that need to be considered are those of sampling and non-sampling errors.

### SAMPLING & NON-SAMPLING ERRORS

#### **1. SAMPLING ERROR**

MCQs

**The difference between the estimate derived from the sample (i.e. the statistic) and the true population value (i.e. the parameter) is technically called the sampling error. For example,**

$$\text{Sampling error} = \bar{X} - \mu$$

Sampling error arises due to the fact that a sample cannot exactly represent the pop, even if it is drawn in a correct manner

## 2. NON-SAMPLING ERROR

Besides sampling errors, there are certain errors which are not attributable to sampling but arise in the process of data collection, even if a complete count is carried out.

Main sources of non sampling errors are:

- The defect in the sampling frame.
- Faulty reporting of facts due to personal preferences.
- Negligence or indifference of the investigators
- Non-response to mail questionnaires.

These (non-sampling) errors can be avoided through

- Following up the non-response,
- Proper training of the investigators.
- Correct manipulation of the collected information,

Let us now consider exactly what is meant by ‘sampling error’: We can say that there are two types of non-response --- partial non-response and total non-response. ‘*Partial non-response*’ implies that the respondent refuses to answer some of the questions. On the other hand, ‘*total non-response*’ implies that the respondent refuses to answer any of the questions. Of course, the problem of late returns and non-response of the kind that I have just mentioned occurs in the case of HUMAN populations. Although refusal of sample units to cooperate is encountered in interview surveys, it is far more of a problem in mail surveys. It is not uncommon to find the response rate to mail questionnaires as low as 15 or 20%. The provision of INFORMATION ABOUT THE PURPOSE OF THE SURVEY helps in stimulating interest, thus increasing the chances of greater response. Particularly if it can be shown that the work will be to the ADVANTAGE of the respondent IN THE LONG RUN.

Similarly, the respondent will be encouraged to reply if a pre-paid and addressed ENVELOPE is sent out with the questionnaire. But in spite of these ways of reducing non-response, we are bound to have some amount of non-response. Hence, a decision has to be taken about how many RECALLS should be made.

The term ‘recall’ implies that we approach the respondent more than once in order to persuade him to respond to our queries.

Another point worth considering is:

How long the process of data collection should be continued? Obviously, no such process can be carried out for an indefinite period of time! In fact, the longer the time period over which the survey is conducted, the greater will be the potential VARIATIONS in attitudes and opinions of the respondents. Hence, a well-defined cut-off date generally needs to be established. Let us now look at the various ways in which we can select a sample from our population. We begin by looking at the difference between non-random and RANDOM sampling. First of all, what do we mean by non-random sampling?

## NONRANDOM SAMPLING

‘Nonrandom sampling’ implies that kind of sampling in which the population units are drawn into the sample by using one’s personal judgment. This type of sampling is also known as purposive sampling. Within this category, one very important type of sampling is known as Quota Sampling.

## QUOTA SAMPLING

In this type of sampling, the selection of the sampling unit from the population is no longer dictated by chance. A sampling frame is not used at all, and the choice of the actual sample units to be interviewed is left to the discretion of the interviewer. However, the interviewer is restricted by quota controls. For example, one particular interviewer may be told to interview ten married women between thirty and forty years of age living in town X, whose husbands are professional workers, and five unmarried professional women of the same age living in the same town. Quota sampling is often used in commercial surveys such as consumer market-research. Also, it is often used in public opinion polls.

## ADVANTAGES OF QUOTA SAMPLING

- There is no need to construct a frame.
- It is a very quick form of investigation.
- Cost reduction

MCQs

**DISADVANTAGES**

- It is a subjective method. One has to choose between objectivity and convenience.
- If random sampling is not employed, it is no longer theoretically possible to evaluate the sampling error.
- (Since the selection of the elements is not based on probability theory but on the personal judgment of the interviewer, hence the precision and the reliability of the estimates can not be determined objectively i.e. in terms of probability.)
- Although the purpose of implementing quota controls is to reduce bias, bias creeps in due to the fact that the interviewer is FREE to select particular individuals within the quotas. (Interviewers usually look for persons who either agree with their points of view or are personally known to them or can easily be contacted.)
- Even if the above is not the case, the interviewer may still be making unsuitable selection of sample units.
- (Although he may put some qualifying questions to a potential respondent in order to determine whether he or she is of the type prescribed by the quota controls, some features must necessarily be decided arbitrarily by the interviewer, the most difficult of these being social class.)

If mistakes are being made, it is almost impossible for the organizers to detect these, because follow-ups are not possible unless a detailed record of the respondents' names, addresses etc. has been kept.

Falsification of returns is therefore more of a danger in quota sampling than in random sampling. In spite of the above limitations, it has been shown by F. Edwards that a well-organized quota survey with well-trained interviewers can produce quite adequate results.

Next, let us consider the concept of random sampling.

**RANDOM SAMPLING**

The theory of statistical sampling rests on the assumption that the selection of the sample units has been carried out in a random manner.

By random sampling we mean sampling that has been done by adopting the lottery method.

**TYPES OF RANDOM SAMPLING**

- Simple Random Sampling
- Stratified Random Sampling
- Systematic Sampling
- Cluster Sampling
- Multi-stage Sampling, etc.

In this course, I will discuss with you the simplest type of random sampling i.e. simple random sampling.

**SIMPLE RANDOM SAMPLING**

In this type of sampling, the chance of any one element of the parent pop being included in the sample is the same as for any other element.

By extension, it follows that, in simple random sampling, the chance of any one sample appearing is the same as for any other. There exists quite a lot of misconception regarding the concept of random sampling. Many a time, haphazard selection is considered to be equivalent to simple random sampling.

For example, a market research interviewer may select women shoppers to find their attitude to brand X of a product by stopping one and then another as they pass along a busy shopping area --- and he may think that he has accomplished simple random sampling!

Actually, there is a strong possibility of bias as the interviewer may tend to ask his questions of young attractive women rather than older housewives, or he may stop women who have packets of brand X prominently on show in their shopping bags!.

In this example, there is no suggestion of INTENTIONAL bias! From experience, it is known that the human being is a poor random selector --- one who is very subject to bias.

Fundamental psychological traits prevent complete objectivity, and no amount of training or conscious effort can eradicate them. As stated earlier, random sampling is that in which population units are selected by the lottery method.

As you know, the traditional method of writing people's names on small pieces of paper, folding these pieces of paper and shuffling them is very cumbersome!

A much more convenient alternative is the use of *RANDOM NUMBERS TABLES*.

A random number table is a page full of digits from zero to 9. These digits are printed on the page in a TOTALLY random manner i.e. there is no systematic pattern of printing these digits on the page.

**ONE THOUSAND RANDOM DIGITS**

2 3 1 5 7 5 4 8 5 9 0 1 8 3 7 2 5 9 9 3 7 6 2 4 9 7 0 8 8 6 9 5 2 3 0 3 6 7 4 4  
 0 5 5 4 5 5 5 0 4 3 1 0 5 3 7 4 3 5 0 8 9 0 6 1 1 8 3 7 4 4 1 0 9 6 2 2 1 3 4 3  
 1 4 8 7 1 6 0 3 5 0 3 2 4 0 4 3 6 2 2 3 5 0 0 5 1 0 0 3 2 2 1 1 5 4 3 8 0 8 3 4  
 3 8 9 7 6 7 4 9 5 1 9 4 0 5 1 7 5 8 5 3 7 8 8 0 5 9 0 1 9 4 3 2 4 2 8 7 1 6 9 5  
 9 7 3 1 2 6 1 7 1 8 9 9 7 5 5 3 0 8 7 0 9 4 2 5 1 2 5 8 4 1 5 4 8 8 2 1 0 5 1 3  
 1 1 7 4 2 6 9 3 8 1 4 4 3 3 9 3 0 8 7 2 3 2 7 9 7 3 3 1 1 8 2 2 6 4 7 0 6 8 5 0  
 4 3 3 6 1 2 8 8 5 9 1 1 0 1 6 4 5 6 2 3 9 3 0 0 9 0 0 4 9 9 4 3 6 4 0 7 4 0 3 6  
 9 3 8 0 6 2 0 4 7 8 3 8 2 6 8 0 4 4 9 1 5 5 7 5 1 1 8 9 3 2 5 8 4 7 5 5 2 5 7 1  
 4 9 5 4 0 1 3 1 8 1 0 8 4 2 9 8 4 1 8 7 6 9 5 3 8 2 9 6 6 1 7 7 7 3 8 0 9 5 2 7  
 3 6 7 6 8 7 2 6 3 3 3 7 9 4 8 2 1 5 6 9 4 1 9 5 9 6 8 6 7 0 4 5 2 7 4 8 3 8 8 0  
 0 7 0 9 2 5 2 3 9 2 2 4 6 2 7 1 2 6 0 7 0 6 5 5 8 4 5 3 4 4 6 7 3 3 8 4 5 3 2 0  
 4 3 3 1 0 0 1 0 8 1 4 4 8 6 3 8 0 3 0 7 5 2 5 5 5 1 6 1 4 8 8 9 7 4 2 9 4 6 4 7  
 6 1 5 7 0 0 6 3 6 0 0 6 1 7 3 6 3 7 7 5 6 3 1 4 8 9 5 1 2 3 3 5 0 1 7 4 6 9 9 3  
 3 1 3 5 2 8 3 7 9 9 1 0 7 7 9 1 8 9 4 1 3 1 5 7 9 7 6 4 4 8 6 2 5 8 4 8 6 9 1 9  
 5 7 0 4 8 8 6 5 2 6 2 7 7 9 5 9 3 6 8 2 9 0 5 2 9 5 6 5 4 6 3 5 0 6 5 3 2 2 5 4  
 0 9 2 4 3 4 4 2 0 0 6 8 7 2 1 0 7 1 3 7 3 0 7 2 9 7 5 7 3 6 0 9 2 9 8 2 7 6 5 0  
 9 7 9 5 5 3 5 0 1 8 4 0 8 9 4 8 8 3 2 9 5 2 2 3 0 8 2 5 2 1 2 2 5 3 2 6 1 5 8 7  
 9 3 7 3 2 5 9 5 7 0 4 3 7 8 1 9 8 8 8 5 5 6 6 7 1 6 6 8 2 6 9 5 9 9 6 4 4 5 6 9  
 7 2 6 2 1 1 1 2 2 5 0 0 9 2 2 6 8 2 6 4 3 5 6 6 6 5 9 4 3 4 7 1 6 8 7 5 1 8 6 7  
 6 1 0 2 0 7 4 4 1 8 4 5 3 7 1 2 0 7 9 4 9 5 9 1 7 3 7 8 6 6 9 9 5 3 6 1 9 3 7 8  
 9 7 8 3 9 8 5 4 7 4 3 3 0 5 5 9 1 7 1 8 4 5 4 7 3 5 4 1 4 4 2 2 0 3 4 2 3 0 0 0  
 8 9 1 6 0 9 7 1 9 2 2 2 2 3 2 9 0 6 3 7 3 5 0 5 5 4 5 4 8 9 8 8 4 3 8 1 6 3 6 1  
 2 5 9 6 6 8 8 2 2 0 6 2 8 7 1 7 9 2 6 5 0 2 8 2 3 5 2 8 6 2 8 4 9 1 9 5 4 8 8 3  
 8 1 4 4 3 3 1 7 1 9 0 5 0 4 9 5 4 8 0 6 7 4 6 9 0 0 7 5 6 7 6 5 0 1 7 1 6 5 4 5  
 1 1 3 2 2 5 4 9 3 1 4 2 3 6 2 3 4 3 8 6 0 8 6 2 4 9 7 6 6 7 4 2 2 4 5 2 3 2 4 5

Actually, Random Number Tables are constructed according to certain mathematical principles so that each digit has the same chance of selection. Of course, nowadays randomness may be achieved electronically. Computers have all those programmes by which we can generate random numbers.

**EXAMPLE**

The following frequency table of distribution gives the ages of a population of 1000 teen-age college students in a particular country.

Select a sample of 10 students using the random numbers table. **Find the sample mean age and compare with the population mean age.**

Student-Population of a College

| Age<br>(X) | No. of Students<br>(f) |
|------------|------------------------|
| 13         | 6                      |
| 14         | 61                     |
| 15         | 270                    |
| 16         | 491                    |
| 17         | 153                    |
| 18         | 15                     |
| 19         | 4                      |
|            | <b>1000</b>            |

How will we proceed to select our sample of size 10 from this population of size 1000?



The first step is to allocate to each student in this population a sampling number. For this purpose, we will begin by constructing a column of cumulative frequencies.

| AGE<br>X | No. of Students<br>f | Cumulative Frequency<br>cf |
|----------|----------------------|----------------------------|
| 13       | 6                    | 6                          |
| 14       | 61                   | 67                         |
| 15       | 270                  | 337                        |
| 16       | 491                  | 828                        |
| 17       | 153                  | 981                        |
| 18       | 15                   | 996                        |
| 19       | 4                    | 1000                       |
|          | 1000                 |                            |

Now that we have the cumulative frequency of each class, we are in a position to allocate the sampling numbers to all the values in a class. As the frequency as well as the cumulative frequency of the first class is 6, we allocate numbers 000 to 005 to the six students who belong to this class.

| AGE<br>X | No. of Students<br>f | cf   | Sampling Numbers |
|----------|----------------------|------|------------------|
| 13       | 6                    | 6    | 000 – 005        |
| 14       | 61                   | 67   |                  |
| 15       | 270                  | 337  |                  |
| 16       | 491                  | 828  |                  |
| 17       | 153                  | 981  |                  |
| 18       | 15                   | 996  |                  |
| 19       | 4                    | 1000 |                  |
|          | 1000                 |      |                  |

As the cumulative frequency of the second class is 67 while that of the first class was 6, therefore we allocate sampling numbers 006 to 066 to the 61 students who belong to this class.

| AGE<br>X | No. of Students<br>f | cf   | Sampling Numbers |
|----------|----------------------|------|------------------|
| 13       | 6                    | 6    | 000 – 005        |
| 14       | 61                   | 67   | 006 – 066        |
| 15       | 270                  | 337  |                  |
| 16       | 491                  | 828  |                  |
| 17       | 153                  | 981  |                  |
| 18       | 15                   | 996  |                  |
| 19       | 4                    | 1000 |                  |
|          | 1000                 |      |                  |

As the cumulative frequency of the third class is 337 while that of the second class was 67, therefore we allocate sampling numbers 067 to 336 to the 270 students who belong to this class.

| AGE<br>X | No. of<br>Students<br>f | cf   | Sampling<br>Numbers |
|----------|-------------------------|------|---------------------|
| 13       | 6                       | 6    | 000 – 005           |
| 14       | 61                      | 67   | 006 – 066           |
| 15       | 270                     | 337  | 067 – 336           |
| 16       | 491                     | 828  |                     |
| 17       | 153                     | 981  |                     |
| 18       | 15                      | 996  |                     |
| 19       | 4                       | 1000 |                     |
|          | 1000                    |      |                     |

Proceeding in this manner, we obtain the column of sampling numbers.

| AGE<br>X  | No. of<br>Students<br>f | cf          | Sampling<br>Numbers |
|-----------|-------------------------|-------------|---------------------|
| <b>13</b> | <b>6</b>                | <b>6</b>    | <b>000 – 005</b>    |
| <b>14</b> | <b>61</b>               | <b>67</b>   | <b>006 – 066</b>    |
| <b>15</b> | <b>270</b>              | <b>337</b>  | <b>067 – 336</b>    |
| <b>16</b> | <b>491</b>              | <b>828</b>  | <b>337 – 827</b>    |
| <b>17</b> | <b>153</b>              | <b>981</b>  | <b>828 – 980</b>    |
| <b>18</b> | <b>15</b>               | <b>996</b>  | <b>981 – 995</b>    |
| <b>19</b> | <b>4</b>                | <b>1000</b> | <b>996 - 999</b>    |
|           | <b>1000</b>             |             |                     |

The column implies that the first student of the first class has been allocated the sampling number 000, the second student has been allocated the sampling 001, and, proceeding in this fashion, the last student i.e. the 1000th student has been allocated the sampling number 999.

The question is: Why did we not allot the number 0001 to the first student and the number 1000 to the 1000th student? The answer is that we could do that but that would have meant that every student would have been allocated a four-digit number, whereas by shifting the number backward by 1, we are able to allocate to every student a three-digit number --- which is obviously simpler.

The next step is to SELECT 10 RANDOM NUMBERS from the random number table. This is accomplished by closing one's eyes and letting one's finger land anywhere on the random number table. In this example, since all our sampling numbers are three-digit numbers, hence we will read three digits that are adjacent to each other at that position where our finger landed. Suppose that we adopt this procedure and our random numbers come out to be 041, 103, 374, 171, 508, 652, 880, 066, 715, 471

Selected Random Numbers:

041, 103, 374, 171, 508, 652, 880, 066, 715, 471

Thus the corresponding ages are:

14, 15, 16, 15, 16, 16, 17, 15, 16, 16

### EXPLANATION

Our first selected random number is 041 which mean that we have to pick up the 42nd student. The cumulative frequency of the first class is 6 whereas the cumulative frequency of the second class is 67. This means that definitely the 42nd student does not belong to the first class but does belong to the second class.

| AGE<br>X | No. of<br>Students<br>f | cf   |
|----------|-------------------------|------|
| 13       | 6                       | 6    |
| 14       | 61                      | 67   |
| 15       | 270                     | 337  |
| 16       | 491                     | 828  |
| 17       | 153                     | 981  |
| 18       | 15                      | 996  |
| 19       | 4                       | 1000 |
|          | 1000                    |      |

The age of each student in this class is 14 years; hence, obviously, the age of the 42nd student is also 14 years. This is how we are able to ascertain the ages of all the students who have been selected in our sampling. You will recall that in this example, our aim was to draw a sample from the population of college students, and to compare the sample's mean age with the population mean age. The population mean age comes out to be 15.785 years.

| AGE<br>X | No. of Students<br>f | fX    |
|----------|----------------------|-------|
| 13       | 6                    | 78    |
| 14       | 61                   | 854   |
| 15       | 270                  | 4050  |
| 16       | 491                  | 7856  |
| 17       | 153                  | 2601  |
| 18       | 15                   | 270   |
| 19       | 4                    | 76    |
|          | 1000                 | 15785 |

The population mean age is :

$$\mu = \frac{\sum fx}{\sum f} = \frac{15785}{1000} = 15.785 \text{ years}$$

The above formula is a slightly modified form of the basic formula that you have done ever-since school-days i.e. the mean is equal to the sum of all the observations divided by the total number of observations.

Next, we compute the sample mean age.

Adding the 10 values and dividing by 10, we obtain:

Ages of students selected in the sample (in years):

14, 15, 16, 15, 16, 16, 17, 15, 16, 16

Hence the sample mean age is: 15.6, comparing the sample mean age of 15.6 years with the population mean age of 15.785 years, we note that the difference is really quite slight, and hence the sampling error is equal to

Sampling Error

$$\begin{aligned}\bar{X} - \mu &= 15.6 - 15.785 \\ &= -0.185 \text{ years}\end{aligned}$$

And the reason for such a small error is that we have adopted the RANDOM sampling method. The basic advantage of random sampling is that the probability is very high that the sample will be a good representative of the population from which it has been drawn, and any quantity computed from the sample will be a good estimate of the corresponding quantity computed from the population! Actually, a sample is supposed to be a MINIATURE REPLICA of the population. As stated earlier, there are various other types of random sampling.

#### **OTHER TYPES OF RANDOM SAMPLING**

- Stratified sampling (if the population is heterogeneous)
- Systematic sampling (practically, more convenient than simple random sampling)
- Cluster sampling (sometimes the sampling units exist in natural clusters)
- Multi-stage sampling

All these designs rest upon random or quasi-random sampling. They are various forms of PROBABILITY sampling --- that in which each sampling unit has a known (but not necessarily equal) probability of being selected.

Because of this knowledge, there exist methods by which the precision and the reliability of the estimates can be calculated OBJECTIVELY.

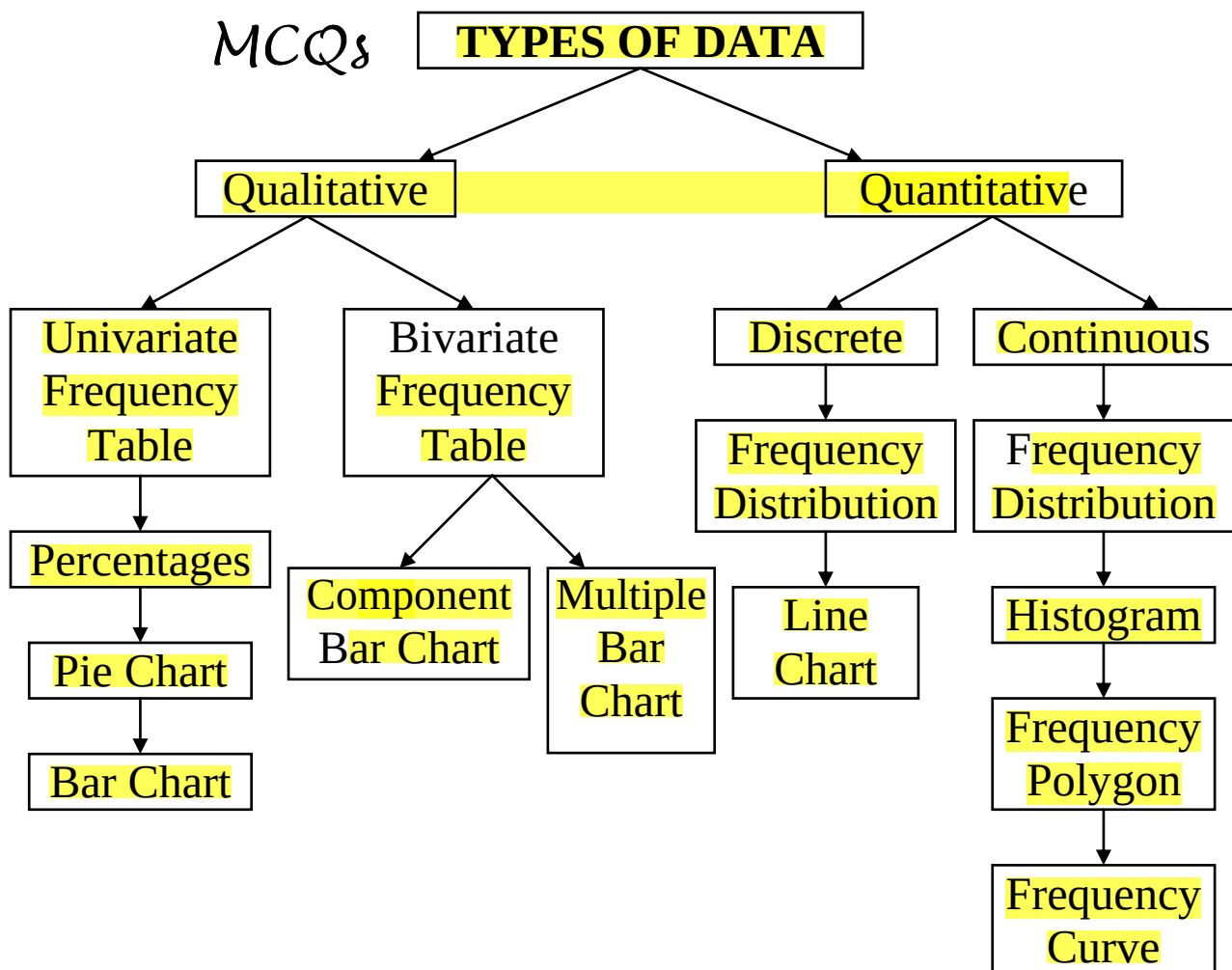
It should be realized that in practice, several sampling techniques are incorporated into each survey design, and only rarely will simple random sample be used, or a multi-stage design be employed, without stratification.

The point to remember is that whatever method be adopted, care should be exercised at every step so as to make the results as reliable as possible.

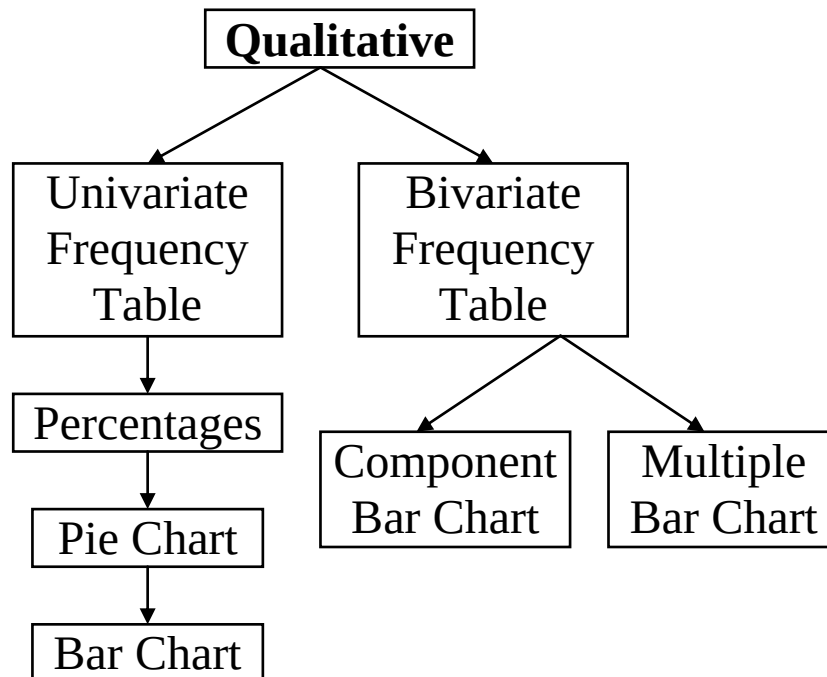
**LECTURE NO.3**

- Tabulation
- Simple bar chart
- Component bar chart
- Multiple bar chart
- Pie chart

As indicated in the last lecture, there are two broad categories of data ... qualitative data and quantitative data. A variety of methods exist for summarizing and describing these two types of data. The tree-diagram below presents an outline of the various techniques



In today's lecture, we will be dealing with various techniques for summarizing and describing qualitative data.



We will begin with the univariate situation, and will proceed to the bivariate situation.

#### **EXAMPLE**

Suppose that we are carrying out a survey of the students of first year studying in a co-educational college of Lahore. Suppose that in all there are 1200 students of first year in this large college. We wish to determine what proportion of these students have come from Urdu medium schools and what proportion has come from English medium schools. So we will interview the students and we will inquire from each one of them about their schooling. As a result, we will obtain a set of data as you can now see on the screen.

We will have an array of observations as follows:

U, U, E, U, E, E, E, U, .....

(U : URDU MEDIUM)

(E : ENGLISH MEDIUM)

Now, the question is what should we do with this data?

Obviously, the first thing that comes to mind is to count the number of students who said “Urdu medium” as well as the number of students who said “English medium”. This will result in the following table:

| Medium of Institution | No. of Students (f) |
|-----------------------|---------------------|
| Urdu                  | 719                 |
| English               | 481                 |
|                       | 1200                |

The technical term for the numbers given in the second column of this table is “frequency”. It means “how frequently something happens?” Out of the 1200 students, 719 stated that they had come from Urdu medium schools. So in this example, the frequency of the first category of responses is 719 whereas the frequency of the second category of responses is 481.

It is evident that this information is not as useful as if we compute the proportion or percentage of students falling in each category. Dividing the cell frequencies by the total frequency and multiplying by 100 we obtain the following:

urdu medium  
 $719/1200$   
 $=59.9$   
 english medium

$481/1200$   
 $= 40.2$

| Medium of Institution | f    | %             |
|-----------------------|------|---------------|
| Urdu                  | 719  | $59.9 = 60\%$ |
| English               | 481  | $40.1 = 40\%$ |
|                       | 1200 |               |

What we have just accomplished is an example of a univariate frequency table pertaining to qualitative data.

Let us now see how we can represent this information in the form of a diagram.

One good way of representing the above information is in the form of a pie chart.

A pie chart consists of a circle which is divided into two or more parts in accordance with the number of distinct categories that we have in our data.

For the example that we have just considered, the circle is divided into two sectors, the larger sector pertaining to students coming from Urdu medium schools and the smaller sector pertaining to students coming from English medium schools.

How do we decide where to cut the circle?

The answer is very simple! All we have to do is to divide the cell frequency by the total frequency and multiply by 360.

This process will give us the exact value of the angle at which we should cut the circle.

### PIE CHART

| Medium of Institution | f    | Angle     |
|-----------------------|------|-----------|
| Urdu                  | 719  | $215.7^0$ |
| English               | 481  | $144.3^0$ |
|                       | 1200 |           |

urdu

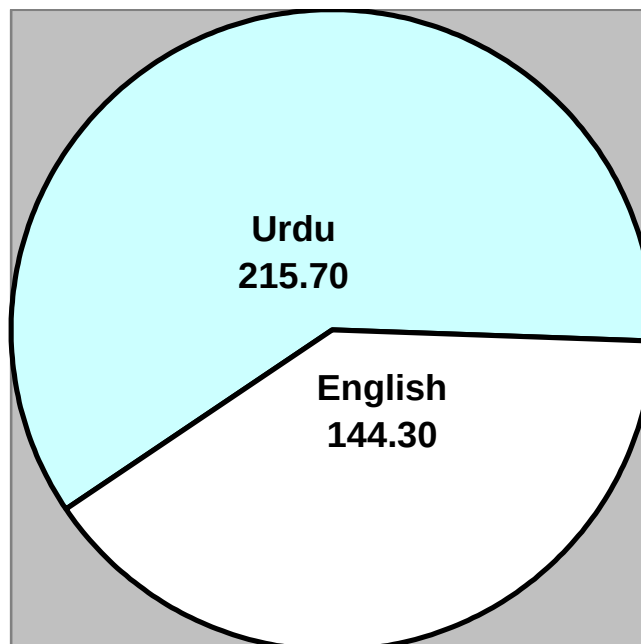
$719/1200 \times 360$

$= 215.7$

English

$481/1200 \times 360$

$= 144.3$



**SIMPLE BAR CHART:**

The next diagram to be considered is the simple bar chart.

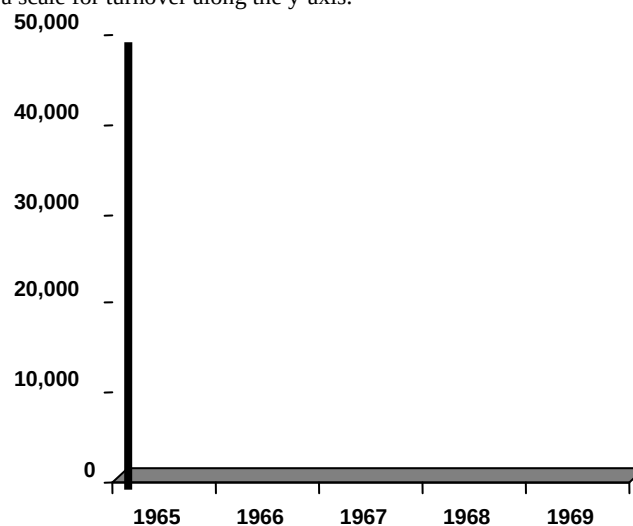
A simple bar chart consists of horizontal or vertical bars of equal width and lengths proportional to values they represent.

As the basis of comparison is one-dimensional, the widths of these bars have no mathematical significance but are taken in order to make the chart look attractive. Let us consider an example.

Suppose we have available to us information regarding the turnover of a company for 5 years as given in the table below:

| Years             | 1965   | 1966   | 1967   | 1968   | 1969   |
|-------------------|--------|--------|--------|--------|--------|
| Turnover (Rupees) | 35,000 | 42,000 | 43,500 | 48,000 | 48,500 |

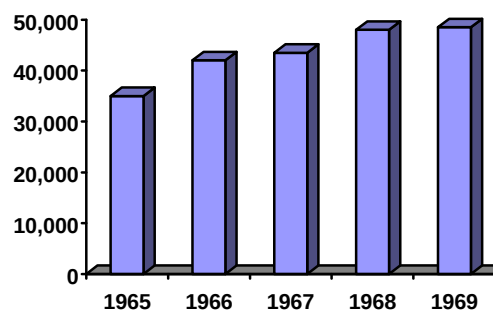
In order to represent the above information in the form of a bar chart, all we have to do is to take the year along the x-axis and construct a scale for turnover along the y-axis.



Next, against each year, we will draw vertical bars of equal width and different heights in accordance with the turn-over figures that we have in our table.

As a result we obtain a simple and attractive diagram as shown below.

When our values do not relate to time, they should be arranged in ascending or descending order before-charting.

**BIVARIATE FREQUENCY TABLE**

What we have just considered was the univariate situation. In each of the two examples, we were dealing with one single variable. In the example of the first year students of a college, our lone variable of interest was 'medium of schooling'. And in the second example, our one single variable of interest was turnover. Now let us expand the discussion a little, and consider the bivariate situation.



Going back to the example of the first year students, suppose that alongwith the enquiry about the Medium of Institution, you are also recording the sex of the student.

Suppose that our survey results in the following information:

| Student No. | Medium | Gender |
|-------------|--------|--------|
| 1           | U      | F      |
| 2           | U      | M      |
| 3           | E      | M      |
| 4           | U      | F      |
| 5           | E      | M      |
| 6           | E      | F      |
| 7           | U      | M      |
| 8           | E      | M      |
| :           | :      | :      |
| :           | :      | :      |

Now this is a bivariate situation; we have two variables, medium of schooling and sex of the student.

In order to summarize the above information, we will construct a table containing a box head and a stub as shown below:

| Sex<br>Med. | M<br>A<br>L<br>E | Female | Total |
|-------------|------------------|--------|-------|
| Urdu        |                  |        |       |
| English     |                  |        |       |
| Total       |                  |        |       |

The top row of this kind of a table is known as the boxhead and the first column of the table is known as stub. Next, we will count the number of students falling in each of the following four categories:

1. Male student coming from an Urdu medium school.
2. Female student coming from an Urdu medium school.
3. Male student coming from an English medium school.
4. Female student coming from an English medium school.

As a result, suppose we obtain the following figures:

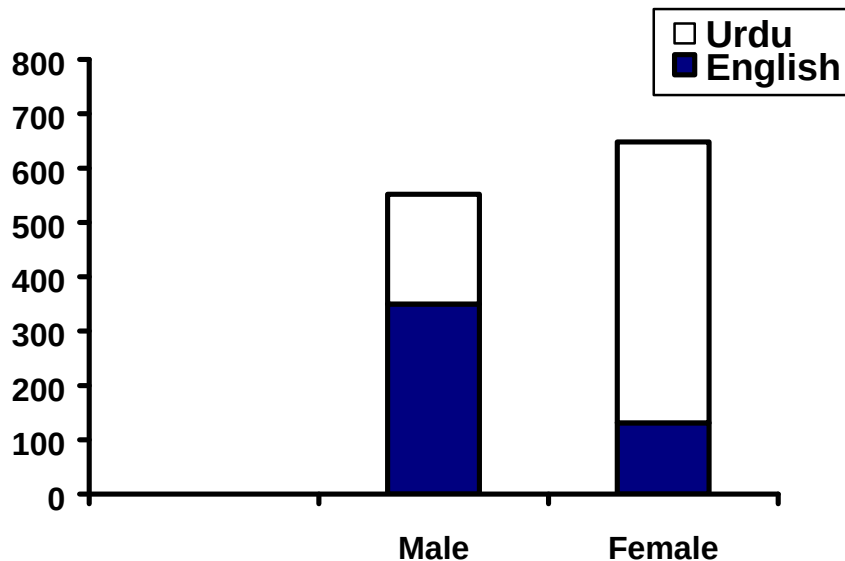
| Sex<br>Med. | M<br>A<br>L<br>E | Female | Total |
|-------------|------------------|--------|-------|
| Urdu        | 202              | 517    | 719   |
| English     | 350              | 131    | 481   |
| Total       | 552              | 648    | 1200  |

What we have just accomplished is an example of a bivariate frequency table pertaining to two qualitative variables.

**COMPONENT BAR CHAR:**

Let us now consider how we will depict the above information diagrammatically.

This can be accomplished by constructing the component bar chart (also known as the subdivided bar chart) as shown below:



In the above figure, each bar has been divided into two parts. The first bar represents the total number of male students whereas the second bar represents the total number of female students.

As far as the medium of schooling is concerned, the lower part of each bar represents the students coming from English medium schools. Whereas the upper part of each bar represents the students coming from the Urdu medium schools. The advantage of this kind of a diagram is that we are able to ascertain the situation of both the variables at a glance.

We can compare the number of male students in the college with the number of female students, and at the same time we can compare the number of English medium students among the males with the number of English medium students among the females.

**MULTIPLE BAR CHARTS**

The next diagram to be considered is the multiple bar charts. Let us consider an example.

Suppose we have information regarding the imports and exports of Pakistan for the years 1970-71 to 1974-75 as shown in the table below:

| Years   | Imports<br>(Crores of Rs.) | Exports<br>(Crores of Rs.) |
|---------|----------------------------|----------------------------|
| 1970-71 | 370                        | 200                        |
| 1971-72 | 350                        | 337                        |
| 1972-73 | 840                        | 855                        |
| 1973-74 | 1438                       | 1016                       |
| 1974-75 | 2092                       | 1029                       |

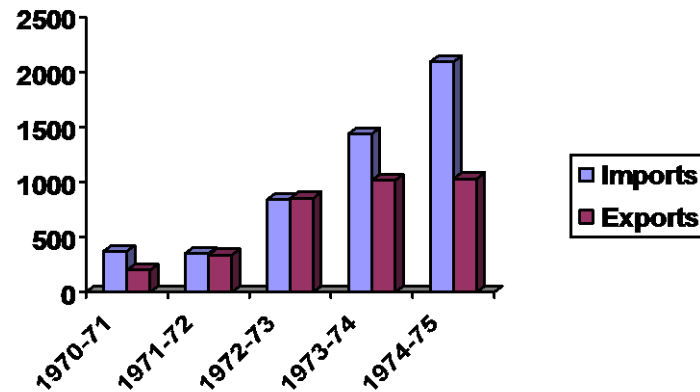
Source: State Bank of Pakistan

A multiple bar chart is a very useful and effective way of presenting this kind of information.

This kind of a chart consists of a set of grouped bars, the lengths of which are proportionate to the values of our variables, and each of which is shaded or colored differently in order to aid identification. With reference to the above example, we obtain the multiple bar chart shown below:

**Multiple Bar Chart Showing  
Imports & Exports  
of Pakistan 1970-71 to 1974-75**

This is a very good device for the comparison of two different kinds of information. mcqs in 2023



If, in addition to information regarding imports and exports, we also had information regarding production, we could have compared them from year to year by grouping the three bars together.

The question is, what is the basic difference between a component bar chart and a multiple bar chart?

The component bar chart should be used when we have available to us information regarding totals and their components.

For example, the total number of male students out of which some are Urdu medium and some are English medium. The number of Urdu medium male students and the number of English medium male students add up to give us the total number of male students.

On the contrary, in the example of exports and imports, the imports and exports do not add up to give us the totality of some one thing!

*Masters*

*Subscribes to Masters*

**LECTURE NO. 4**

In THIS Lecture, we will discuss the frequency distribution of a continuous variable & the graphical ways of representing data pertaining to a continuous variable i.e. **histogram, frequency polygon and frequency curve**.

You will recall that in Lecture No. 1, it was mentioned that a continuous variable takes values over a continuous interval (e.g. a normal Pakistani adult male's height may lie anywhere between 5.25 feet and 6.5 feet).

Hence, in such a situation, the method of constructing a frequency distribution is somewhat different from the one that was discussed in the last lecture.

**EXAMPLE:**

Suppose that the Environmental Protection Agency of a developed country performs extensive tests on all new car models in order to determine their mileage rating. Suppose that the following 30 measurements are obtained by conducting such tests on a particular new car model.

| <b>EPA MILEAGE RATINGS ON 30 CARS<br/>(MILES PER GALLON)</b> |      |      |
|--------------------------------------------------------------|------|------|
| 36.3                                                         | 42.1 | 44.9 |
| 30.1                                                         | 37.5 | 32.9 |
| 40.5                                                         | 40.0 | 40.2 |
| 36.2                                                         | 35.6 | 35.9 |
| 38.5                                                         | 38.8 | 38.6 |
| 36.3                                                         | 38.4 | 40.5 |
| 41.0                                                         | 39.0 | 37.0 |
| 37.0                                                         | 36.7 | 37.1 |
| 37.1                                                         | 34.8 | 33.9 |
| 39.9                                                         | 38.1 | 39.8 |

EPA: Environmental Protection Agency

There are a few steps in the **construction of a frequency distribution** for this type of a variable.

**CONSTRUCTION OF A FREQUENCY DISTRIBUTION****Step-1**

**Identify the smallest and the largest measurements in the data set.**

**In our example:**

**Smallest value ( $X_0$ ) = 30.1,**

**Largest Value ( $X_m$ ) = 44.9,**

**Step-2**

**Find the range** which is defined as the **difference between the largest value and the smallest value**

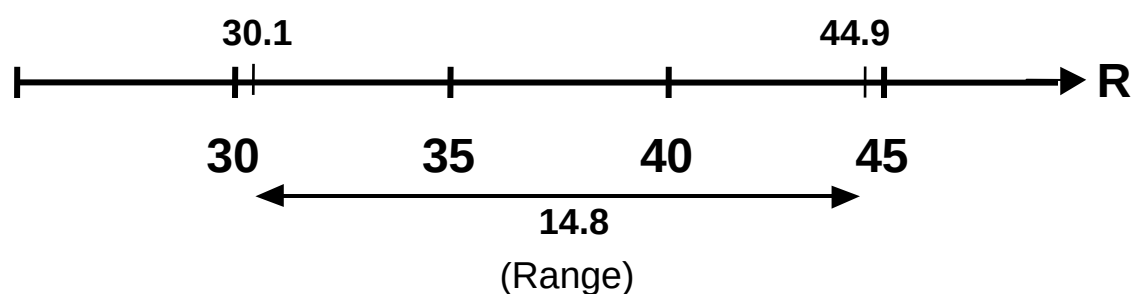
In our example:

$$\text{Range} = X_m - X_0$$

$$= 44.9 - 30.1$$

$$= 14.8$$

Let us now look at the graphical picture of what we have just computed.

**Step-3**

Decide on the number of classes into which the data are to be grouped.

(By classes, we mean small sub-intervals of the total interval which, in this example, is 14.8 units long.) There are no hard and fast rules for this purpose. The decision will depend on the size of the data. When the data are sufficiently large, the number of classes is usually taken between 10 and 20. In this example, suppose that we decide to form 5 classes (as there are only 30 observations).

**Step-4**

Divide the range by the chosen number of classes in order to obtain the approximate value of the class interval i.e. the width of our classes.

Class interval is usually denoted by  $h$ . Hence, in this example

$$\text{Class interval} = h = 14.8 / 5 = 2.96$$

Rounding the number 2.96, we obtain 3, and hence we take  $h = 3$ .

This means that our big interval will be divided into small sub-intervals, each of which will be 3 units long.

**Step-5**

Decide the lower class limit of the lowest class. Where should we start from?

The answer is that we should start constructing our classes from a number equal to or slightly less than the smallest value in the data.

In this example, smallest value = 30.1

So we may choose the lower class limit of the lowest class to be 30.0.

**Step-6**

Determine the lower class limits of the successive classes by adding  $h = 3$  successively. Hence, we obtain the following table:

| Class Number | Lower Class Limit |
|--------------|-------------------|
| 1            | 30.0              |
| 2            | $30.0 + 3 = 33.0$ |
| 3            | $33.0 + 3 = 36.0$ |
| 4            | $36.0 + 3 = 39.0$ |
| 5            | $39.0 + 3 = 42.0$ |

**Step-7**

Determine the upper class limit of every class. The upper class limit of the highest class should cover the largest value in the data. It should be noted that the upper class limits will also have a difference of  $h$  between them. Hence, we obtain the upper class limits that are visible in the third column of the following table.

| Class Number | Lower Class Limit | Upper Class Limit |
|--------------|-------------------|-------------------|
| 1            | 30.0              | 32.9              |
| 2            | $30.0 + 3 = 33.0$ | $32.9 + 3 = 35.9$ |
| 3            | $33.0 + 3 = 36.0$ | $35.9 + 3 = 38.9$ |
| 4            | $36.0 + 3 = 39.0$ | $38.9 + 3 = 41.9$ |
| 5            | $39.0 + 3 = 42.0$ | $41.9 + 3 = 44.9$ |

Hence we obtain the following classes:

| Classes     |
|-------------|
| 30.0 – 32.9 |
| 33.0 – 35.9 |
| 36.0 – 38.9 |
| 39.0 – 41.9 |
| 42.0 – 44.9 |

The question arises: why did we not write 33 instead of 32.9? Why did we not write 36 instead of 35.9? and so on.

The reason is that if we wrote 30 to 33 and then 33 to 36, we would have trouble when tallying our data into these classes. Where should I put the value 33? Should I put it in the first class, or should I put it in the second class? By writing 30.0 to 32.9 and 33.0 to 35.9, we avoid this problem. And the point to be noted is that the class interval is still 3, and not 2.9 as it appears to be. This point will be better understood when we discuss the concept of class boundaries ... which will come a little later in today's lecture.

#### Step-8

After forming the classes, distribute the data into the appropriate classes and find the frequency of each class, in this example:

| Class       | Tally        | Frequency |
|-------------|--------------|-----------|
| 30.0 – 32.9 |              | 2         |
| 33.0 – 35.9 |              | 4         |
| 36.0 – 38.9 | <br>    <br> | 14        |
| 39.0 – 41.9 | <br>         | 8         |
| 42.0 – 44.9 |              | 2         |
|             | <b>Total</b> | <b>30</b> |

This is a simple example of the frequency distribution of a continuous or, in other words, measurable variable.

#### CLASS BOUNDARIES:

The true class limits of a class are known as its class boundaries. In this example:

| Class Limit | Class Boundaries | Frequency |
|-------------|------------------|-----------|
| 30.0 – 32.9 | 29.95 – 32.95    | 2         |
| 33.0 – 35.9 | 32.95 – 35.95    | 4         |
| 36.0 – 38.9 | 35.95 – 38.95    | 14        |
| 39.0 – 41.9 | 38.95 – 41.95    | 8         |
| 42.0 – 44.9 | 41.95 – 44.95    | 2         |
|             | <b>Total</b>     | <b>30</b> |

It should be noted that the difference between the upper class boundary and the lower class boundary of any class is equal to the class interval  $h = 3$ .

32.95 minus 29.95 is equal to 3, 35.95 minus 32.95 is equal to 3, and so on.

**A key point in this entire discussion is that the class boundaries should be taken up to one decimal place more than the given data.** In this way, the possibility of an observation falling exactly on the boundary is avoided. (The observed value will either be greater than or less than a particular boundary and hence will conveniently fall in its appropriate class).

Next, we consider the concept of the relative frequency distribution and the percentage frequency distribution.

Next, we consider the concept of the relative frequency distribution and the percentage frequency distribution.

This concept has already been discussed when we considered the frequency distribution of a discrete variable.

Dividing each frequency of a frequency distribution by the total number of observations, we obtain the relative frequency distribution.

Multiplying each relative frequency by 100, we obtain the percentage of frequency distribution.

In this way, we obtain the relative frequencies and the percentage frequencies shown below

| <b>Class Limit</b> | <b>Frequency</b> | <b>Relative Frequency</b>         | <b>%age Frequency</b> |
|--------------------|------------------|-----------------------------------|-----------------------|
| <b>30.0 – 32.9</b> | <b>2</b>         | <b><math>2/30 = 0.067</math></b>  | <b>6.7</b>            |
| <b>33.0 – 35.9</b> | <b>4</b>         | <b><math>4/30 = 0.133</math></b>  | <b>13.3</b>           |
| <b>36.0 – 38.9</b> | <b>14</b>        | <b><math>14/30 = 0.467</math></b> | <b>46.7</b>           |
| <b>39.0 – 41.9</b> | <b>8</b>         | <b><math>8/30 = 0.267</math></b>  | <b>26.7</b>           |
| <b>42.0 – 44.9</b> | <b>2</b>         | <b><math>2/30 = 0.067</math></b>  | <b>6.7</b>            |
|                    | <b>30</b>        |                                   |                       |

The term ‘relative frequencies’ simply means that we are considering the frequencies of the various classes relative to the total number of observations.

The advantage of constructing a relative frequency distribution is that comparison is possible between two sets of data having similar classes.

For example, suppose that the Environment Protection Agency perform tests on two car models A and B, and obtains the frequency distributions shown below:

| MILEAGE     | FREQUENCY |         |
|-------------|-----------|---------|
|             | Model A   | Model B |
| 30.0 – 32.9 | 2         | 7       |
| 33.0 – 35.9 | 4         | 10      |
| 36.0 – 38.9 | 14        | 16      |
| 39.0 – 41.9 | 8         | 9       |
| 42.0 – 44.9 | 2         | 8       |
|             | 30        | 50      |

In order to be able to compare the performance of the two car models, we construct the relative frequency distributions in the percentage form:

| MILEAGE   | Model A                   | Model B                 |
|-----------|---------------------------|-------------------------|
| 30.0-32.9 | $2/30 \times 100 = 6.7$   | $7/50 \times 100 = 14$  |
| 33.0-35.9 | $4/30 \times 100 = 13.3$  | $10/50 \times 100 = 20$ |
| 36.0-38.9 | $14/30 \times 100 = 46.7$ | $16/50 \times 100 = 32$ |
| 39.0-41.9 | $8/30 \times 100 = 26.7$  | $9/50 \times 100 = 18$  |
| 42.0-44.9 | $2/30 \times 100 = 6.7$   | $8/50 \times 100 = 16$  |

From the table it is clear that whereas 6.7% of the cars of model A fall in the mileage group 42.0 to 44.9, as many as 16% of the cars of model B fall in this group. Other comparisons can similarly be made.

Let us now turn to the visual representation of a continuous frequency distribution. In this context, we will discuss three different types of graphs i.e. the histogram, the frequency polygon, and the frequency curve.

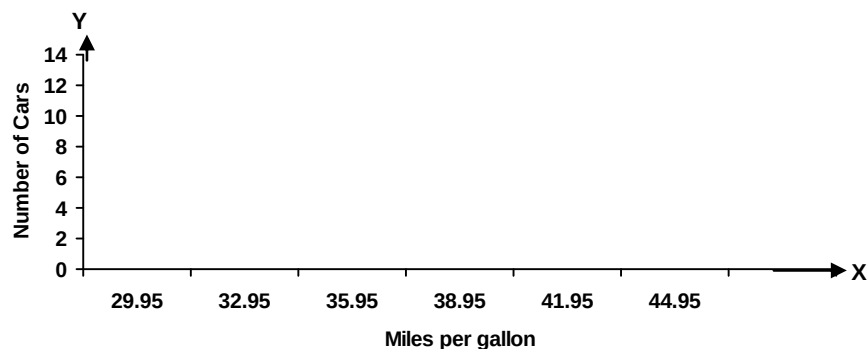
### **HISTOGRAM:**

A histogram consists of a set of adjacent rectangles whose bases are marked off by class boundaries along the X-axis, and whose heights are proportional to the frequencies associated with the respective classes.

It will be recalled that, in the last lecture, we were considering the mileage ratings of the cars that had been inspected by the Environment Protection Agency. Our frequency table came out as shown below:

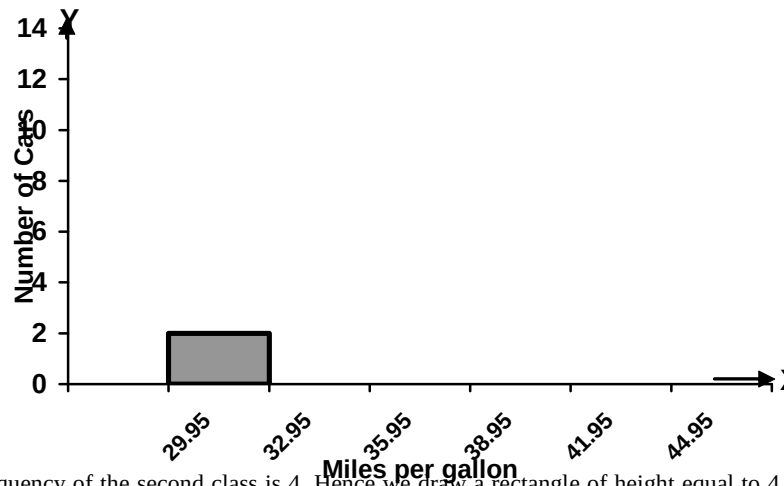
| Class Limit | Class Boundaries | Frequency |
|-------------|------------------|-----------|
| 30.0 – 32.9 | 29.95 – 32.95    | 2         |
| 33.0 – 35.9 | 32.95 – 35.95    | 4         |
| 36.0 – 38.9 | 35.95 – 38.95    | 14        |
| 39.0 – 41.9 | 38.95 – 41.95    | 8         |
| 42.0 – 44.9 | 41.95 – 44.95    | 2         |
|             | Total            | 30        |

In accordance with the procedure that I just mentioned, we need to take the class boundaries along the X axis. We obtain

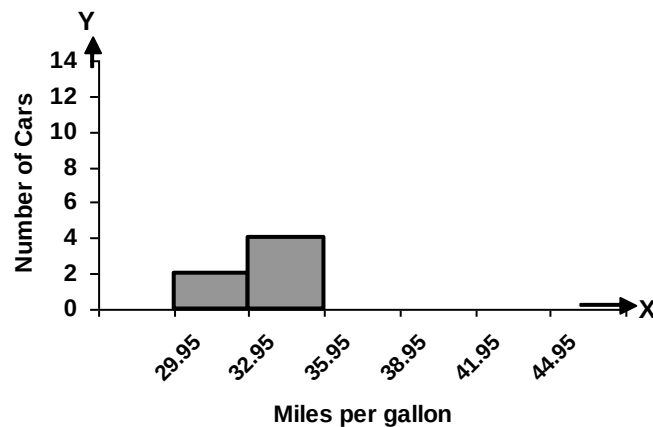




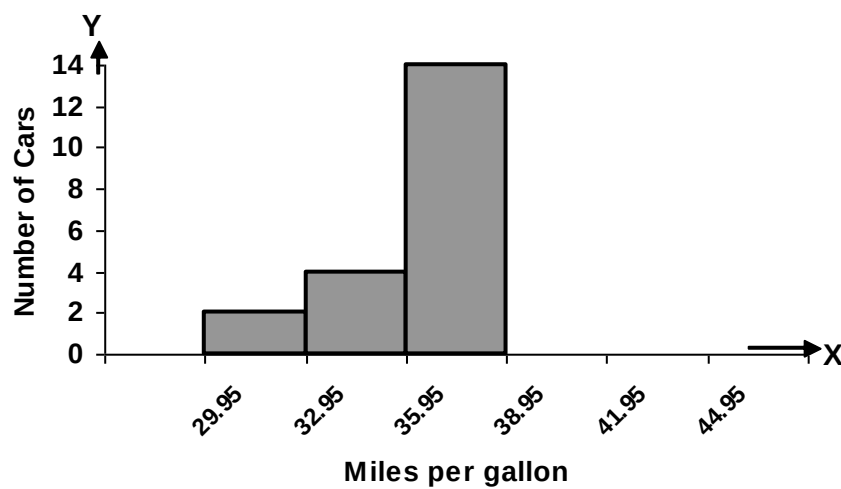
Now, as seen in the frequency table, the frequency of the first class is 2. As such, we will draw a rectangle of height equal to 2 units and obtain the following figure:



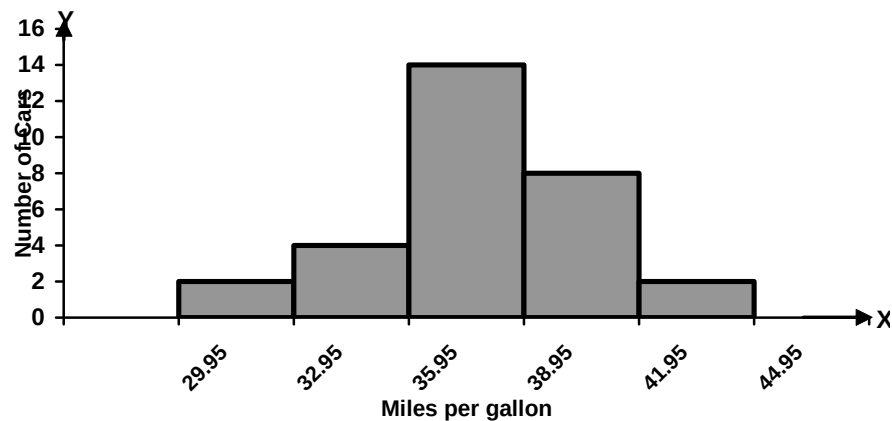
The frequency of the second class is 4. Hence we draw a rectangle of height equal to 4 units against the second class, and thus obtain the following situation:



The frequency of the third class is 14. Hence we draw a rectangle of height equal to 14 units against the third class, and thus obtain the following picture:



Continuing in this fashion, we obtain the following attractive diagram:



This diagram is known as the histogram, and it gives an indication of the overall pattern of our frequency distribution.

#### **FREQUENCY POLYGON:**

A frequency polygon is obtained by plotting the class frequencies against the mid-points of the classes, and connecting the points so obtained by straight line segments. In our example of the EPA mileage ratings, the classes are

| Class Boundaries |
|------------------|
| 29.95 – 32.95    |
| 32.95 – 35.95    |
| 35.95 – 38.95    |
| 38.95 – 41.95    |
| 41.95 – 44.95    |

These mid-points are denoted by X.

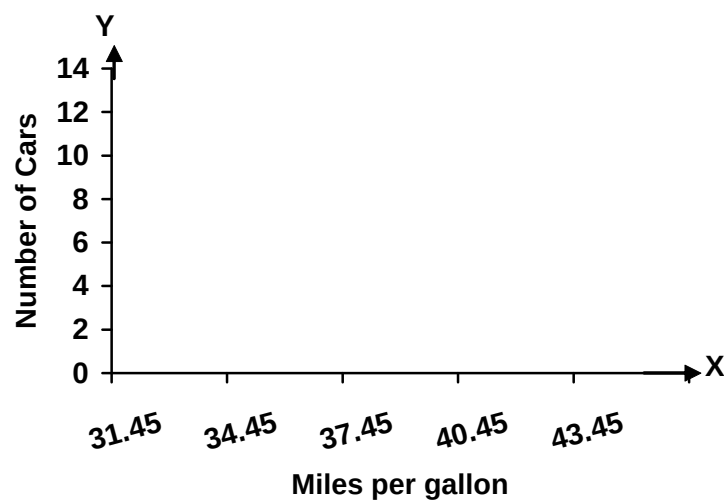
Now let us add two classes to my frequency table, one class in the very beginning, and one class at the very end.

| Class Boundaries | Mid-Point (X) | Frequency (f) |
|------------------|---------------|---------------|
| 26.95 – 29.95    | 28.45         |               |
| 29.95 – 32.95    | 31.45         | 2             |
| 32.95 – 35.95    | 34.45         | 4             |
| 35.95 – 38.95    | 37.45         | 14            |
| 38.95 – 41.95    | 40.45         | 8             |
| 41.95 – 44.95    | 43.45         | 2             |
| 44.95 – 47.95    | 46.45         |               |

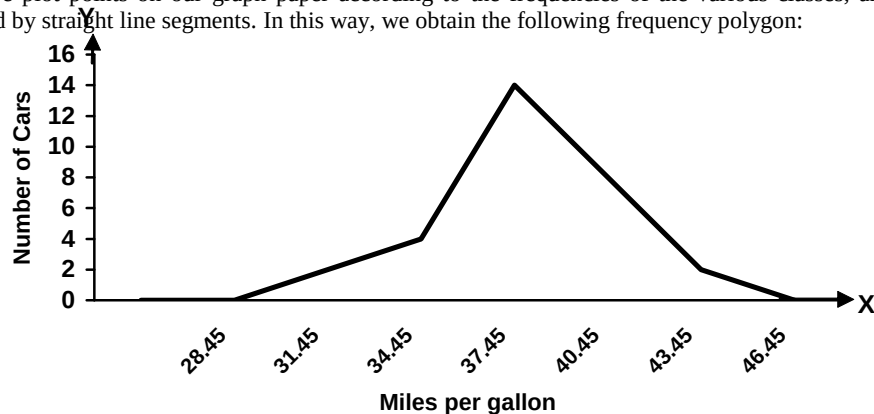
The frequency of each of these two classes is 0, as in our data set, no value falls in these classes.

| Class Boundaries | Mid-Point (X) | Frequency (f) |
|------------------|---------------|---------------|
| 26.95 – 29.95    | 28.45         | 0             |
| 29.95 – 32.95    | 31.45         | 2             |
| 32.95 – 35.95    | 34.45         | 4             |
| 35.95 – 38.95    | 37.45         | 14            |
| 38.95 – 41.95    | 40.45         | 8             |
| 41.95 – 44.95    | 43.45         | 2             |
| 44.95 – 47.95    | 46.45         | 0             |

Now, in order to construct the frequency polygon, the mid-points of the classes are taken along the X-axis and the frequencies along the Y-axis, as shown below:

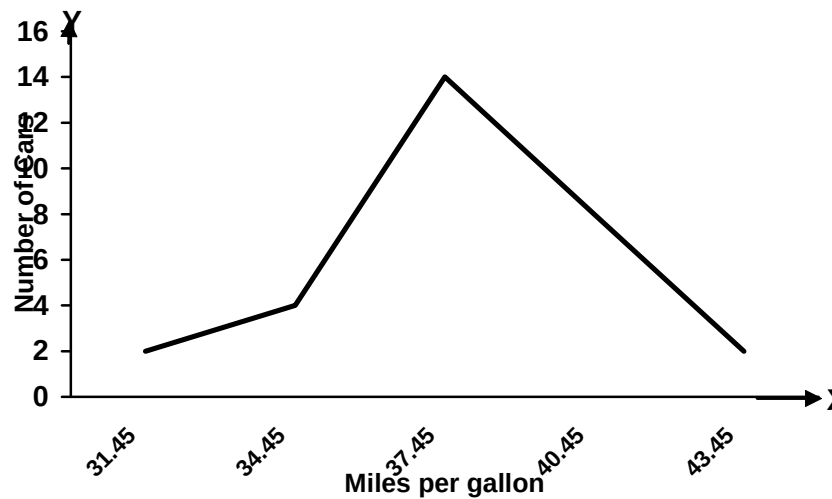


Next, we plot points on our graph paper according to the frequencies of the various classes, and join the points so obtained by straight line segments. In this way, we obtain the following frequency polygon:



It is well-known that the term ‘polygon’ implies a many-sided closed figure. As such, we want our frequency polygon to be a closed figure. This is exactly the reason why we added two classes to our table, each having zero frequency. Because of the frequency being zero, the line segment touches the X-axis both at the beginning and at the end, and our figure becomes a closed figure.

Had we not carried out this step, our graph would have been as follows:

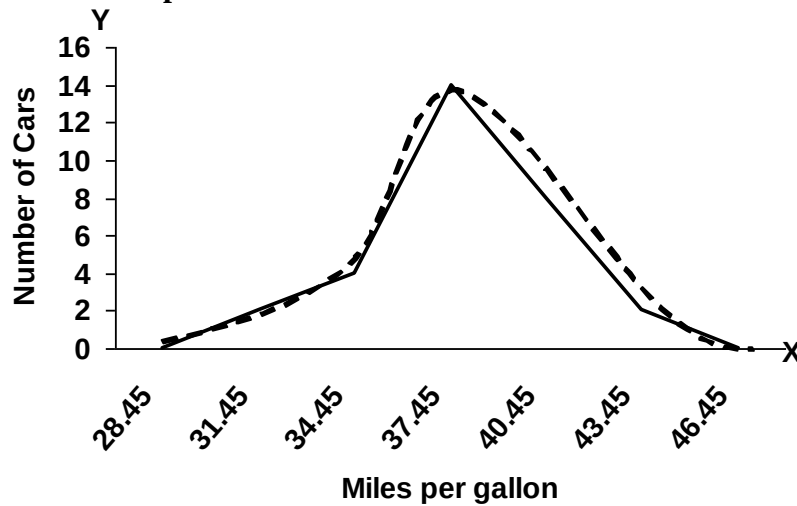


And since this graph is touching the X-axis, hence it cannot be called a frequency polygon (because it is not a closed figure)!

#### **FREQUENCY CURVE:**

When the frequency polygon is **smoothed**, we obtain what may be called the frequency curve.

In our example:



## LECTURE NO. 5

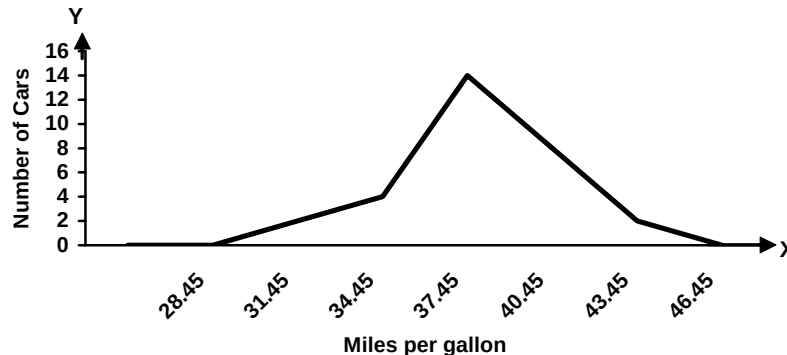
Today's lecture is in continuation with the last lecture, and today we will begin with various types of frequency curves that are encountered in practice. Also, we will discuss the cumulative frequency distribution and cumulative frequency polygon for a continuous variable.

**FREQUENCY POLYGON:**

A frequency polygon is obtained by plotting the class frequencies against the mid-points of the classes, and connecting the points so obtained by straight line segments. In our example of the EPA mileage ratings, the classes were:

| Class Boundaries | Mid-Point (X) | Frequency (f) |
|------------------|---------------|---------------|
| 26.95 – 29.95    | 28.45         |               |
| 29.95 – 32.95    | 31.45         | 2             |
| 32.95 – 35.95    | 34.45         | 4             |
| 35.95 – 38.95    | 37.45         | 14            |
| 38.95 – 41.95    | 40.45         | 8             |
| 41.95 – 44.95    | 43.45         | 2             |
| 44.95 – 47.95    | 46.45         |               |

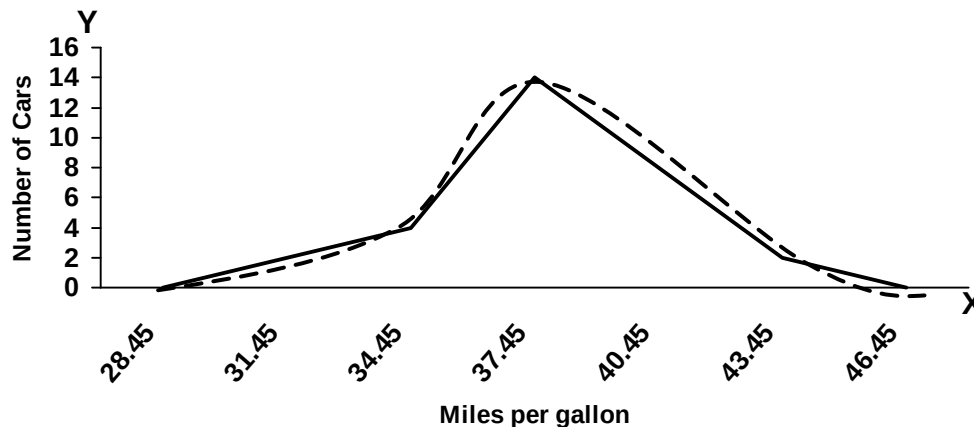
And our frequency polygon came out to be:



Also, it was mentioned that, when the frequency polygon is smoothed, we obtain what may be called the

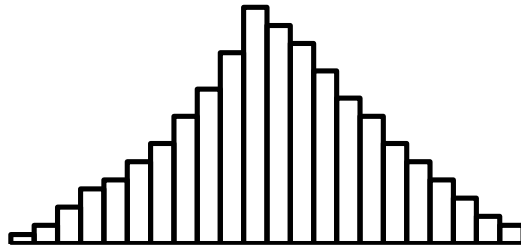
**FREQUENCY CURVE**

In our example:



In the above figure, the dotted line represents the frequency curve. It should be noted that it is not necessary that our frequency curve must touch all the points. The purpose of the frequency curve is simply to display the overall pattern of the distribution. Hence we draw the curve by the free-hand method, and hence it does not have to touch all the plotted points. It should be realized that the frequency curve is actually a theoretical concept.

If the class interval of a histogram is made very small, and the number of classes is very large, the rectangles of the histogram will be narrow as shown below:



The smaller the class interval and the larger the number of classes, the narrower the rectangles will be. In this way, the histogram approaches a smooth curve as shown below:



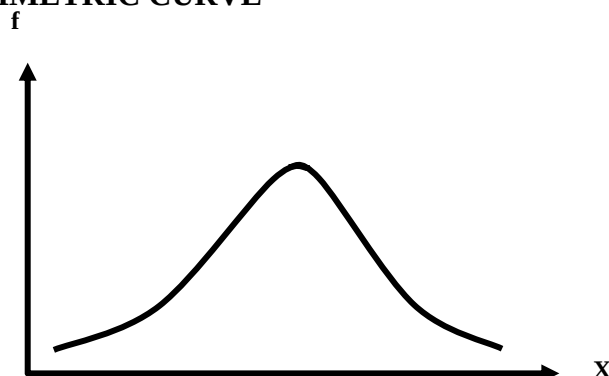
In spite of the fact that the frequency curve is a theoretical concept, it is useful in analyzing real-world problems. The reason is that very close approximations to theoretical curves are often generated in the real world so close that it is quite valid to utilize the properties of various types of mathematical curves in order to aid analysis of the real-world problem at hand.

#### VARIOUS TYPES OF FREQUENCY CURVES

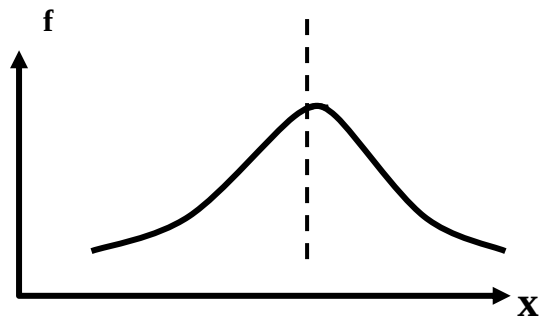
- the symmetrical frequency curve
- the moderately skewed frequency curve
- the extremely skewed frequency curve
- the U-shaped frequency curve

Let us discuss them one by one. First of all, the symmetrical frequency curve is of the following shape:

#### THE SYMMETRIC CURVE

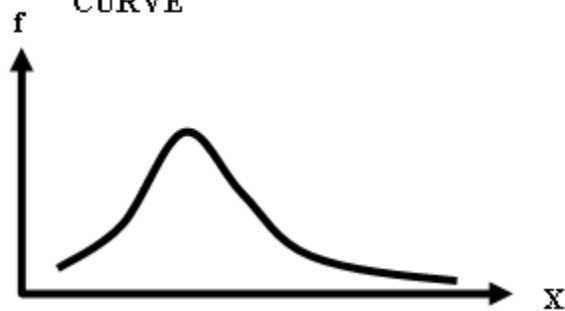


If we place a vertical mirror in the centre of this graph, the left hand side will be the mirror image of the right hand side.

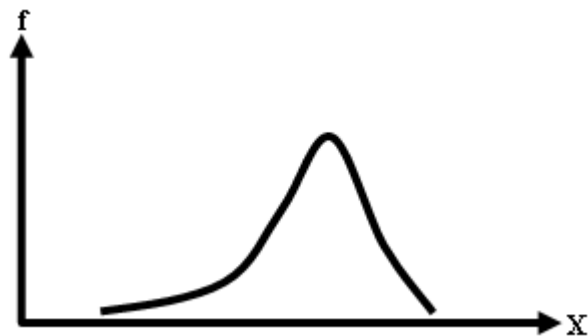


Next, we consider the moderately skewed frequency curve. We have the positively skewed curve and the negatively skewed curve. The positively skewed curve is that one whose right tail is longer than its left tail, as shown below

#### THE POSITIVELY SKEWED CURVE



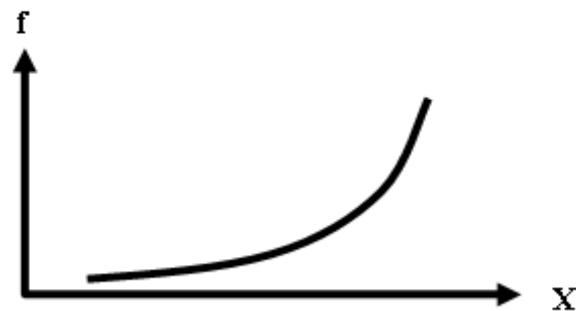
On the other hand, the negatively skewed frequency curve is the one for which the left tail is longer than the right tail.



Both of these that we have just considered are moderately positively and negatively skewed.

Sometimes, we have the extreme case when we obtain the EXTREMELY skewed frequency curve. An extremely negatively skewed curve is of the type shown below:

**THE EXTREMELY  
NEGATIVELY SKEWED  
(J-SHAPED) CURVE**



This is the case when the maximum frequency occurs at the end of the frequency table.

For example, if we think of the death rates of adult males of various age groups starting from age 20 and going up to age 79 years, we might obtain something like this:

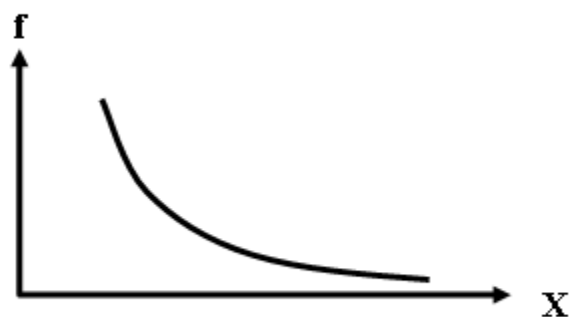
**DEATH RATES BY AGE GROUP**

| Age Group | No. of deaths per thousand |
|-----------|----------------------------|
| 20 – 29   | 2.1                        |
| 30 – 39   | 4.3                        |
| 40 – 49   | 5.7                        |
| 50 – 59   | 8.9                        |
| 60 – 69   | 12.4                       |
| 70 – 79   | 16.7                       |

This will result in a J-shaped distribution similar to the one shown above.

Similarly, the extremely positively skewed distribution is known as the REVERSE J-shaped distribution.

**THE EXTREMELY POSITIVELY  
SKEWED (REVERSE J-SHAPED)  
CURVE**





A relatively LESS frequently encountered frequency distribution is the U-shaped distribution.



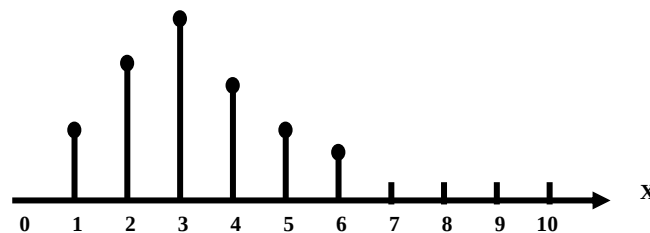
If we consider the example of the death rates not for only the adult population but for the population of ALL the age groups, we will obtain the **U-shaped distribution**.

Out of all these curves, the MOST frequently encountered frequency distribution is the moderately skewed frequency distribution. There are thousands of natural and social phenomena which yield the moderately skewed frequency distribution. Suppose that we walk into a school and collect data of the weights, heights, marks, shoulder-lengths, finger-lengths or any other such variable pertaining to the children of any one class.

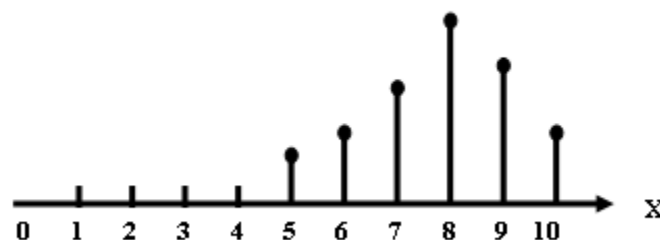
If we construct a frequency distribution of this data, and draw its histogram and its frequency curve, we will find that our data will generate a moderately skewed distribution. Until now, we have discussed the various possible shapes of the frequency distribution of a continuous variable. Similar shapes are possible for the frequency distribution of a discrete variable.

## VARIOUS TYPES OF DISCRETE FREQUENCY DISTRIBUTION

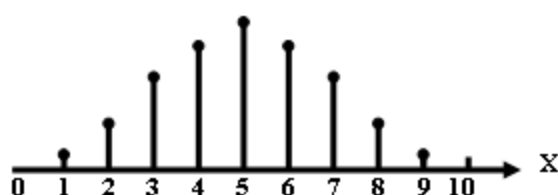
### I. Positively Skewed Distribution



### II. Negatively Skewed Distribution



### III. Symmetric Distribution



Let us now consider another aspect of the frequency distribution i.e.

#### CUMULATIVE FREQUENCY DISTRIBUTION

As in the case of the frequency distribution of a discrete variable, if we start adding the frequencies of our frequency table column-wise, we obtain the column of cumulative frequencies.

In our example, we obtain the cumulative frequencies shown below:

#### CUMULATIVE FREQUENCY DISTRIBUTION

| Class Boundaries | Frequency | Cumulative Frequency |
|------------------|-----------|----------------------|
| 29.95 – 32.95    | 2         | 2                    |
| 32.95 – 35.95    | 4         | 2+4 = 6              |
| 35.95 – 38.95    | 14        | 6+14 = 20            |
| 38.95 – 41.95    | 8         | 20+8 = 28            |
| 41.95 – 44.95    | 2         | 28+2 = 30            |
|                  | 30        |                      |

In the above table, 2+4 gives 6, 6+14 gives 20, and so on.

The question arises: “What is the purpose of making this column?”

You will recall that, when we were discussing the frequency distribution of a discrete variable, any particular cumulative frequency meant that we were counting the number of observations starting from the very first value of X and going up to THAT particular value of X against which that particular cumulative frequency was falling.

In case of a the distribution of a continuous variable, each of these cumulative frequencies represents the total frequency of a frequency distribution from the lower class boundary of the lowest class to the UPPER class boundary of THAT class whose cumulative frequency we are considering.

In the above table, the total number of cars showing mileage less than 35.95 miles per gallon is 6, the total number of car showing mileage less than 41.95 miles per gallon is 28, etc.

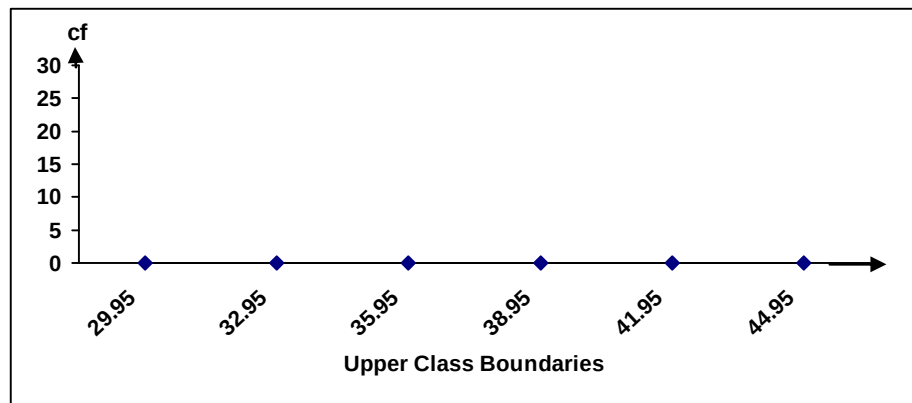
#### CUMULATIVE FREQUENCY DISTRIBUTION

| Class Boundaries | Frequency | Cumulative Frequency |
|------------------|-----------|----------------------|
| 29.95 – 32.95    | 2         | 2                    |
| 32.95 – 35.95    | 4         | 2+4 = 6              |
| 35.95 – 38.95    | 14        | 6+14 = 20            |
| 38.95 – 41.95    | 8         | 20+8 = 28            |
| 41.95 – 44.95    | 2         | 28+2 = 30            |
|                  | 30        |                      |

Such a cumulative frequency distribution is called a “less than” type of a cumulative frequency distribution. The graph of a cumulative frequency distribution is called a

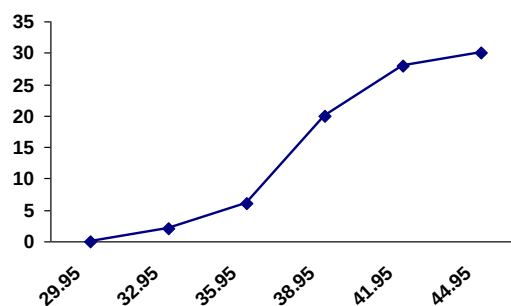
**CUMULATIVE FREQUENCY POLYGON or OGIVE**

A “less than” type ogive is obtained by marking off the upper class boundaries of the various classes along the X-axis and the cumulative frequencies along the y-axis, as shown below:



The cumulative frequencies are plotted on the graph paper against the upper class boundaries, and the points so obtained are joined by means of straight line segments.

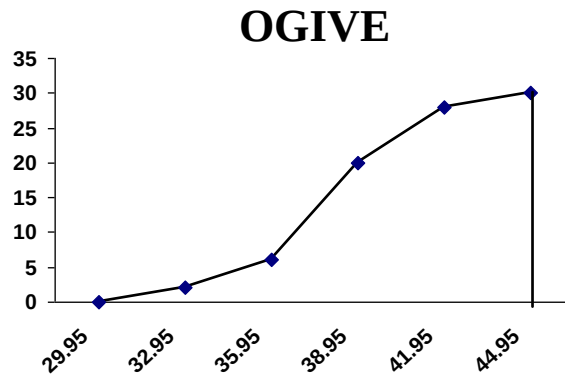
Hence we obtain the cumulative frequency polygon shown below:

**Cumulative Frequency Polygon or OGIVE**

It should be noted that this graph is touching the X-axis on the left-hand side. This is achieved by ADDING a class having zero frequency in the beginning of our frequency distribution, as shown below:

| Class Boundaries | Frequency | Cumulative Frequency |
|------------------|-----------|----------------------|
| 26.95 – 29.95    | 0         | 0                    |
| 29.95 – 32.95    | 2         | $0+2 = 2$            |
| 32.95 – 35.95    | 4         | $2+4 = 6$            |
| 35.95 – 38.95    | 14        | $6+14 = 20$          |
| 38.95 – 41.95    | 8         | $20+8 = 28$          |
| 41.95 – 44.95    | 2         | $28+2 = 30$          |
|                  | 30        |                      |

Since the frequency of the first class is zero, hence the cumulative frequency of the first class will also be zero, and hence, automatically, the cumulative frequency polygon will touch the X-axis from the left hand side. If we want our cumulative frequency polygon to be closed from the right-hand side also, we can achieve this by connecting the last point on our graph paper with the X-axis by means of a vertical line, as shown below:



In the example of EPA mileage ratings, all the data-values were correct to one decimal place. Let us now consider another example:

**EXAMPLE:**

For a sample of 40 pizza products, the following data represent cost of a slice in dollars (S Cost).

| PRODUCT                    | S cost |
|----------------------------|--------|
| Pizza Hut Hand Tossed      | 1.51   |
| Domino's Deep Dish         | 1.53   |
| Pizza Hut Pan Pizza        | 1.51   |
| Domino's Hand Tossed       | 1.90   |
| Little Caesars Pan! Pizza! | 1.23   |

| PRODUCT                              | S Cost |
|--------------------------------------|--------|
| Boboli crust with Boboli sauce       | 1.00   |
| Jack's Super Cheese                  | 0.69   |
| Pappalo's Three Cheese               | 0.75   |
| Tombstone Original Extra Cheese      | 0.81   |
| Master Choice Gourmet Four Cheese    | 0.90   |
| Celeste Pizza For One                | 0.92   |
| Totino's Party                       | 0.64   |
| The New Weight Watchers Extra Cheese | 1.54   |
| Jeno's Crisp'N Tasty                 | 0.72   |
| Stouffer's French Bread 2-Cheese     | 1.15   |

| PRODUCT                     | S Cost |
|-----------------------------|--------|
| Ellio's 9-slice             | 0.52   |
| Kroger                      | 0.72   |
| Healthy Choice French Bread | 1.50   |
| Lean Cuisine French Bread   | 1.49   |
| DiGiorno Rising Crust       | 0.87   |
| Tombstone Special Order     | 0.81   |
| Pappalo's                   | 0.73   |
| Jack's New More Cheese!     | 0.64   |
| Tombstone Original          | 0.77   |
| Red Baron Premium           | 0.80   |

| PRODUCT                           | Scost |
|-----------------------------------|-------|
| Tony's Italian Style Pastry Cruse | 0.83  |
| Red Baron Deep Dish Singles       | 1.13  |
| Totino's Party                    | 0.62  |
| The New Weight Watchers           | 1.52  |
| Jeno's Crisp'N Tasty              | 0.71  |
| Stouffer's French Bread           | 1.14  |
| Celeste Pizza For One             | 1.11  |
| Tombstone For One French Bread    | 1.11  |
| Healthy Choice French Bread       | 1.46  |
| Lean Cuisine French Bread         | 1.71  |

| PRODUCT                             | Scost |
|-------------------------------------|-------|
| Little Caesars Pizza! Pizza!        | 1.28  |
| Pizza Hut Stuffed Crust             | 1.23  |
| DiGiorno Rising Crust Four Cheese   | 0.90  |
| Tombstone Speical Order Four Cheese | 0.85  |
| Red Baron Premium 4-Cheese          | 0.80  |

Source: "Pizza," Copyright 1997 by Consumers Union of United States, Inc., Yonkers, N.Y. 10703.

Example taken from

"Business Statistics – A First Course" by Mark L. Berenson & David M. Levine (International Edition), Prentice-Hall International, Inc., Copyright © 1998.

In order to construct the frequency distribution of the above data, the first thing to note is that, in this example, all our data values are correct to two decimal places. As such, we should construct the class limits correct to TWO decimal places, and the class boundaries correct to three decimal places.

As in the last example, first of all, let us find the maximum and the minimum values in our data, and compute the RANGE.

Minimum value  $X_0 = 0.52$

Maximum value  $X_m = 1.90$

Hence:

$$\text{Range} = 1.90 - 0.52 = 1.38$$

Desired number of classes = 8

Hence:

Class interval  $h = \text{RANGE} / \text{No. of classes}$

$$= 1.38 / 8 = 0.1725 \simeq 0.20$$

Lower limit of the first class = 0.51

Hence, our successive class limits come out to be:

| Class Limits |
|--------------|
| 0.51 – 0.70  |
| 0.71 – 0.90  |
| 0.91 – 1.10  |
| 1.11 – 1.30  |
| 1.31 – 1.50  |
| 1.51 – 1.70  |
| 1.71 – 1.90  |

Stretching the class limits to the left and to the right, we obtain class boundaries as shown below:

| Class Limits | Class Boundaries |
|--------------|------------------|
| 0.51 – 0.70  | 0.505 – 0.705    |
| 0.71 – 0.90  | 0.705 – 0.905    |
| 0.91 – 1.10  | 0.905 – 1.105    |
| 1.11 – 1.30  | 1.105 – 1.305    |
| 1.31 – 1.50  | 1.305 – 1.505    |
| 1.51 – 1.70  | 1.505 – 1.705    |
| 1.71 – 1.90  | 1.705 – 1.905    |

By tallying the data-values in the appropriate classes, we will obtain a frequency distribution similar to the one that we obtained in the examples of the EPA mileage ratings.

By constructing the histogram of this data-set, we will be able to decide whether our distribution is symmetric, positively skewed or negatively skewed. This may please be attempted as an exercise.

## LECTURE NO. 6

This plot was introduced by the famous statistician **John Tukey in 1977**. A frequency table has the disadvantage that the identity of individual observations is lost in grouping process. To overcome this drawback, John Tukey (1977) introduced this particular technique (known as the **Stem-and-Leaf Display**).

This technique offers a quick and novel way for simultaneously sorting and displaying data sets where each number in the data set is divided into two parts, a Stem and a Leaf.

A stem is the leading digit(s) of each number and is used in sorting, while a leaf is the rest of the number or the trailing digit(s) and shown in display. A vertical line separates the leaf (or leaves) from the stem.

For example, the number 243 could be split in two ways:

| Leading Digit | Trailing Digits | OR | Leading Digit | Trailing Digit |
|---------------|-----------------|----|---------------|----------------|
| 2             | 43              |    | 24            | 3              |
| Stem          | Leaf            |    | Stem          | Leaf           |

How do we construct a stem and leaf display when we have a whole set of values? This is explained by way of the following example:

**EXAMPLE:**

The ages of 30 patients admitted to a certain hospital during a particular week were as follows:

48, 31, 54, 37, 18, 64, 61, 43, 40, 71, 51, 12, 52, 65, 53, 42, 39, 62, 74, 48, 29, 67, 30, 49, 68, 35, 57, 26, 27, 58.

Construct a stem-and-leaf display from the data and list the data in an array.

A scan of the data indicates that the observations range (in age) from 12 to 74. We use the first (or leading) digit as the stem and the second (or trailing) digit as the leaf. The first observation is 48, which has a stem of 4 and a leaf of 8, the second a stem of 3 and a leaf of 1, etc. Placing the leaves in the order in which they APPEAR in the data, we get the stem-and-leaf display as shown below:

| Stem<br>(Leading Digit) | Leaf<br>(Trailing Digit) |
|-------------------------|--------------------------|
| 1                       | 8 2                      |
| 2                       | 9 6 7                    |
| 3                       | 1 7 9 0 5                |
| 4                       | 8 3 0 2 8 9              |
| 5                       | 4 1 2 3 7 8              |
| 6                       | 4 1 5 2 7 8              |
| 7                       | 1 4                      |

But it is a common practice to ARRANGE the trailing digits in each row from smallest to highest. In this example, in order to obtain an array, we associate the leaves in order of size with the stems as shown below:

**DATA IN THE FORM OF AN ARRAY (in ascending order):**

12, 18, 26, 27, 29, 30, 31, 35, 37, 39, 40, 42, 43, 48, 48, 49, 51, 52, 53, 54, 57, 58, 61, 62, 64, 65, 67, 68, 71, 74.

Hence we obtain the stem and leaf plot shown below:

**STEM AND LEAF DISPLAY**

| Stem<br>(Leading Digit) | Leaf<br>(Trailing Digit) |
|-------------------------|--------------------------|
| 1                       | 2 8                      |
| 2                       | 6 7 9                    |
| 3                       | 0 1 5 7 9                |
| 4                       | 0 2 3 8 8 9              |
| 5                       | 1 2 3 4 7 8              |
| 6                       | 1 2 4 5 7 8              |
| 7                       | 1 4                      |

The stem-and-leaf table provides a useful description of the data set and, if we so desire, can easily be converted to a frequency table. In this example, the frequency of the class 10-19 is 2, the frequency of the class 20-29 is 3, and the frequency of the class 30-39 is 5, and so on.

**STEM AND LEAF DISPLAY**

| Stem<br>(Leading Digit) | Leaf<br>(Trailing Digit) |
|-------------------------|--------------------------|
| 1                       | 2 8                      |
| 2                       | 6 7 9                    |
| 3                       | 0 1 5 7 9                |
| 4                       | 0 2 3 8 8 9              |
| 5                       | 1 2 3 4 7 8              |
| 6                       | 1 2 4 5 7 8              |
| 7                       | 1 4                      |

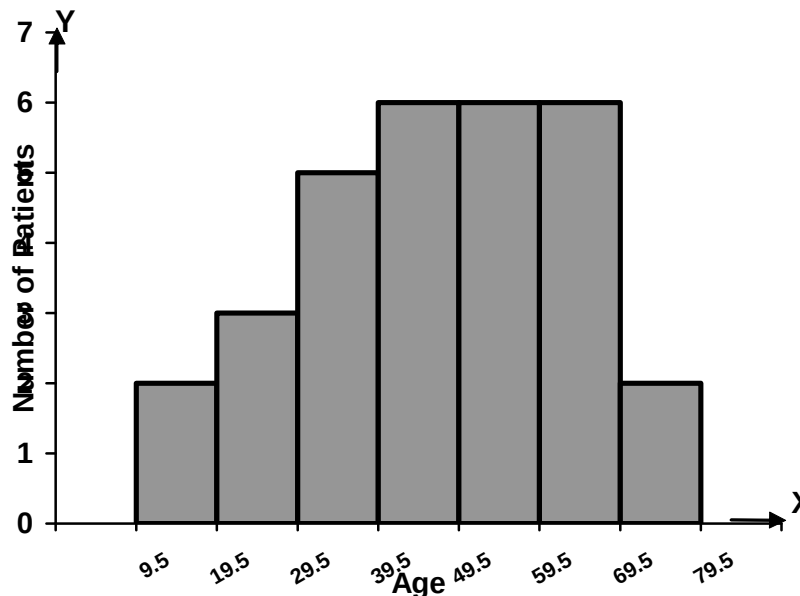
Hence, this stem and leaf plot conveniently converts into the frequency distribution shown below:

**FREQUENCY DISTRIBUTION**

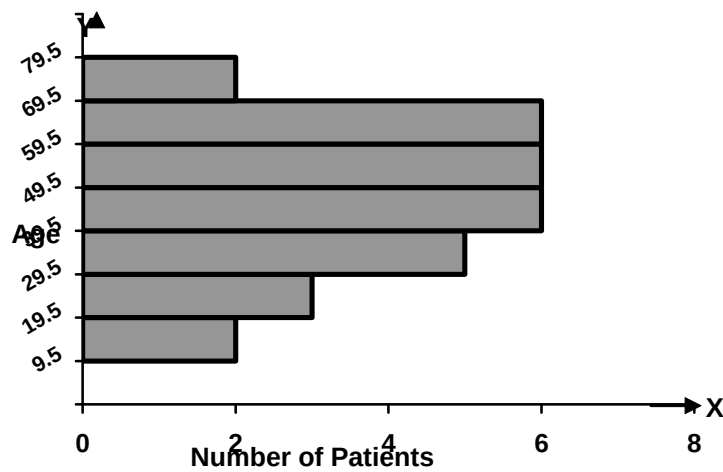
| Class Limits | Class Boundaries | Tally Marks | Frequency |
|--------------|------------------|-------------|-----------|
| 10 – 19      | 9.5 – 19.5       | //          | 2         |
| 20 – 29      | 19.5 – 29.5      | ///         | 3         |
| 30 – 39      | 29.5 – 39.5      | ###         | 5         |
| 40 – 49      | 39.5 – 49.5      | ### /       | 6         |
| 50 – 59      | 49.5 – 59.5      | ### /       | 6         |
| 60 – 69      | 59.5 – 69.5      | ### /       | 6         |
| 70 – 79      | 69.5 – 79.5      | //          | 2         |



Converting this frequency **distribution into a histogram**, we obtain:



If we rotate this histogram by 90 degrees, we will obtain:



Let us re-consider the stem and leaf plot that we obtained a short while ago.

#### STEM AND LEAF DISPLAY

| Stem<br>(Leading Digit) | Leaf<br>(Trailing Digit) |
|-------------------------|--------------------------|
| 7                       | 1 4                      |
| 6                       | 1 2 4 5 7 8              |
| 5                       | 1 2 3 4 7 8              |
| 4                       | 0 2 3 8 8 9              |
| 3                       | 0 1 5 7 9                |
| 2                       | 6 7 9                    |
| 1                       | 2 8                      |

It is noteworthy that the shape of the stem and leaf display is exactly like the shape of our histogram. Let us now consider another example.

### EXAMPLE

Construct a stem-and-leaf display for the data of mean annual death rates per thousand at ages 20-65 given below:

7.5, 8.2, 7.2, 8.9, 7.8, 5.4, 9.4, 9.9, 10.9, 10.8, 7.4, 9.7, 11.6, 12.6, 5.0, 10.2, 9.2, 12.0, 9.9, 7.3, 7.3, 8.4, 10.3, 10.1, 10.0, 11.1, 6.5, 12.5, 7.8, 6.5, 8.7, 9.3, 12.4, 10.6, 9.1, 9.7, 9.3, 6.2, 10.3, 6.6, 7.4, 8.6, 7.7, 9.4, 7.7, 12.8, 8.7, 5.5, 8.6, 9.6, 11.9, 10.4, 7.8, 7.6, 12.1, 4.6, 14.0, 8.1, 11.4, 10.6, 11.6, 10.4, 8.1, 4.6, 6.6, 12.8, 6.8, 7.1, 6.6, 8.8, 8.8, 10.7, 10.8, 6.0, 7.9, 7.3, 9.3, 9.3, 8.9, 10.1, 3.9, 6.0, 6.9, 9.0, 8.8, 9.4, 11.4, 10.9

Using the decimal part in each number as the leaf and the rest of the digits as the stem, we get the ordered stem-and-leaf display shown below:

### STEM AND LEAF DISPLAY

| Stem | Leaf                          |
|------|-------------------------------|
| 3    | 9                             |
| 4    | 6 6                           |
| 5    | 0 4 5                         |
| 6    | 0 0 2 2 5 5 6 6 6 8 9         |
| 7    | 1 3 3 3 4 4 5 6 7 7 8 8 8 9   |
| 8    | 1 1 2 4 6 6 7 7 8 8 8 9 9     |
| 9    | 0 1 2 3 3 3 3 4 4 4 6 7 7 9 9 |
| 10   | 0 1 1 2 3 3 4 4 6 6 7 8 8 9 9 |
| 11   | 1 4 4 6 6 9                   |
| 12   | 0 1 4 5 6 8 8                 |
| 14   | 0                             |

### EXERCISE

- The above data may be converted into a stem and leaf plot (so as to verify that the one shown above is correct).
- Various variations of the stem and leaf display may be studied on your own.

The next concept that we are going to consider is the concept of the central tendency of a data-set. In this context, the first thing to note is that in any data-based study, our data is always going to be variable, and hence, first of all, we will need to describe the data that is available to us.

**DESCRIPTION OF VARIABLE DATA:**

Regarding any statistical enquiry, primarily we need some means of describing the situation with which we are confronted. A concise numerical description is often preferable to a lengthy tabulation, and if this form of description also enables us to form a mental image of the data and interpret its significance, so much the better.

**MEASURES OF CENTRAL TENDENCY**  
**AND**  
**MEASURES OF DISPERSION**

- Averages enable us to measure the central tendency of variable data
- Measures of dispersion enable us to measure its variability.

**AVERAGES (I.E. MEASURES OF CENTRAL TENDENCY)**

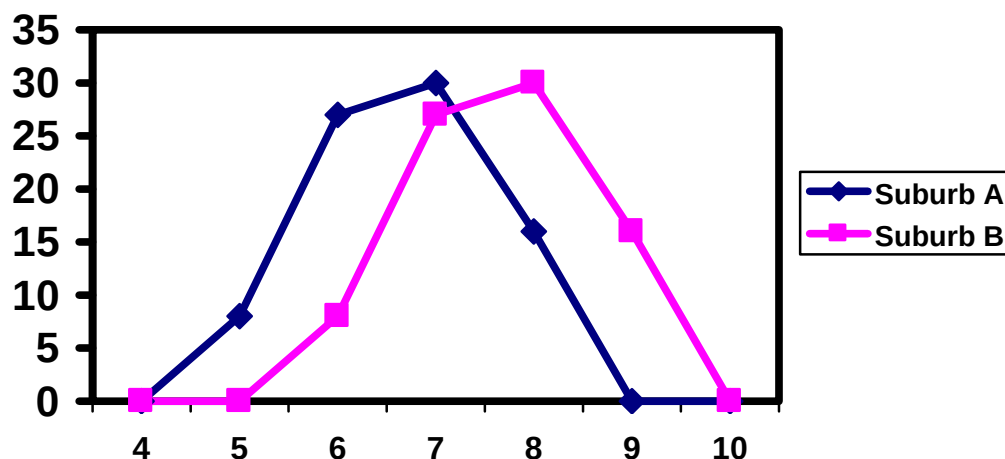
An average is a single value which is intended to represent a set of data or a distribution as a whole. It is more or less CENTRAL value ROUND which the observations in the set of data or distribution usually tend to cluster.

As a measure of central tendency (i.e. an average) indicates the location or general position of the distribution on the X-axis, it is also known as a measure of location or position.

Let us consider an example: Suppose that we have the following two frequency distributions:

**EXAMPLE:**

Looking at these two frequency distributions, we should ask ourselves what exactly is the distinguishing feature? If we draw the frequency polygon of the two frequency distributions, we obtain



Inspection of these frequency polygons shows that they have exactly the same shape. It is their position relative to the horizontal axis (X-axis) which distinguishes them.

If we compute the mean number of rooms per house for each of the two suburbs, we will find that the average number of rooms per house in A is 6.67 while in B it is 7.67.

This difference of 1 is equivalent to the difference in position of the two frequency polygons.

Our interpretation of the above situation would be that there are LARGER houses in suburb B than in suburb A, to the extent that there are on the *average*.

**VARIOUS TYPES OF AVERAGES:**

There are several types of averages each of which has a use in specifically defined circumstances.

The most common types of averages are:

- The arithmetic mean,
- The geometric mean,

- The harmonic mean
- The median, and
- The mode

The **Arithmetic, Geometric and Harmonic means** are averages that are mathematical in character, and give an indication of the magnitude of the observed values.

The **Median** indicates the middle position while the mode provides information about the most frequent value in the distribution or the set of data. THE MODE:

The **Mode** is defined as that value which occurs most frequently in a set of data i.e. it indicates the most common result.

#### EXAMPLE:

Suppose that the marks of eight students in a particular test are as follows:

2, 7, 9, 5, 8, 9, 10, 9

Obviously, the most common mark is 9. In other words, Mode = 9.

#### MODE IN CASE OF RAW DATA PERTAINING TO A CONTINUOUS VARIABLE

In case of a set of values (pertaining to a continuous variable) that have not been grouped into a frequency distribution (i.e. in case of raw data pertaining to a continuous variable), the mode is obtained by counting the number of times each value occurs.

#### EXAMPLE:

Suppose that the government of a country collected data regarding the percentages of revenues spent on Research and Development by 49 different companies, and obtained the following figures:

**Percentage of Revenues Spent on Research and Development**

| Compan<br>y | Percentage | Compan<br>y | Percentage |
|-------------|------------|-------------|------------|
| 1           | 13.5       | 14          | 9.5        |
| 2           | 8.4        | 15          | 8.1        |
| 3           | 10.5       | 16          | 13.5       |
| 4           | 9.0        | 17          | 9.9        |
| 5           | 9.2        | 18          | 6.9        |
| 6           | 9.7        | 19          | 7.5        |
| 7           | 6.6        | 20          | 11.1       |
| 8           | 10.6       | 21          | 8.2        |
| 9           | 10.1       | 22          | 8.0        |
| 10          | 7.1        | 23          | 7.7        |
| 11          | 8.0        | 24          | 7.4        |
| 12          | 7.9        | 25          | 6.5        |
| 13          | 6.8        | 26          | 9.5        |

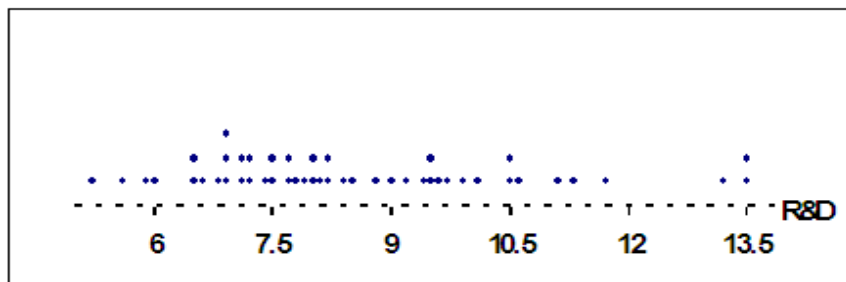
| Compan<br>y | Percentage | Compan<br>y | Percentage |
|-------------|------------|-------------|------------|
| 27          | 8.2        | 39          | 6.5        |
| 28          | 6.9        | 40          | 7.5        |
| 29          | 7.2        | 41          | 7.1        |
| 30          | 8.2        | 42          | 13.2       |
| 31          | 9.6        | 43          | 7.7        |
| 32          | 7.2        | 44          | 5.9        |
| 33          | 8.8        | 45          | 5.2        |
| 34          | 11.3       | 46          | 5.6        |
| 35          | 8.5        | 47          | 11.7       |
| 36          | 9.4        | 48          | 6.0        |
| 37          | 10.5       | 49          | 7.8        |
| 38          | 6.9        |             |            |

We can represent this data by means of a plot that is called dot plot.

### DOT PLOT:

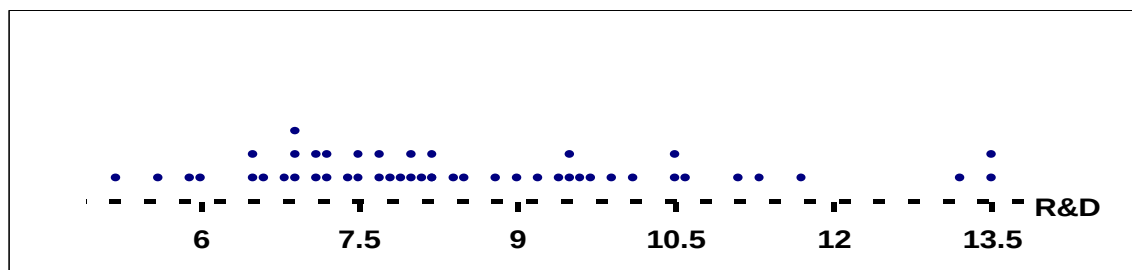
The horizontal axis of a dot plot contains a scale for the quantitative variable that we want to represent. The numerical value of each measurement in the data set is located on the horizontal scale by a dot. When data values repeat, the dots are placed above one another, forming a pile at that particular numerical location. In this example

**Dot Plot**



The above material has been taken from  
*"Statistics for Business and Economics"* by James T. McClave, P. George Benson and Terry Sincich (Seventh Edition), © 1998 Prentice-Hal International, Inc.

As is obvious from the above diagram, the value 6.9 occurs 3 times whereas all the other values are occurring either once or twice. Hence the modal value is 6.9.



**Dot Plot**

$\hat{X}$

$$= 6.9$$

Also, this dot plot shows that

- almost all of the R&D percentages are falling between 6% and 12%,
- most of the percentages are falling between 7% and 9%.

### **THE MODE IN CASE OF A DISCRETE FREQUENCY DISTRIBUTION:**

In case of a discrete frequency distribution, identification of the mode is immediate; one simply finds that value which has the highest frequency.

#### **EXAMPLE:**

An airline found the following numbers of passengers in fifty flights of a forty-seated plane

| No. of Passengers<br>X | No. of Flights<br>f |
|------------------------|---------------------|
| 28                     | 1                   |
| 33                     | 1                   |
| 34                     | 2                   |
| 35                     | 3                   |
| 36                     | 5                   |
| 37                     | 7                   |
| 38                     | 10                  |
| 39                     | 13                  |
| 40                     | 8                   |
| <b>Total</b>           | <b>50</b>           |

Highest Frequency  $f_m = 13$

Occurs against the X value 39

Hence: Mode =  $x = 39$

The mode is obviously 39 passengers and the company should be quite satisfied that a 40 seater is the correct-size aircraft for this particular route.

### **THE MODE IN CASE OF THE FREQUENCY DISTRIBUTION OF A CONTINUOUS VARIABLE**

In case of grouped data, the modal group is easily recognizable (the one that has the highest frequency).

At what point within the modal group does the mode lie?

The answer is contained in the following formula:

**Mode:**

$$\hat{X} = l + \frac{f_m - f_1}{(f_m - f_1) + (f_m - f_2)} \times h$$

**Where**

$l$  = lower class boundary of the modal class,

$f_m$  = frequency of the modal class,

$f_1$  = frequency of the class preceding the  
modal class

$f_2$  = frequency of the class following modal

class

h = length of class interval of the modal class

Going back to the example of EPA mileage ratings, we have:

**EPA MILEAGE RATINGS**

| Mileage Rating | Class Boundaries | No. of Cars |
|----------------|------------------|-------------|
| 30.0 – 32.9    | 29.95 – 32.95    | 2           |
| 33.0 – 35.9    | 32.95 – 35.95    | 4 = $f_1$   |
| 36.0 – 38.9    | 35.95 – 38.95    | 14 = $f_m$  |
| 39.0 – 41.9    | 38.95 – 41.95    | 8 = $f_2$   |
| 42.0 – 44.9    | 41.95 – 44.95    | 2           |

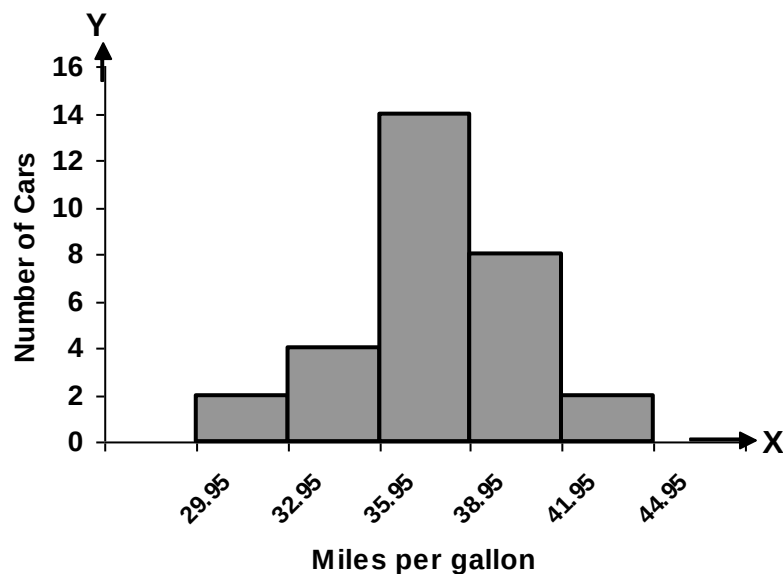
It is evident that the third class is the modal class. The mode lies somewhere between 35.95 and 38.95.

In order to apply the formula for the mode, we note that  $f_m = 14$ ,  $f_1 = 4$  and  $f_2 = 8$ .

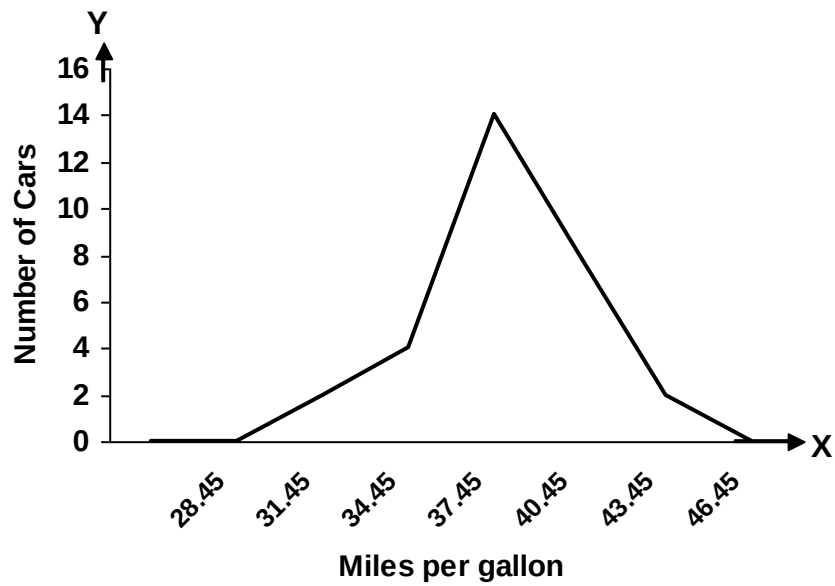
Hence we obtain:

$$\begin{aligned}
 \hat{X} &= 35.95 + \frac{14 - 4}{(14 - 4) + (14 - 8)} \times 3 \\
 &= 35.95 + \frac{10}{10 + 6} \times 3 \\
 &= 35.95 + 1.875 \\
 &= 37.825
 \end{aligned}$$

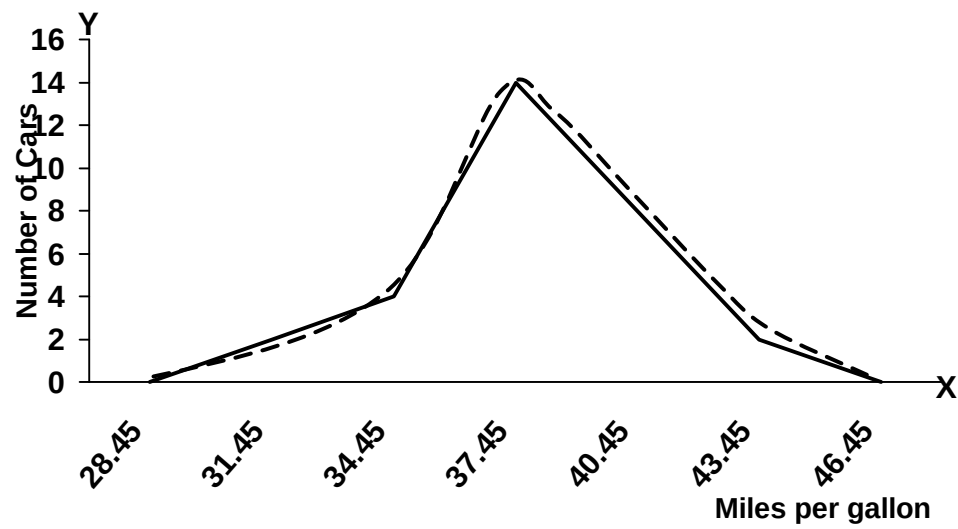
Let us now perceive the mode by considering the graphical representation of our frequency distribution. You will recall that, for the example of EPA Mileage Ratings, the histogram was as shown below:



The frequency polygon of the same distribution was:

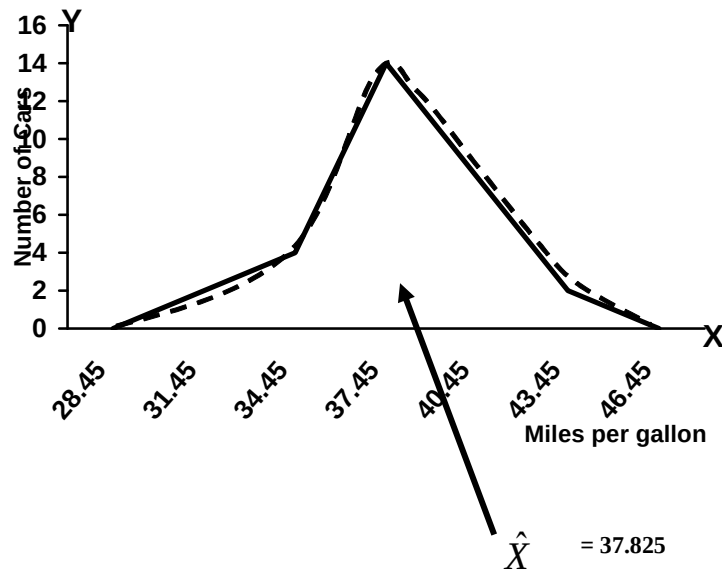


And the frequency curve was as indicated by the dotted line in the following figure:





In this example, the mode is 37.825, and if we locate this value on the X-axis, we obtain the following picture:



Since, in most of the situations the mode exists somewhere in the middle of our data-values, hence it is thought of as a measure of central tendency.

## LECTURE NO. 7

In general, it was noted that, for most of the frequency distributions, the mode lies somewhere in the middle of our frequency distribution, and hence is eligible to be called a measure of central tendency. The mode has some very desirable properties.

### DESIRABLE PROPERTIES OF THE MODE:

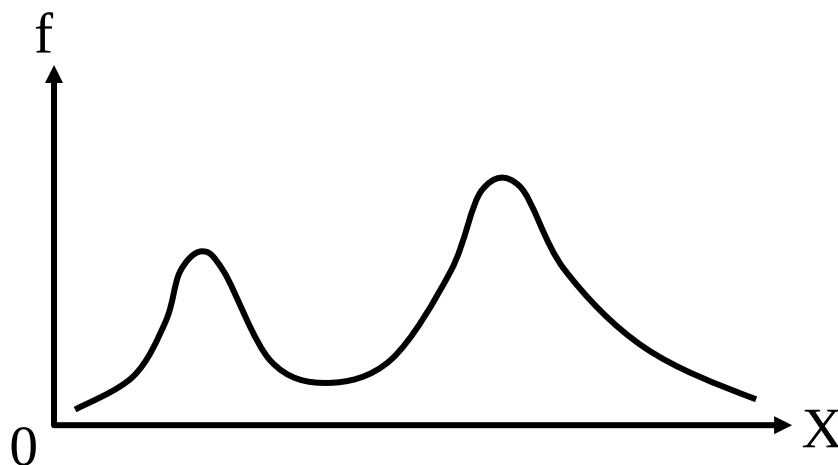
- The mode is easily understood and easily ascertained in case of a discrete frequency distribution.
- It is not affected by a few very high or low values.

The question arises, “When should we use the mode?” The answer to this question is that the mode is a valuable concept in certain situations such as the one described below:

Suppose the manager of a men’s clothing store is asked about the average size of hats sold. He will probably think not of the arithmetic or geometric mean size, or indeed the median size. Instead, he will in all likelihood quote that particular size which is sold **most often**. This average is of far more use to him as a businessman than the arithmetic mean, geometric mean or the median. The modal size of all clothing is the size which the businessman must stock in the greatest quantity and variety in comparison with other sizes. Indeed, in most inventory (stock level) problems, one needs the mode more often than any other measure of central tendency. It should be noted that in some situations there may be no mode in a simple series where no value occurs more than once.

On the other hand, sometimes a frequency distribution contains two modes in which case it is called a bi-modal distribution as shown below:

### THE BI-MODAL FREQUENCY DISTRIBUTION



The next measure of central tendency to be discussed is the arithmetic mean.

### THE ARITHMETIC MEAN

The arithmetic mean is the statistician’s term for what the layman knows as the average. It can be thought of as that value of the variable series which is *numerically* MOST representative of the whole series. Certainly, this is the most widely used average in statistics. Easiest In addition, it is probably the to calculate.

Its formal definition is:

“The arithmetic mean or simply the mean is a value obtained by dividing the sum of all the observations by their number.”

$$\bar{X} = \frac{\text{Sum of all the observations}}{\text{Number of the observations}}$$

$$\bar{X} = \frac{\sum_{i=1}^n x_i}{n}$$

Where n represents the number of observations in the sample that has been the ith observation in the sample (i = 1, 2, 3, ..., n), and  $\bar{X}$  represents the mean of the sample.

For simplicity, the above formula can be written as

$$\bar{X} = \frac{\sum x}{n}$$

In other words, it is not necessary to insert the subscript 'i'.)

#### **EXAMPLE:**

Information regarding the receipts of a news agent for seven days of a particular week are given below

| Day        | Receipt of News Agent |
|------------|-----------------------|
| Monday     | £ 9.90                |
| Tuesday    | £ 7.75                |
| Wednesday  | £ 19.50               |
| Thursday   | £ 32.75               |
| Friday     | £ 63.75               |
| Saturday   | £ 75.50               |
| Sunday     | £ 50.70               |
| Week Total | £ 259.85              |

Mean sales per day in this week:

$$= £ 259.85/7 = £ 37.12 \text{ (To the nearest penny).}$$

#### **INTERPRETATION:**

The mean, £ 37.12, represents the amount (in pounds sterling) that would have been obtained on each day if the same amount were to be obtained on each day. The above example pertained to the computation of the arithmetic mean in case of ungrouped data i.e. raw data.

Let us now consider the case of data that has been grouped into a frequency distribution. When data pertaining to a continuous variable has been grouped into a frequency distribution, the frequency distribution is used to calculate the approximate values of descriptive measures --- as the identity of the observations is lost.

To calculate the approximate value of the mean, the observations in each class are assumed to be identical with the class midpoint  $X_i$ .

The mid-point of every class is known as its class-mark. In other words, the midpoint of a class 'marks' that class. As was just mentioned, the observations in each class are assumed to be identical with the midpoint i.e. the class-mark. (This is based on the assumption that the observations in the group are evenly scattered between the two extremes of the class interval).

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**FREQUENCY DISTRIBUTION**

| Mid Point<br>X | Frequency<br>f |
|----------------|----------------|
| X <sub>1</sub> | f <sub>1</sub> |
| X <sub>2</sub> | f <sub>2</sub> |
| X <sub>3</sub> | f <sub>3</sub> |
| ⋮              | ⋮              |
| ⋮              | ⋮              |
| ⋮              | ⋮              |
| X <sub>k</sub> | f <sub>k</sub> |

In case of a frequency distribution, the arithmetic mean is defined as:  
ARITHMETIC MEAN

$$\bar{X} = \frac{\sum_{i=1}^k f_i X_i}{\sum_{i=1}^k f_i} = \frac{\sum_{i=1}^k f_i X_i}{n}$$

For simplicity, the above formula can be written as

$$\bar{X} = \frac{\sum fX}{\sum f} = \frac{\sum fX}{n} \quad (\text{The subscript 'i' can be dropped.})$$

Let us understand this point with the help of an example:

Going back to the example of EPA mileage ratings, that we dealt with when discussing the formation of a frequency distribution. The frequency distribution that we obtained was:

**EPA MILEAGE RATINGS OF 30 CARS OF A CERTAIN MODEL**

| Class<br>(Mileage Rating) | Frequency<br>(No. of Cars) |
|---------------------------|----------------------------|
| 30.0 – 32.9               | 2                          |
| 33.0 – 35.9               | 4                          |
| 36.0 – 38.9               | 14                         |
| 39.0 – 41.9               | 8                          |
| 42.0 – 44.9               | 2                          |
| Total                     | 30                         |

The first step is to compute the mid-point of every class.

(You will recall that the concept of the mid-point has already been discussed in an earlier lecture.)

**CLASS-MARK (MID-POINT):**

The mid-point of each class is obtained by adding the sum of the two limits of the class and dividing by 2.

Hence, in this example, our mid-points are computed in this manner:

30.0 plus 32.9 divided by 2 is equal to 31.45,

33.0 plus 35.9 divided by 2 is equal to 34.45,

And so on.

| Class<br>(Mileage Rating) | Class-mark<br>(Midpoint)<br>X |
|---------------------------|-------------------------------|
| 30.0 – 32.9               | 31.45                         |
| 33.0 – 35.9               | 34.45                         |
| 36.0 – 38.9               | 37.45                         |
| 39.0 – 41.9               | 40.45                         |
| 42.0 – 44.9               | 43.45                         |

In order to compute the arithmetic mean, we first need to construct the column of  $fX$ , as shown below:

| Class-mark<br>(Midpoint)<br>X | Frequency<br>f | fX     |
|-------------------------------|----------------|--------|
| 31.45                         | 2              | 62.9   |
| 34.45                         | 4              | 137.8  |
| 37.45                         | 14             | 524.3  |
| 40.45                         | 8              | 323.6  |
| 43.45                         | 2              | 86.9   |
|                               | 30             | 1135.5 |

Applying the formula

$$\bar{X} = \frac{\sum fX}{\sum f},$$

We obtain

$$\bar{X} = \frac{1135.5}{30} = 37.85$$

#### **INTERPRETATION:**

The average mileage rating of the 30 cars tested by the Environmental Protection Agency is 37.85 – on the average, these cars run 37.85 miles per gallon. An important concept to be discussed at this point is the concept of grouping error.

#### **GROUPING ERROR:**

“Grouping error” refers to the error that is introduced by the assumption that all the values falling in a class are equal to the mid-point of the class interval. In reality, it is highly improbable to have a class for which all the values lying in that class are equal to the mid-point of that class. This is why the mean that we calculate from a frequency distribution does not give exactly the same answer as what we would get by computing the mean of our raw data.

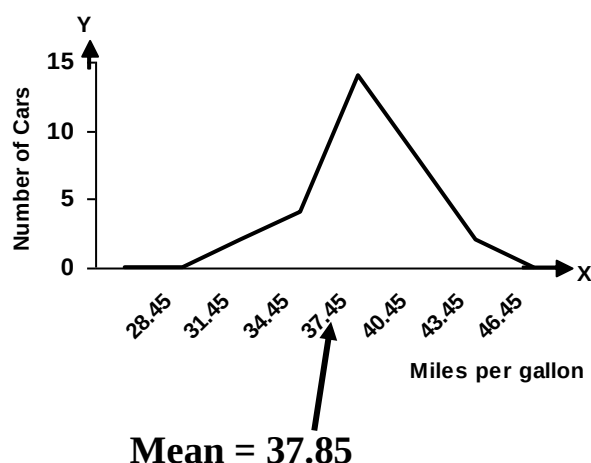
As indicated earlier, a frequency distribution is used to calculate the approximate values of various descriptive measures. (The word ‘approximate’ is being used because of the grouping error that was just discussed.) This grouping error arises in the computation of many descriptive measures such as the geometric mean, harmonic mean, mean deviation and standard deviation. But, experience has shown that in the calculation of the arithmetic mean, this error is usually small and never serious. Only a slight difference occurs between the true answer that we would get from the raw data, and the answer that we get from the data that has been grouped in the form of a frequency distribution.

In this example, if we calculate the arithmetic mean directly from the 30 EPA mileage ratings, we obtain:

Arithmetic mean computed from raw data of the EPA mileage ratings:

$$\begin{aligned}\bar{X} &= \frac{36.3 + 30.1 + \dots + 33.9 + 39.8}{30} \\ &= \frac{1134.7}{30} = 37.82\end{aligned}$$

The difference between the true value of i.e. 37.82 and the value obtained from the frequency distribution i.e. 37.85 is indeed very slight. The arithmetic mean is predominantly used as a measure of central tendency. The question is, “Why is it that the arithmetic mean is known as a measure of central tendency?” The answer to this question is that we have just obtained i.e. 37.85 falls more or less in the centre of our frequency distribution.



As indicated earlier, the arithmetic mean is predominantly used as a measure of central tendency. It has many desirable properties:

#### DESIRABLE PROPERTIES OF THE ARITHMETIC MEAN

- Best understood average in statistics.
- Relatively easy to calculate
- Takes into account every value in the series.

But there is one limitation to the use of the arithmetic mean:

As we are aware, every value in a data-set is included in the calculation of the mean, whether the value be high or low. Where there are a few very high or very low values in the series, their effect can be to drag the arithmetic mean towards them. This may make the mean unrepresentative.

#### EXAMPLE:

Example of the Case Where the Arithmetic Mean Is Not a Proper Representative of the Data:

Suppose one walks down the main street of a large city centre and counts the number of floors in each building.

Suppose, the following answers are obtained:

5, 4, 3, 4, 5, 4, 3, 4, 5, 20, 5, 6, 32, 8, 27

The mean number of floors is 9 even though 12 out of 15 of the buildings have 6 floors or less.

The three skyscraper blocks are having a disproportionate effect on the arithmetic mean (Some other average in this case would be more representative). The concept that we just considered was the concept of the simple arithmetic mean.

Let us now discuss the concept of the weighted arithmetic mean.

Consider the following example:

#### EXAMPLE:

Suppose that in a particular high school, there are:-

|     |   |            |
|-----|---|------------|
| 100 | – | freshmen   |
| 80  | – | sophomores |
| 70  | – | juniors    |
| 50  | – | seniors    |

And suppose that on a given day, 15% of freshmen, 5% of sophomores, 10% of juniors, 2% of seniors are *absent*.

The problem is that: What percentage of students is absent for the school as a whole on that particular day?

Now a student is likely to attempt to find the answer by adding the percentages and dividing by 4

i.e.

$$\frac{15+5+10+2}{4} = \frac{32}{4} = 8$$

But the fact of the matter is that the above calculation gives a wrong answer. In order to figure out why this is a wrong calculation, consider the following: As we have already noted, 15% of the freshmen are absent on this particular day. Since, in all, there are 100 freshmen in the school, hence the total number of freshmen who are absent is also 15.

But as far as the sophomores are concerned, the total number of them in the school is 80, and if 5% of them are absent on this particular day, this means that the total number of sophomores who are absent is only 4. Proceeding in this manner, we obtain the following table.

| Category of Student | Number of Students in the school | Number of Students who are absent |
|---------------------|----------------------------------|-----------------------------------|
| Freshman            | 100                              | 15                                |
| Sophomore           | 80                               | 4                                 |
| Junior              | 70                               | 7                                 |
| Senior              | 50                               | 1                                 |
| TOTAL               | 300                              | 27                                |

Dividing the total number of students who are absent by the total number of students enrolled in the school, and multiplying by 100, we obtain:

$$\frac{27}{300} \times 100 = 9$$

Thus its very clear that previous result was not correct. This situation leads us to a very important observation, i.e. here our figures pertaining to absenteeism in various categories of students cannot be regarded as having equal weightage.

When we have such a situation, the concept of “weighing” applies i.e. every data value in the data set is assigned a certain weight according to a suitable criterion. In this way, we will have a weighted series of data instead of an un-weighted one. In this example, the number of students enrolled in each category acts as the weight for the number of absences pertaining to that category i.e.

| Category of Student | Percentage of Students who are absent<br>$X_i$ | Number of students enrolled in the school<br>(Weights)<br>$W_i$ | $W_i X_i$<br>(Weighted $X_i$ ) |
|---------------------|------------------------------------------------|-----------------------------------------------------------------|--------------------------------|
| Freshman            | 15                                             | 100                                                             | $100 \times 15 = 1500$         |
| Sophomore           | 5                                              | 80                                                              | $80 \times 5 = 400$            |
| Junior              | 10                                             | 70                                                              | $70 \times 10 = 700$           |
| Senior              | 2                                              | 50                                                              | $50 \times 2 = 100$            |
|                     | Total                                          | $\Sigma W_i = 300$                                              | $\Sigma W_i X_i = 2700$        |

The formula for the weighted arithmetic mean is:

#### **WEIGHTED MEAN**

$$\bar{X}_w = \frac{\sum W_i X_i}{\sum W_i}$$

And, in this example, the weighted mean is equal to:

$$\begin{aligned}\bar{X}_w &= \frac{\sum W_i X_i}{\sum W_i} \\ &= \frac{2700}{300} \\ &= 9\end{aligned}$$

Thus we note that, in this example, the weighted mean yields exactly the same as the answer that we obtained earlier.

As obvious, the weighing process leads us to a correct answer under the situation where we have data that cannot be regarded as being such that each value should be given equal weightage.

An important point to note here is the *criterion* for assigning weights. Weights can be assigned in a number of ways depending on the situation and the problem domain.

The next measure of central tendency that we will discuss is the median.

Let us understand this concept with the help of an example.

Let us return to the problem of the ‘average’ number of floors in the buildings at the centre of a city. We saw that the arithmetic mean was distorted towards the few extremely high values in this series and became unrepresentative.

We could more appropriately and easily employ the median as the ‘average’ in these circumstances.

### **MEDIAN**

**The median is the middle value of the series when the variable values are placed in order of magnitude.**

The median is defined as a “value which divides a set of data into two halves, one half comprising of observations greater than and the other half smaller than it. More precisely, the median is a value at or below which 50% of the data lie.”

The median value can be ascertained by inspection in many series. For instance, in this very example, the data that we obtained was:

#### **EXAMPLE-1**

The average number of floors in the buildings at the centre of a city:

5, 4, 3, 4, 5, 4, 3, 4, 5, 6, 32, 8, 27

Arranging these values in ascending order, we obtain

3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 6, 8, 20, 27, 32

Picking up the middle value, we obtain the median equal to 5.

#### **INTERPRETATION**

The median number of floors is 5. Out of those 15 buildings, 7 have upto 5 floors and 7 have 5 floors or more. We noticed earlier that the arithmetic mean was distorted toward the few extremely high values in the series and hence became unrepresentative. The median = 5 is much more representative of this series.

| <b><u>Height of buildings (number of floors)</u></b> |                   |
|------------------------------------------------------|-------------------|
| 3                                                    |                   |
| 3                                                    |                   |
| 4                                                    |                   |
| 4                                                    | 7 lower           |
| 4                                                    |                   |
| 5                                                    |                   |
| 5                                                    |                   |
| 5                                                    | 5 = median height |
| 5                                                    |                   |
| 5                                                    |                   |
| 6                                                    |                   |
| 8                                                    | 7 higher          |
| 20                                                   |                   |
| 27                                                   |                   |
| 32                                                   |                   |



**EXAMPLE 2**

| Retail price of motor-car (£)<br>(several makes and sizes) |                |
|------------------------------------------------------------|----------------|
| 415                                                        |                |
| 480                                                        |                |
| 525                                                        | 4 above        |
| 608                                                        |                |
| <b>719</b>                                                 | = median price |
| 1,090                                                      |                |
| 2,059                                                      |                |
| 4,000                                                      | 4 above        |
| 6,000                                                      |                |

A slight complication arises when there are even numbers of observations in the series, for now there are two middle values.

The expedient of taking the arithmetic mean of the two is adopted as explained below:

**EXAMPLE-3**

| Number of passengers travelling on a<br>bus at six Different times during the day |                |
|-----------------------------------------------------------------------------------|----------------|
| 4                                                                                 |                |
| 9                                                                                 |                |
| <b>14</b>                                                                         | = median value |
| <b>18</b>                                                                         |                |
| 23                                                                                |                |
| 47                                                                                |                |
| Median = $\frac{14 + 18}{2} = 16$ passengers                                      |                |

**EXAMPLE -4:**

The number of passengers traveling on a bus at six different times during a day is as follows:

5, 14, 47, 34, 18, 23

Find the median.

**Solution:**

Arranging the values in ascending order, we obtain

5, 14, 18, 23, 34, 47

As before, a slight complication has arisen because of the fact that there are even numbers of observations in the series and, as such, there are two middle values. As before, we take the arithmetic mean of the two middle values.

Hence we obtain:

**Median:**

$$\tilde{X} = \frac{18 + 23}{2} = 20.$$

A very important point to be noted here is that we must arrange the data in ascending order before searching for the two middle values. All the above examples pertained to raw data. Let us now consider the case of grouped data.

We begin by discussing the case of discrete data grouped into a frequency table.

As stated earlier, a discrete frequency distribution is no more than a concise representation of a simple series pertaining to a discrete variable, so that the same approach as the one discussed just now would seem relevant.

**EXAMPLE OF A DISCRETE FREQUENCY DISTRIBUTION**

| <b>Comprehensive School</b> |                   |
|-----------------------------|-------------------|
| Number of pupils per class  | Number of Classes |
| 23                          | 1                 |
| 24                          | 0                 |
| 25                          | 1                 |
| 26                          | 3                 |
| 27                          | 6                 |
| 28                          | 9                 |
| 29                          | 8                 |
| 30                          | 10                |
| 31                          | 7                 |
|                             | <hr/> 45          |

In order to locate the middle value, the best thing is to first of all construct a column of cumulative frequencies:

| <b>Comprehensive School</b> |                   |                      |
|-----------------------------|-------------------|----------------------|
| Number of pupils per class  | Number of Classes | Cumulative Frequency |
| X                           | f                 | cf                   |
| 23                          | 1                 | 1                    |
| 24                          | 0                 | 1                    |
| 25                          | 1                 | 2                    |
| 26                          | 3                 | 5                    |
| 27                          | 6                 | 11                   |
| 28                          | 9                 | 20                   |
| 29                          | 8                 | 28                   |
| 30                          | 10                | 38                   |
| 31                          | 7                 | 45                   |
|                             | <hr/> 45          |                      |

In this school, there are 45 classes in all, so that we require as the median that class-size below which there are 22 classes and above which also there are 22 classes.

In other words, we must find the 23rd class in an ordered list. We could simply count down noticing that there is 1 class of 23 children, 2 classes with up to 25 children, 5 classes with up to 26 children. Proceeding in this manner, we find that 20 classes contain up to 28 children whereas 28 classes contain up to 29 children. This means that the 23rd class --- the one that we are looking for --- is the one which contains exactly 29 children.

| <b>Comprehensive School</b> |                   |                      |
|-----------------------------|-------------------|----------------------|
| Number of pupils per class  | Number of Classes | Cumulative Frequency |
| X                           | f                 | cf                   |
| 23                          | 1                 | 1                    |
| 24                          | 0                 | 1                    |
| 25                          | 1                 | 2                    |
| 26                          | 3                 | 5                    |
| 27                          | 6                 | 11                   |
| 28                          | 9                 | 20                   |
| 29                          | 8                 | 28                   |
| 30                          | 10                | 38                   |
| 31                          | 7                 | 45                   |
|                             | <hr/> 45          |                      |

Median number of pupils per class:

$$\tilde{X} = 29$$

This means that 29 is the middle size of the class. In other words, 22 classes are such which contain 29 or less than 29 children, and 22 classes are such which contain 29 or more than 29 children.

## LECTURE NO. 8

- Median in case of a frequency distribution of a continuous variable
- Median in case of an open-ended frequency distribution
- Empirical relation between the mean, median and the mode
- Quantiles (quartiles, deciles & percentiles)
- Graphic location of quantiles.

**MEDIAN IN CASE OF A FREQUENCY DISTRIBUTION OF A CONTINUOUS VARIABLE:**

In case of a frequency distribution, the median is given by the formula

$$\tilde{X} = l + \frac{h}{f} \left( \frac{n}{2} - c \right)$$

Where

$l$  = lower class boundary of the median class (i.e. that class for which the cumulative frequency is just in excess of  $n/2$ ).

$h$  = class interval size of the median class

$f$  = frequency of the median class

$n = \sum f$  (the total number of observations)

$c$  = cumulative frequency of the class preceding the median class

**Note:**

This formula is based on the assumption that the observations in each class are evenly distributed between the two class limits.

**EXAMPLE:**

Going back to the example of the EPA mileage ratings, we have

| Mileage Rating | No. of Cars | Class Boundaries | Cumulative Frequency |
|----------------|-------------|------------------|----------------------|
| 30.0 – 32.9    | 2           | 29.95 – 32.95    | 2                    |
| 33.0 – 35.9    | 4           | 32.95 – 35.95    | 6                    |
| 36.0 – 38.9    | 14          | 35.95 – 38.95    | 20                   |
| 39.0 – 41.9    | 8           | 38.95 – 41.95    | 28                   |
| 42.0 – 44.9    | 2           | 41.95 – 44.95    | 30                   |

In this example,  $n = 30$  and  $n/2 = 15$ .

Thus the third class is the median class. The median lies somewhere between 35.95 and 38.95. Applying the above formula, we obtain

$$\begin{aligned} \tilde{X} &= 35.95 + \frac{3}{14}(15 - 6) \\ &= 35.95 + 1.93 \\ &= 37.88 \\ &\simeq 37.9 \end{aligned}$$

**INTERPRETATION**

This result implies that half of the cars have mileage less than or up to 37.88 miles per gallon whereas the other half of the cars has mileage greater than 37.88 miles per gallon. As discussed earlier, the median is preferable to the arithmetic

mean when there are a few very high or low figures in a series. It is also exceedingly valuable when one encounters a frequency distribution having open-ended class intervals.

The concept of open-ended frequency distribution can be understood with the help of the following example.

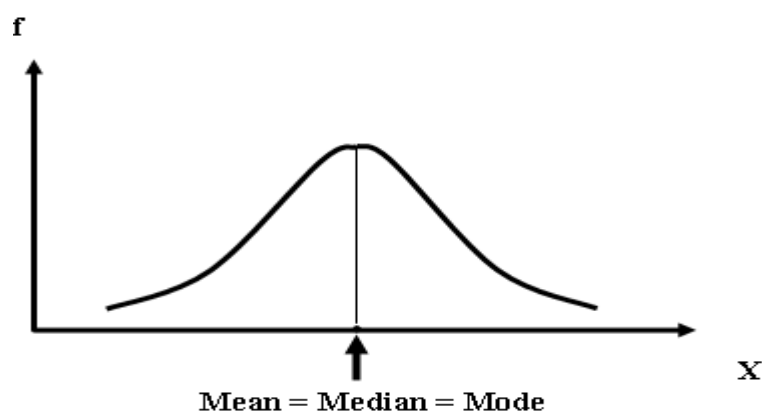
**Example:**

| WAGES OF WORKERS<br>IN A FACTORY |                   |
|----------------------------------|-------------------|
| Monthly Income<br>(in Rupees)    | No. of<br>Workers |
| Less than 2000/-                 | 100               |
| 2000/- to 2999/-                 | 300               |
| 3000/- to 3999/-                 | 500               |
| 4000/- to 4999/-                 | 250               |
| 5000/- and above                 | 50                |
| Total                            | 1200              |

In this example, both the first class and the last class are open-ended classes. This is so because of the fact that we do not have exact figures to begin the first class or to end the last class. The advantage of computing the median in the case of an open-ended frequency distribution is that, except in the unlikely event of the median falling within an open-ended group occurring in the beginning of our frequency distribution, there is no need to estimate the upper or lower boundary. This is so because of the fact that, if the median is falling in an intermediate class, then, obviously, the first class is not being involved in its computation. The next concept that we will discuss is the empirical relation between the mean, median and the mode. This is a concept which is not based on a rigid mathematical formula; rather, it is based on observation. In fact, the word 'empirical' implies 'based on observation'.

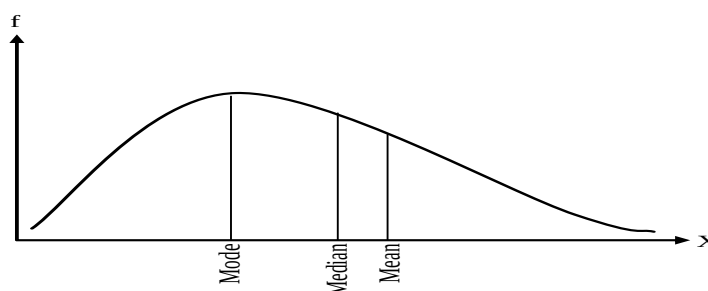
This concept relates to the relative positions of the mean, median and the mode in case of a hump-shaped distribution. In a single-peaked frequency distribution, the values of the mean, median and mode coincide if the frequency distribution is absolutely symmetrical.

### THE SYMMETRIC CURVE



But in the case of a skewed distribution, the mean, median and mode do not all lie on the same point. They are pulled apart from each other, and the empirical relation explains the way in which this happens. Experience tells us that in a unimodal curve of moderate skewness, the median is usually sandwiched between the mean and the mode.

The second point is that, in the case of many real-life data-sets, it has been observed that the distance between the mode and the median is approximately double of the distance between the median and the mean, as shown below:



This diagrammatic picture is equivalent to the following algebraic expression:

$$\text{Median} - \text{Mode} = 2 (\text{Mean} - \text{Median}) \text{ ---- (1)}$$

The above-mentioned point can also be expressed in the following way:

$$\text{Mean} - \text{Mode} = 3 (\text{Mean} - \text{Median}) \text{ ---- (2)}$$

Equation (1) as well as equation (2) yields the approximate relation given below:

### EMPIRICAL RELATION BETWEEN THE MEAN, MEDIAN AND THE MODE

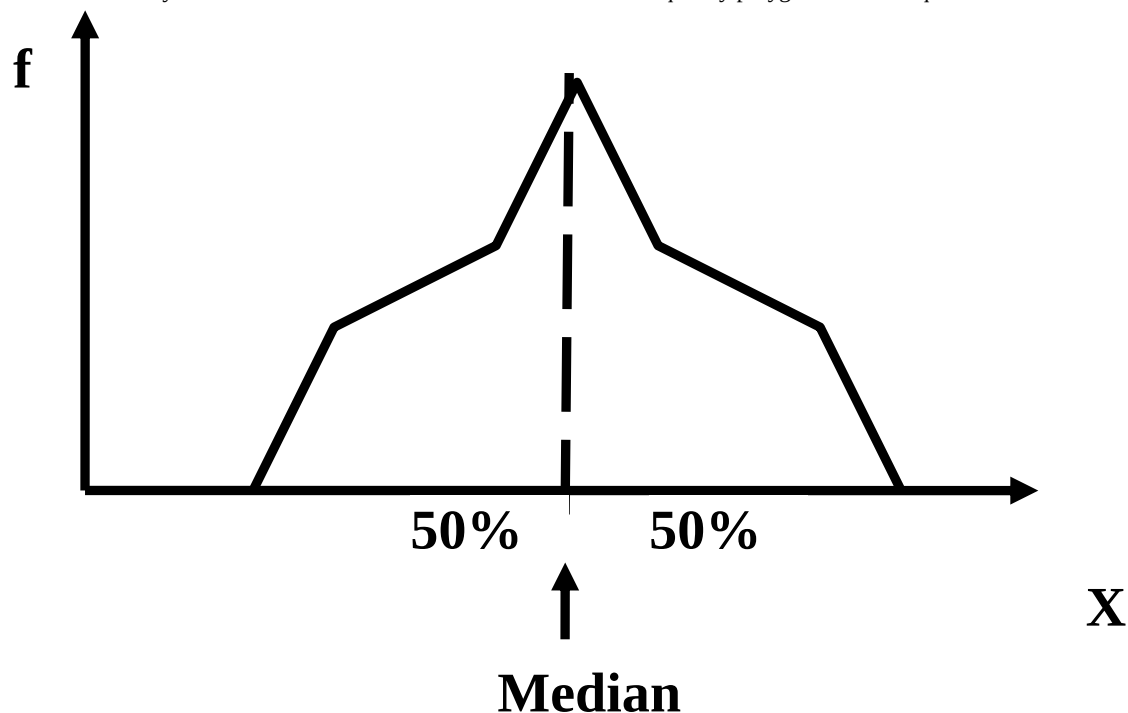
$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$$

An exactly similar situation holds in case of a moderately negatively skewed distribution.

An important point to note is that this empirical relation does not hold in case of a J-shaped or an extremely skewed distribution.

Let us now extend the concept of partitioning of the frequency distribution by taking up the concept of quantiles (i.e. quartiles, deciles and percentiles).

We have already seen that the median divides the area under the frequency polygon into two equal halves:



A further split to produce quarters, tenths or hundredths of the total area under the frequency polygon is equally possible, and may be extremely useful for analysis. (We are often interested in the highest 10% of some group of values or the middle 50% another).

### QUANTILES

The quartiles, together with the median, achieve the division of the total area into four equal parts.

The first, second and third quartiles are given by the formulae:

#### 1. FIRST QUARTILE

$$Q_1 = l + \frac{h}{f} \left( \frac{n}{4} - c \right)$$

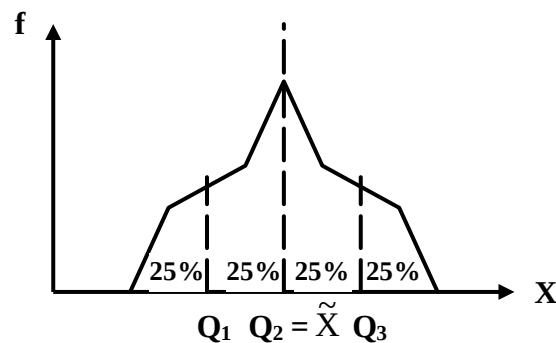
#### 2. SECOND QUARTILE (I.E. MEDIAN)

$$Q_2 = l + \frac{h}{f} \left( \frac{2n}{4} - c \right) = l + \frac{h}{f} (n/2 - c)$$

**3. THIRD QUARTILE**

$$Q_3 = l + \frac{h}{f} \left( \frac{3n}{4} - c \right)$$

It is clear from the formula of the second quartile that the second quartile is the same as the median.

**DECILES & PERCENTILES**

The deciles and the percentiles give the division of the total area into 10 and 100 equal parts respectively. The formula for the first decile is

$$D_1 = l + \frac{h}{f} \left( \frac{n}{10} - c \right)$$

The formulae for the subsequent deciles are

$$D_2 = l + \frac{h}{f} \left( \frac{2n}{10} - c \right)$$

$$D_3 = l + \frac{h}{f} \left( \frac{3n}{10} - c \right)$$

and so on.

It is easily seen that the 5th decile is the same quantity as the median. The formula for the first percentile is

$$P_1 = l + \frac{h}{f} \left( \frac{n}{100} - c \right)$$

The formulae for the subsequent percentiles are

$$P_2 = l + \frac{h}{f} \left( \frac{2n}{100} - c \right)$$

$$P_3 = l + \frac{h}{f} \left( \frac{3n}{100} - c \right)$$

and so on.

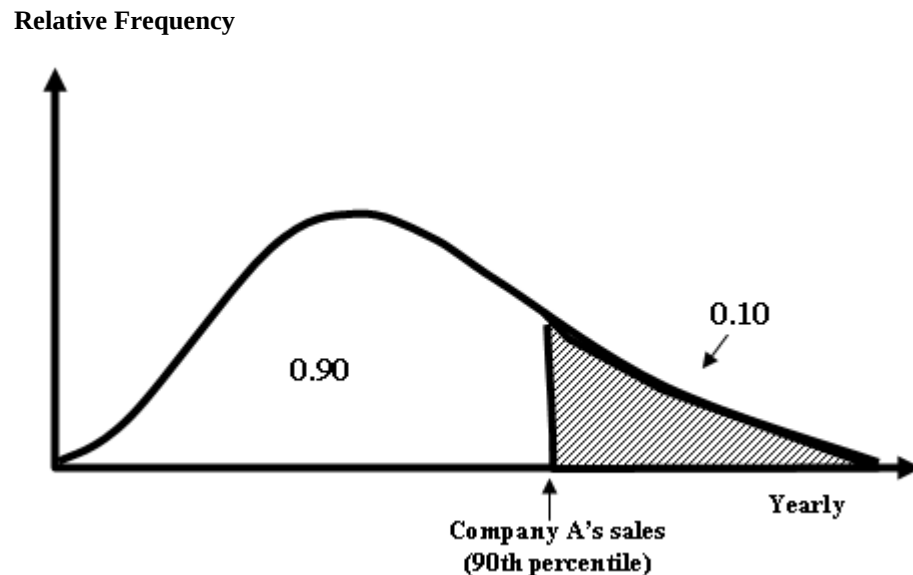
Again, it is easily seen that the 50th percentile is the same as the median, the 25th percentile is the same as the 1st quartile, the 75th percentile is the same as the 3rd quartile, the 40th percentile is the same as the 4th decile, and so on.

All these measures i.e. the median, quartiles, deciles and percentiles are collectively called quantiles. The question is, “What is the significance of this concept of partitioning? Why is it that we wish to divide our frequency distribution into two, four, ten or hundred parts?”

The answer to the above questions is: In certain situations, we may be interested in describing the *relative* quantitative location of a particular measurement within a data set. Quantiles provide us with an easy way of achieving this. Out of these various quantiles, one of the most frequently used is percentile ranking. Let us understand this point with the help of an example.

#### **EXAMPLE**

If oil company ‘A’ reports that its yearly sales are at the 90th percentile of all companies in the industry, the implication is that 90% of all oil companies have yearly sales *less* than company A’s, and only 10% have yearly sales exceeding company A’s, this is demonstrated in the following figure:



It is evident from the above example that the concept of percentile ranking is quite a useful concept, but it should be kept in mind that percentile rankings are of practical value only for large data sets.

It is evident from the above example that the concept of percentile ranking is quite a useful concept, but it should be kept in mind that percentile rankings are of practical value only for large data sets. The next concept that we will discuss is the graphic location of quantiles.

Let us go back to the example of the EPA mileage ratings of 30 cars that was discussed in an earlier lecture.

#### **EXAMPLE**

Suppose that the Environmental Protection Agency of a developed country performs extensive tests on all new car models in order to determine their mileage rating. Suppose that the following 30 measurements are obtained by conducting such tests on a particular new car model.

| EPA MILEAGE RATINGS ON 30 CARS (MILES PER GALLON) |      |      |
|---------------------------------------------------|------|------|
| 36.3                                              | 42.1 | 44.9 |
| 30.1                                              | 37.5 | 32.9 |
| 40.5                                              | 40.0 | 40.2 |
| 36.2                                              | 35.6 | 35.9 |
| 38.5                                              | 38.8 | 38.6 |
| 36.3                                              | 38.4 | 40.5 |
| 41.0                                              | 39.0 | 37.0 |
| 37.0                                              | 36.7 | 37.1 |
| 37.1                                              | 34.8 | 33.9 |
| 39.9                                              | 38.1 | 39.8 |

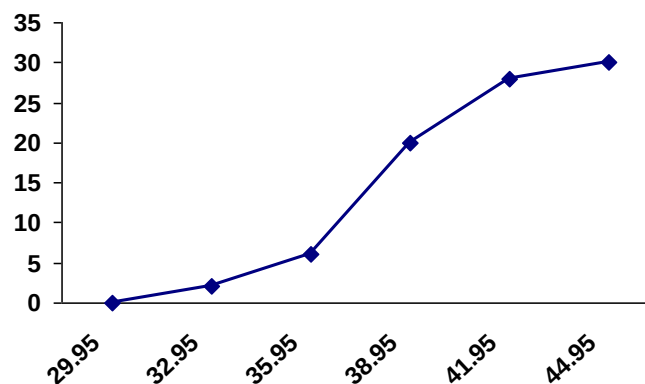
When the above data was converted to a frequency distribution, we obtained:

| Class Limit | Frequency |
|-------------|-----------|
| 30.0 – 32.9 | 2         |
| 33.0 – 35.9 | 4         |
| 36.0 – 38.9 | 14        |
| 39.0 – 41.9 | 8         |
| 42.0 – 44.9 | 2         |
|             | 30        |

Also, we considered the graphical representation of this distribution.

The cumulative frequency polygon of this distribution came out to be as shown in the following figure:

### **Cumulative Frequency Polygon or OGIVE**



This ogive enables us to find the median and any other quantile that we may be interested in very conveniently. And this process is known as the graphic location of quantiles.

Let us begin with the graphical location of the median:

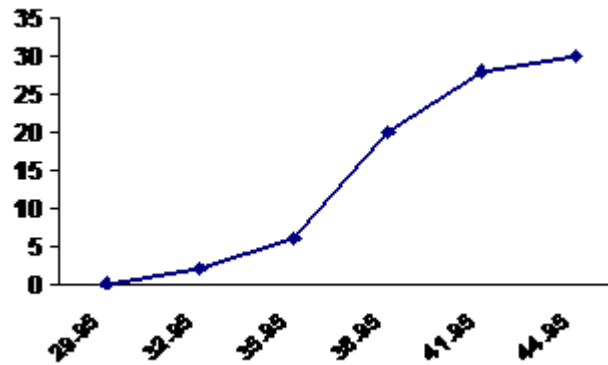
Because of the fact that the median is that value before which half of the data lies, the first step is to divide the total number of observations  $n$  by 2.

In this example:

$$\frac{n}{2} = \frac{30}{2} = 15$$

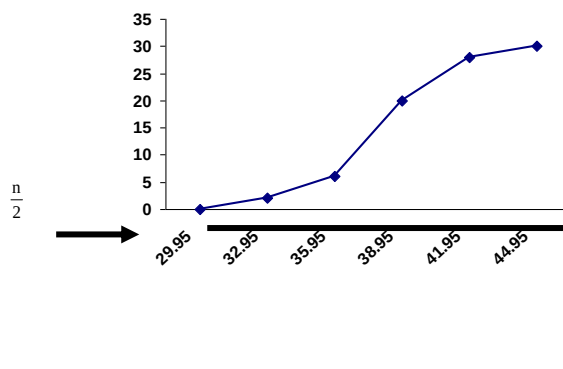
The next step is to locate this number 15 on the y-axis of the cumulative frequency polygon.





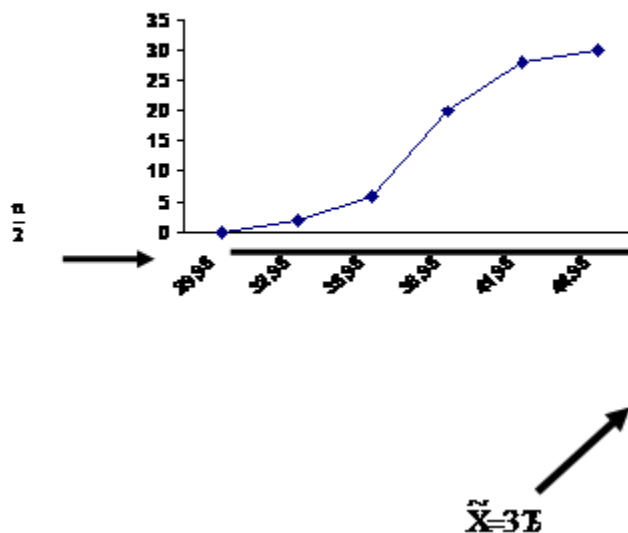
Lastly, we drop a vertical line from the cumulative frequency polygon down to the x-axis.

### Cumulative Frequency Polygon or OGIVE



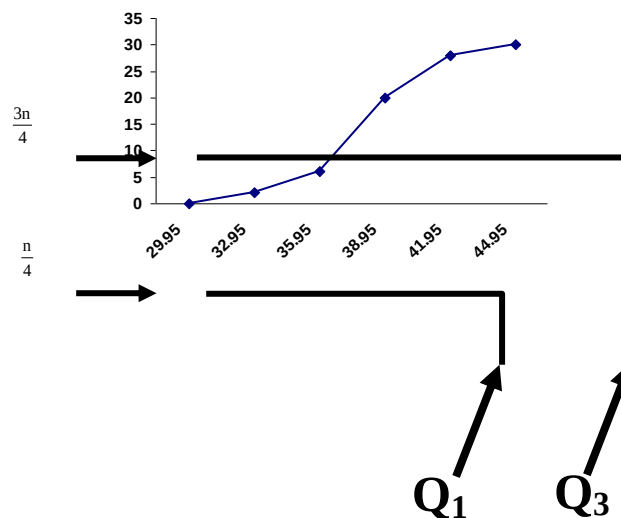
Now, if we read the x-value where our perpendicular touches the x-axis, students, we find that this value is approximately the same as what we obtained from our formula.

### Cumulative Frequency Polygon or OGIVE



It is evident from the above example that the cumulative frequency polygon is a very useful device to find the value of the median very quickly. In a similar way, we can locate the quartiles, deciles and percentiles. To obtain the first quartile, the horizontal line will be drawn against the value  $n/4$ , and for the third quartile, the horizontal line will be drawn against the value  $3n/4$ .

### Cumulative Frequency Polygon or OGIVE



For the deciles, the horizontal lines will be against the values  $n/10$ ,  $2n/10$ ,  $3n/10$ , and so on. And for the percentiles, the horizontal lines will be against the values  $n/100$ ,  $2n/100$ ,  $3n/100$ , and so on.

The graphic location of the quartiles as well as of a few deciles and percentiles for the data-set of the EPA mileage ratings may be taken up as an exercise:

This brings us to the end of our discussion regarding quantiles which are sometimes also known as fractiles --- this terminology because of the fact that they divide the frequency distribution into various parts or fractions.

**LECTURE NO. 9**

- Geometric mean
- Harmonic mean
- Relation between the arithmetic, geometric and harmonic means
- Some other measures of central tendency

**GEOMETRIC MEAN**

The geometric mean,  $G$ , of a set of  $n$  positive values  $X_1, X_2, \dots, X_n$  is defined as the positive  $n$ th root of their product.

$$G = \sqrt[n]{X_1 X_2 \dots X_n}$$

(Where  $X_i > 0$ )

When  $n$  is large, the computation of the geometric mean becomes laborious as we have to extract the  $n$ th root of the product of all the values.

The arithmetic is simplified by the use of logarithms.

Taking logarithms to the base 10, we get

$$\log G = \frac{1}{n} [\log X_1 + \log X_2 + \dots + \log X_n]$$

Hence

$$G = \text{antilog} \left[ \frac{\sum \log X}{n} \right]$$

**EXAMPLE**

Find the geometric mean of numbers:

45, 32, 37, 46, 39, 36, 41, 48, 36

**Solution:**

We need to compute the numerical value of

$$= \sqrt[9]{45 \times 32 \times 37 \times 46 \times 39 \times 36 \times 41 \times 48 \times 36}$$

But, obviously, it is a bit cumbersome to find the ninth root of a quantity. So we make use of logarithms, as shown below:

| X       | log X  |
|---------|--------|
| 45      | 1.6532 |
| 32      | 1.5052 |
| 37      | 1.5682 |
| 46      | 1.6628 |
| 39      | 1.5911 |
| 36      | 1.5563 |
| 41      | 1.6128 |
| 48      | 1.6812 |
| 36      | 1.5563 |
| 14.3870 |        |

$$\log G = \frac{\sum \log X}{n}$$

$$= \frac{14.3870}{9} = 1.5986$$

$$\text{Hence } G = \text{antilog } 1.5986 \\ = 39.68$$

The above example pertained to the computation of the geometric mean in case of raw data. Next, we consider the computation of the geometric mean in the case of grouped data.

**GEOMETRIC MEAN FOR GROUPED DATA**

In case of a frequency distribution having  $k$  classes with midpoints  $X_1, X_2, \dots, X_k$  and the corresponding frequencies  $f_1, f_2, \dots, f_k$  (such that  $\sum f_i = n$ ), the geometric mean is given by

$$G = \sqrt[n]{X_1^{f_1} X_2^{f_2} \dots X_k^{f_k}}$$

Each value of  $X$  thus has to be multiplied by itself  $f$  times, and the whole procedure becomes quite a formidable task! In terms of logarithms, the formula becomes

$$\log G = \frac{1}{n} [f_1 \log X_1 + f_2 \log X_2 + \dots + f_k \log X_k]$$

$$= \frac{\sum f \log X}{n}$$

Hence

$$G = \text{anti log} \left[ \frac{\sum f \log X}{n} \right]$$

Obviously, the above formula is much easier to handle. Let us now apply it to an example. Going back to the example of the EPA mileage ratings, we have:

| Mileage Rating | No. of Cars | Class-mark (midpoint) $X$ | $\log X$ | $f \log X$ |
|----------------|-------------|---------------------------|----------|------------|
| 30.0 - 32.9    | 2           | 31.45                     | 1.4976   | 2.9952     |
| 33.0 - 35.9    | 4           | 34.45                     | 1.5372   | 6.1488     |
| 36.0 - 38.9    | 14          | 37.45                     | 1.5735   | 22.0290    |
| 39.0 - 41.9    | 8           | 40.45                     | 1.6069   | 12.8552    |
| 42.0 - 44.9    | 2           | 43.45                     | 1.6380   | 3.2760     |
|                | 30          |                           |          | 47.3042    |

$$G = \text{antilog} \frac{47.3042}{30}$$

$$= \text{antilog } 1.5768 = 37.74$$

This means that, if we use the geometric mean to measure the central tendency of this data set, then the central value of the mileage of those 30 cars comes out to be 37.74 miles per gallon.

The question is, “When should we use the geometric mean?”

The answer to this question is that when relative changes in some variable quantity are averaged, we prefer the geometric mean.

**EXAMPLE**

Suppose it is discovered that a firm's turnover has increased during 4 years by the following amounts:

| Year | Turnover | Percentage Compared With Year Earlier |
|------|----------|---------------------------------------|
| 1958 | £ 2,000  | —                                     |
| 1959 | £ 2,500  | 125                                   |
| 1960 | £ 5,000  | 200                                   |
| 1961 | £ 7,500  | 150                                   |
| 1962 | £ 10,500 | 140                                   |

The yearly increase is shown in a percentage form in the right-hand column i.e. the turnover of 1959 is 125 percent of the turnover of 1958, the turnover of 1960 is 200 percent of the turnover of 1959, and so on. The firm's owner may be interested in knowing his average rate of turnover growth.

If the arithmetic mean is adopted he finds his answer to be:

Arithmetic Mean:

$$\frac{125 + 200 + 150 + 140}{4} = 153.75$$

i.e. we are concluding that the turnover for any year is 153.75% of the turnover for the previous year. In other words, the turnover in each of the years considered appears to be 53.75 per cent higher than in the previous year.

If this percentage is used to calculate the turnover from 1958 to 1962 inclusive, we obtain:

$$153.75\% \text{ of } £ 2,000 = £ 3,075$$

$$153.75\% \text{ of } £ 3,075 = £ 4,728$$

$$153.75\% \text{ of } £ 4,728 = £ 7,269$$

$$153.75\% \text{ of } £ 7,269 = £ 11,176$$

Whereas the actual turnover figures were

| Year | Turnover |
|------|----------|
| 1958 | £ 2,000  |
| 1959 | £ 2,500  |
| 1960 | £ 5,000  |
| 1961 | £ 7,500  |
| 1962 | £ 10,500 |

It seems that both the individual figures and, more important, the total at the end of the period, are incorrect. Using the arithmetic mean has exaggerated the 'average' annual rate of increase in the turnover of this firm. Obviously, we would like to rectify this false impression. The geometric mean enables us to do so:

Geometric mean of the turnover figures:

$$\sqrt[4]{(125 \times 200 \times 150 \times 140)} = \sqrt[4]{525000000} = 151.37\%$$

Now, if we utilize this particular value to obtain the individual turnover figures, we find that:

$$151.37\% \text{ of } £2,000 = £3,027$$

$$151.37\% \text{ of } £3,027 = £4,583$$

$$151.37\% \text{ of } £4,583 = £6,937$$

$$151.37\% \text{ of } £6,937 = £10,500$$

So that the turnover figure of 1962 is exactly the same as what we had in the original data.

### INTERPRETATION

If the turnover of this company were to increase annually at a constant rate, then the annual increase would have been 51.37 percent. (On the average, each year's turnover is 51.37% higher than that in the previous year.) The above example clearly indicates the significance of the geometric mean in a situation when relative changes in a variable quantity are to be averaged.

But we should bear in mind that such situations are not encountered too often, and that the occasion to calculate the geometric mean arises less frequently than the arithmetic mean. (The most frequently used measure of central tendency is the arithmetic mean.)

The next measure of central tendency that we will discuss is the harmonic mean.

### HARMONIC MEAN

The harmonic mean is defined as the reciprocal of the arithmetic mean of the reciprocals of the values.

In case of raw data:

$$H.M. = \frac{n}{\sum \left( \frac{1}{x} \right)}$$

In case of grouped data (data grouped into a frequency distribution):

$$H.M. = \frac{n}{\sum f \left( \frac{1}{X} \right)}$$

(Where X represents the midpoints of the various classes)

### EXAMPLE

Suppose a car travels 100 miles with 10 stops, each stop after an interval of 10 miles. Suppose that the speeds at which the car travels these 10 intervals are 30, 35, 40, 40, 45, 40, 50, 55, 55 and 30 miles per hours respectively.

What is the average speed with which the car traveled the total distance of 100 miles?

If we find the arithmetic mean of the 10 speeds, we obtain:

Arithmetic mean of the 10 speeds:

$$\frac{30 + 35 + \dots + 30}{10} = \frac{420}{10} = 42 \text{ miles per hour.}$$

But, if we study the problem carefully, we find that the above answer is incorrect.

By definition, the average speed is the speed with which the car would have traveled the 100 mile distance if it had maintained a constant speed throughout the 10 intervals of 10 miles each.

$$\text{Average speed} = \frac{\text{Total distance travelled}}{\text{Total time taken}}$$

Now, total distance traveled = 100 miles. Total time taken will be computed as shown below:

| Interval          | Distance | Speed = $\frac{\text{Distance}}{\text{Time}}$ | Time = $\frac{\text{Distance}}{\text{Speed}}$ |
|-------------------|----------|-----------------------------------------------|-----------------------------------------------|
| 1                 | 10 miles | 30 mph                                        | 10/30 = 0.3333 hrs                            |
| 2                 | 10 miles | 35 mph                                        | 10/35 = 0.2857 hrs                            |
| 3                 | 10 miles | 40 mph                                        | 10/40 = 0.2500 hrs                            |
| 4                 | 10 miles | 40 mph                                        | 10/40 = 0.2500 hrs                            |
| 5                 | 10 miles | 45 mph                                        | 10/45 = 0.2222 hrs                            |
| 6                 | 10 miles | 40 mph                                        | 10/40 = 0.2500 hrs                            |
| 7                 | 10 miles | 50 mph                                        | 10/50 = 0.2000 hrs                            |
| 8                 | 10 miles | 55 mph                                        | 10/55 = 0.1818 hrs                            |
| 9                 | 10 miles | 55 mph                                        | 10/55 = 0.1818 hrs                            |
| 10                | 10 miles | 30 mph                                        | 10/30 = 0.333 hrs                             |
| Total = 100 miles |          | Total Time = 2.4881 hrs                       |                                               |

Hence

$$\text{Average speed} = \frac{100}{2.4881} = 40.2 \text{ mph}$$

which is not the same as 42 miles per hour.

Let us now try the harmonic mean to find the average speed of the car.

$$H.M. = \frac{n}{\sum \left( \frac{1}{X} \right)}$$

where n is the no. of terms)

We have:

| X  | 1/X                         |
|----|-----------------------------|
| 30 | 1/30 = 0.0333               |
| 35 | 1/35 = 0.0286               |
| 40 | 1/40 = 0.0250               |
| 40 | 1/40 = 0.0250               |
| 45 | 1/45 = 0.0222               |
| 40 | 1/40 = 0.0250               |
| 50 | 1/50 = 0.0200               |
| 55 | 1/55 = 0.0182               |
| 55 | 1/55 = 0.0182               |
| 30 | 1/30 = 0.0333               |
|    | $\sum \frac{1}{X} = 0.2488$ |

$$H.M. = \frac{n}{\sum \frac{1}{X}}$$

$$= \frac{10}{0.2488}$$

$$= 40.2 \text{ mph}$$

Hence it is clear that the harm gives the totally correct result

The key question is, “When should we compute the harmonic mean of a data set?” The answer to this question will be easy to understand if we consider the following rules:

#### **RULES**

- When values are given as x per y where x is constant and y is variable, the Harmonic Mean is the appropriate average to use.
- When values are given as x per y where y is constant and x is variable, the Arithmetic Mean is the appropriate average to use.
- When relative changes in some variable quantity are to be averaged, the geometric mean is the appropriate average to use.

We have already discussed the geometric and the harmonic means. Let us now try to understand Rule No. 1 with the help of an example:

#### **EXAMPLE**

If 10 students have obtained the following marks (in a test) out of 20:

13, 11, 9, 9, 6, 5, 19, 17, 12, 9

**Then the average marks (by the formula of the arithmetic mean) are:**

$$\frac{13 + 11 + 9 + 9 + 6 + 5 + 19 + 17 + 12 + 9}{10}$$

$$= \frac{110}{10} = 11$$

This is equivalent to

$$\frac{\frac{13}{20} + \frac{11}{20} + \frac{9}{20} + \frac{9}{20} + \frac{6}{20} + \frac{5}{20} + \frac{19}{20} + \frac{17}{20} + \frac{12}{20} + \frac{9}{20}}{10}$$

$$= \frac{\frac{110}{20}}{10} = \frac{110}{10 \times 20} = \frac{11}{20}$$

(i.e. the average marks of this group of students are 11 out of 20). In the above example, the point to be noted was that all the marks were expressible as  $x$  per  $y$  where the denominator  $y$  was constant i.e. equal to 20, and hence, it was appropriate to compute the arithmetic mean.

Let us now consider a mathematical relationship exists between these three measures of central tendency.

### **RELATION BETWEEN ARITHMETIC, GEOMETRIC AND HARMONIC MEANS**

#### **Arithmetic Mean $\geq$ Geometric Mean $\geq$ Harmonic Mean**

We have considered the five most well-known measures of central tendency i.e. arithmetic mean, median, mode, geometric mean and harmonic mean. It is interesting to note that there are some other measures of central tendency as well. Two of these are the mid range, and the mid quartile range.

Let us consider these one by one:

#### **MID-RANGE**

If there are  $n$  observations with  $x_0$  and  $x_m$  as their smallest and largest observations respectively, then their mid-range is defined as

$$\text{mid - range} = \frac{x_0 + x_m}{2}$$

It is obvious that if we add the smallest value with the largest, and divide by 2, we will get a value which is more or less in the middle of the data-set.

#### **MID-QUARTILE RANGE**

If  $x_1, x_2 \dots x_n$  are  $n$  observations with  $Q_1$  and  $Q_3$  as their first and third quartiles respectively, then their mid-quartile range is defined as

$$\text{mid - quartile range} = \frac{Q_1 + Q_3}{2}$$

Similar to the case of the mid-range, if we take the arithmetic mean of the upper and lower quartiles, we will obtain a value that is somewhere in the middle of the data-set. The mid-quartile range is also known as the mid-hinge.

Let us now revise briefly the core concept of central tendency: Masses of data are usually expressed in the form of frequency tables so that it becomes easy to comprehend the data. Usually, a statistician would like to go a step ahead and to compute a number that will represent the data in some definite way.

**Any such single number that represents a whole set of data is called 'Average'.**

Technically speaking, there are many kinds of averages (i.e. there are several ways to compute them). These quantities that represent the data-set are called "measures of central tendency".



**LECTURE NO. 10**

- Concept of dispersion
- Absolute and relative measures of dispersion
- Range
- Coefficient of dispersion
- Quartile deviation
- Coefficient of quartile deviation

Let us begin the concept of DISPERSION.

Just as variable series differ with respect to their location on the horizontal axis (having different 'average' values); similarly, they differ in terms of the amount of variability which they exhibit. Let us understand this point with the help of an example:

**EXAMPLE**

In a technical college, it may well be the case that the ages of a group of first-year students are quite consistent, e.g. 17, 18, 18, 19, 18, 19, 18, 17, 18 and 18 years.

A class of evening students undertaking a course of study in their spare time may show just the opposite situation, e.g. 35, 23, 19, 48, 32, 24, 29, 37, 58, 18, 21 and 30.

It is very clear from this example that the variation that exists between the various values of a data-set is of substantial importance. We obviously need to be aware of the amount of variability present in a data-set if we are to come to useful conclusions about the situation under review. This is perhaps best seen from studying the two frequency distributions given below.

**EXAMPLE**

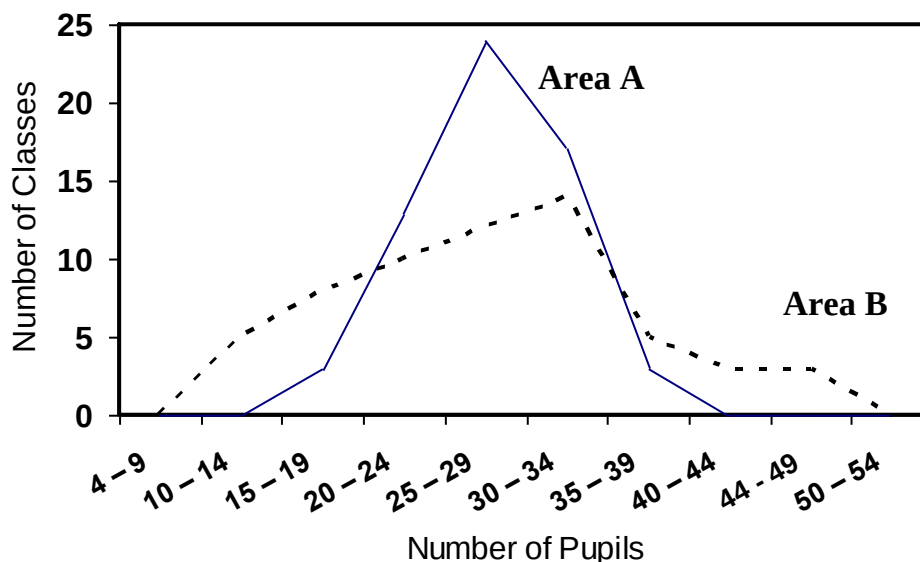
The sizes of the classes in two comprehensive schools in different areas are as follows:

| Number of Pupils | Number of Classes |        |
|------------------|-------------------|--------|
|                  | Area A            | Area B |
| 10 – 14          | 0                 | 5      |
| 15 – 19          | 3                 | 8      |
| 20 – 24          | 13                | 10     |
| 25 – 29          | 24                | 12     |
| 30 – 34          | 17                | 14     |
| 35 – 39          | 3                 | 5      |
| 40 – 44          | 0                 | 3      |
| 45 - 49          | 0                 | 3      |
|                  | 60                | 60     |

If the arithmetic mean size of class is calculated, we discover that the answer is identical: 27.33 pupils in both areas.  
Average class-size of each school

$$\bar{X} = 27.33$$

Even though these two distributions share a common average, it can readily be seen that they are entirely DIFFERENT. And the graphs of the two distributions (given below) clearly indicate this fact.



The question which must be posed and answered is ‘In what way can these two situations be distinguished?’ We need a measure of variability or DISPERSION to accompany the relevant measure of position or ‘average’ used. The word ‘relevant’ is important here for we shall find one measure of dispersion which expresses the scatter of values round the arithmetic mean, another the scatter of values round the median, and so forth. Each measure of dispersion is associated with a particular ‘average’.

### **ABSOLUTE VERSUS RELATIVE MEASURES OF DISPERSION**

There are two types of measurements of dispersion: absolute and relative.

**An absolute measure of dispersion is one that measures the dispersion in terms of the same units or in the square of units, as the units of the data.**

For example, if the units of the data are rupees, meters, kilograms, etc., the units of the measures of dispersion will also be rupees, meters, kilograms, etc.

On the other hand, **relative measure of dispersion is one that is expressed in the form of a ratio, co-efficient of percentage and is independent of the units of measurement.**

**A relative measure of dispersion is useful for comparison of data of different nature.** A measure of central tendency together with a measure of dispersion gives an adequate description of data. We will be discussing FOUR measures of dispersion i.e. the range, the quartile deviation, the mean deviation, and the standard deviation.

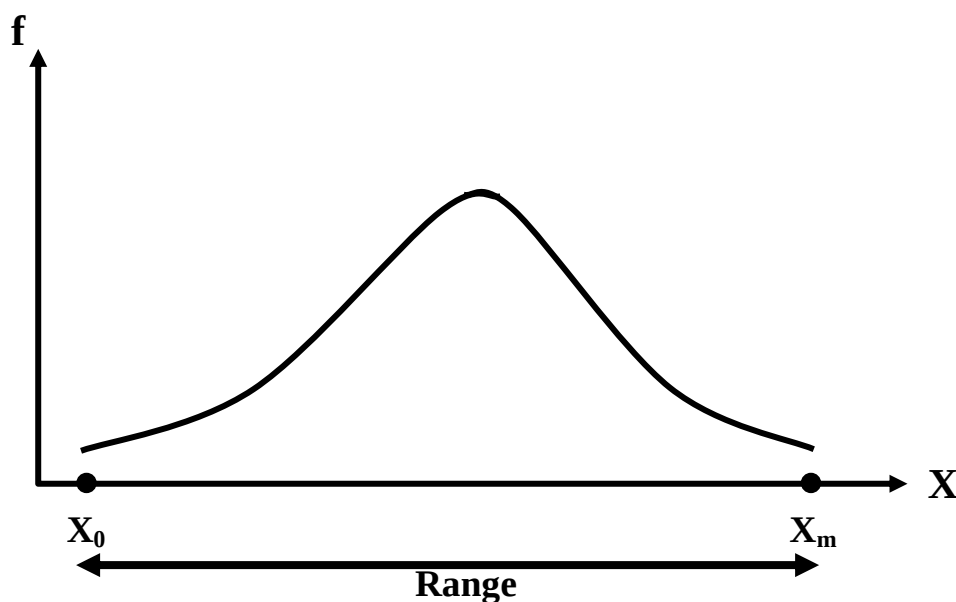
### **RANGE**

**The range is defined as the difference between the two extreme values of a data-set, i.e.  $R = X_m - X_0$  where  $X_m$  represents the highest value and  $X_0$  the lowest.**

Evidently, the calculation of the range is a simple question of MENTAL arithmetic.

The simplicity of the concept does not necessarily invalidate it, but in general it gives no idea of the DISTRIBUTION of the observations between the two ends of the series. For this reason it is used principally as a supplementary aid in the description of variable data, in conjunction with other measures of dispersion. When the data are grouped into a frequency distribution, the range is estimated by finding the difference between the upper boundary of the highest class and the lower boundary of the lowest class.

We now consider the graphical representation of the range:



Obviously, the greater the difference between the largest and the smallest values, the greater will be the range. As stated earlier, the range is a simple concept and is easy to compute. However, because of the fact that it is computed from only the two extreme values in a data-set, it has two serious disadvantages.

- It ignores all the INFORMATION available from the intermediate observations.
- It might give a MISLEADING picture of the spread in the data.

From THIS point of view, it is an unsatisfactory measure of dispersion. However, it is APPROPRIATELY used in statistical quality control charts of manufactured products, daily temperatures, stock prices, etc. It is interesting to note that the range can also be viewed in the following way.

It is twice of the arithmetic mean of the deviations of the smallest and largest values round the mid-range i.e.

$$\begin{aligned}
 & \frac{(\text{Midrange} - X_0) + (X_m - \text{Midrange})}{2} \\
 &= \frac{\text{Midrange} - X_0 + X_m - \text{Midrange}}{2} \\
 &= \frac{X_m - X_0}{2}
 \end{aligned}$$

Because of what has been just explained, the range can be regarded as that measure of dispersion which is associated with the mid-range. As such, the range may be employed to indicate dispersion when the mid-range has been adopted as the most appropriate average.

The range is an *absolute* measure of dispersion. Its *relative* measure is known as the CO-EFFICIENT OF DISPERSION, and is defined by the relation given below:

\ **COEFFICIENT OF DISPERSION**

$$\begin{aligned}
 &= \frac{\frac{1}{2}(\text{Range})}{\text{Mid - Range}} \\
 &= \frac{\frac{X_m - X_0}{2}}{\frac{X_m + X_0}{2}} = \frac{X_m - X_0}{X_m + X_0}
 \end{aligned}$$

This is a pure (i.e. dimensionless) number and is used for the purposes of COMPARISON. (This is so because a pure number can be compared with another pure number.)

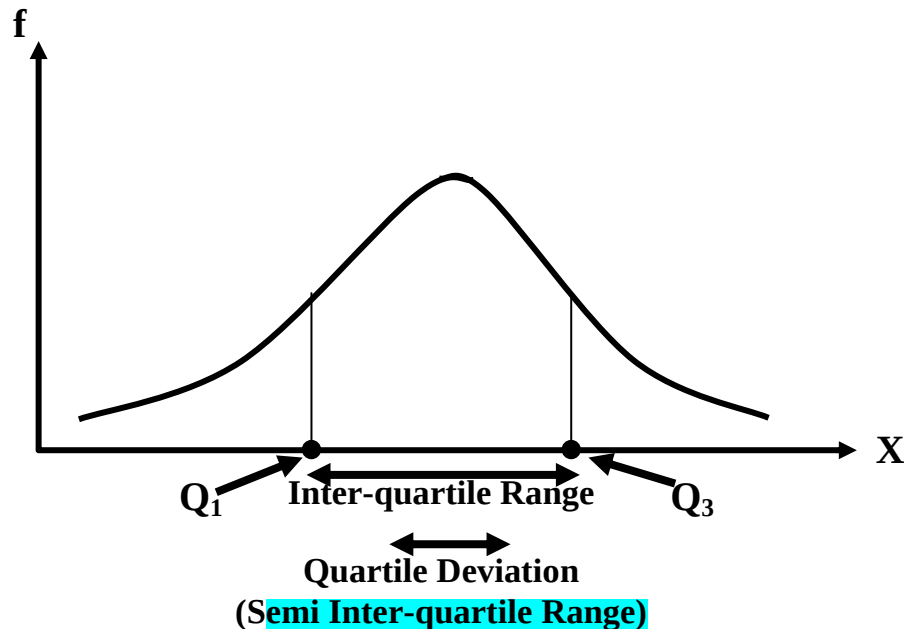
For example, if the coefficient of dispersion for one data-set comes out to be 0.6 whereas the coefficient of dispersion for another data-set comes out to be 0.4, then it is obvious that there is greater amount of dispersion in the first data-set as compared with the second.

### QUARTILE DEVIATION

The quartile deviation is defined as half of the difference between the third and first quartiles i.e.

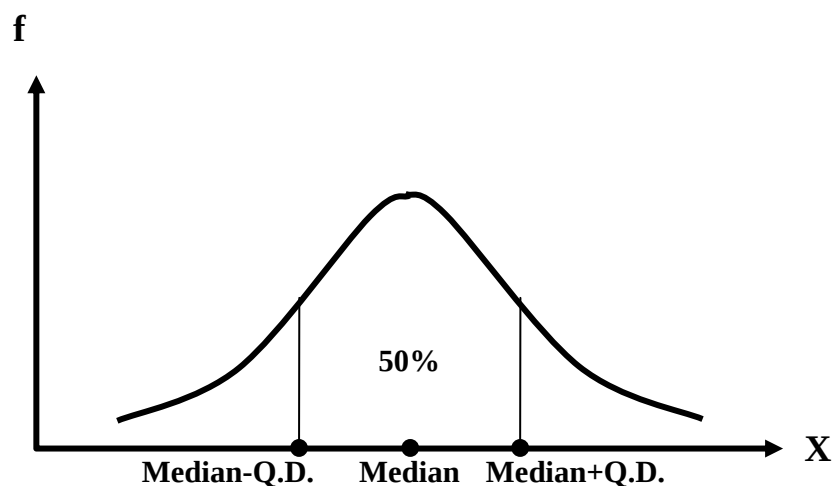
$$Q.D. = \frac{Q_3 - Q_1}{2}$$

It is also known as **semi-interquartile range**. Let us now consider the graphical representation of the quartile deviation:



Although simple to compute, it is NOT an extremely satisfactory measure of dispersion because it takes into account the spread of only two values of the variable round the median, and this gives no idea of the rest of the dispersion within the distribution.

The quartile deviation has an attractive feature that the range “Median + Q.D.” contains approximately 50% of the data. This is illustrated in the figure given below:



Let us now apply the concept of quartile deviation to the following example:

**EXAMPLE**

The shareholding structure of two companies is given below:

|                          | Company<br>X | Company<br>Y |
|--------------------------|--------------|--------------|
| 1 <sup>st</sup> quartile | 60 shares    | 165 shares   |
| Median                   | 185 shares   | 185 shares   |
| 3 <sup>rd</sup> quartile | 270 shares   | 210 shares   |

The quartile deviation for company X is

$$\frac{270 - 60}{2} = 105 \text{ Shares}$$

For company Y, it is

$$\frac{210 - 165}{2} = 22 \text{ Shares}$$

A comparison of the above two results indicate that there is a considerable concentration of shareholders about the MEDIAN number of shares in company Y, whereas in company X, there does not exist this kind of a concentration around the median. (In company X, there is approximately the SAME numbers of small, medium and large shareholders.)

From the above example, it is obvious that the larger the quartile deviation, the greater is the scatter of values within the series. The quartile deviation is superior to range as it is not affected by extremely large or small observations. It is simple to understand and easy to calculate.

The mean deviation can also be viewed in another way: It is the arithmetic mean of the deviations of the first and third quartiles round the median i.e.

$$\begin{aligned} & \frac{(M - Q_1) + (Q_3 - M)}{2} \\ &= \frac{M - Q_1 + Q_3 - M}{2} \\ &= \frac{Q_3 - Q_1}{2} \end{aligned}$$

Because of what has been just explained, the quartile deviation is regarded as that measure of dispersion which is associated with the median. As such, the quartile deviation should always be employed to indicate dispersion when the median has been adopted as the most appropriate average.

The quartile deviation is also an *absolute* measure of dispersion. Its *relative* measure called the CO-EFFICIENT OF QUARTILE DEVIATION or of Semi-Inter-quartile Range, is defined by the relation:

**COEFFICIENT OF QUARTILE DEVIATION**

$$\begin{aligned} &= \frac{\text{Quartile Deviation}}{\text{Mid - Quartile Range}} \\ &= \frac{\frac{Q_3 - Q_1}{2}}{\frac{Q_3 + Q_1}{2}} = \frac{Q_3 - Q_1}{Q_3 + Q_1} \end{aligned}$$

The Coefficient of Quartile Deviation is a pure number and is used for *COMPARING* the variation in two or more sets of data.

The next two measures of dispersion to be discussed are the Mean Deviation and the Standard Deviation. In this regard, the first thing to note is that, whereas the range as well as the quartile deviation are two such measures of dispersion which are NOT based on all the values, the mean deviation and the standard deviation are two such measures of dispersion that involve each and every data-value in their computation.

The range measures the dispersion of the data-set around the *mid-range*, whereas the quartile deviation measures the dispersion of the data-set around the *median*.

How are we to decide upon the amount of dispersion round the *arithmetic mean*?

It would seem reasonable to compute the *DISTANCE* of each observed value in the series from the arithmetic mean of the series.

But the problem is that the sum of the deviations of the values from the mean is ZERO! (No matter what the amount of dispersion in a data-set is, this quantity will *always* be zero, and hence it cannot be used to measure the dispersion in the data-set.)

Then, the question arises, 'HOW will we be able to measure the dispersion present in our data-set?' In an attempt to answer this question, we might look at the numerical differences between the mean and the data values WITHOUT considering whether these are positive or negative. By ignoring the sign of the deviations we will achieve a NON-ZERO sum, and averaging these absolute differences, again, we obtain a non-zero quantity which can be used as a measure of dispersion. (The larger this quantity, the greater is the dispersion in the data-set).

This quantity is known as the **MEAN DEVIATION**.

Let us denote these absolute differences

by 'modulus of d'

or 'mod d'. Then, the mean deviation is given by

### MEAN DEVIATION

$$M.D. = \frac{\sum |d|}{n}$$

As the absolute deviations of the observations from their mean are being averaged, therefore the complete name of this measure is Mean Absolute Deviation --- but generally, it is simply called "Mean Deviation". In the next lecture, this concept will be discussed in detail. (The case of raw data as well as the case of grouped data will be considered.) Next, we will discuss the most important and the most widely used measure of dispersion i.e. the Standard Deviation.

## LECTURE NO. 11

- Mean Deviation
- Standard Deviation and Variance
- Coefficient of variation

First, we will discuss it for the case of raw data, and then we will go on to the case of a frequency distribution. The first thing to note is that, whereas the range as well as the quartile deviation are two such measures of dispersion which are NOT based on all the values, the mean deviation and the standard deviation are two such measures of dispersion that involve each and every data-value in their computation.

You must have noted that the range was measuring the dispersion of the data-set around the mid-range, whereas the quartile deviation was measuring the dispersion of the data-set around the median.

How are we to decide upon the amount of dispersion round the arithmetic mean? It would seem reasonable to compute the DISTANCE of each observed value in the series from the arithmetic mean of the series.

Let us do this for a simple data-set shown below:

THE NUMBER OF FATALITIES IN MOTORWAY ACCIDENTS IN ONE WEEK

| Day       | Number of fatalities<br>X |
|-----------|---------------------------|
| Sunday    | 4                         |
| Monday    | 6                         |
| Tuesday   | 2                         |
| Wednesday | 0                         |
| Thursday  | 3                         |
| Friday    | 5                         |
| Saturday  | 8                         |
| Total     | 28                        |

Let us do this for a simple data-set shown below:

THE NUMBER OF FATALITIES IN MOTORWAY ACCIDENTS IN ONE WEEK

| Day       | Number of fatalities<br>X |
|-----------|---------------------------|
| Sunday    | 4                         |
| Monday    | 6                         |
| Tuesday   | 2                         |
| Wednesday | 0                         |
| Thursday  | 3                         |
| Friday    | 5                         |
| Saturday  | 8                         |
| Total     | 28                        |

The arithmetic mean number of fatalities per day is

$$\bar{X} = \frac{\sum X}{n} = \frac{28}{7} = 4$$

In order to determine the distances of the data-values from the mean, we subtract our value of the arithmetic mean from each daily figure, and this gives us the deviations that occur in the third column of the table below

| Day       | Number of fatalities<br>$X$ | $X - \bar{X}$ |
|-----------|-----------------------------|---------------|
| Sunday    | 4                           | 0             |
| Monday    | 6                           | + 2           |
| Tuesday   | 2                           | - 2           |
| Wednesday | 0                           | - 4           |
| Thursday  | 3                           | - 1           |
| Friday    | 5                           | + 1           |
| Saturday  | 8                           | + 4           |
| TOTAL     | 28                          | 0             |

The deviations are negative when the daily figure is less than the mean (4 accidents) and positive when the figure is higher than the mean. It does seem, however, that our efforts for computing the dispersion of this data set have been in vain, for we find that the total amount of dispersion obtained by summing the  $(x - \bar{x})$  column comes out to be zero! In fact, this should be no surprise, for it is a basic property of the arithmetic means that: The sum of the deviations of the values from the mean is zero. The question arises:

How will we measure the dispersion that is actually present in our data-set?

Our problem might at first sight seem irresolvable, for by this criterion it appears that no series has any dispersion. Yet we know that this is absolutely incorrect, and we must think of some other way of handling this situation. Surely, we might look at the numerical difference between the mean and the daily fatality figures without considering whether these are positive or negative. Let us denote these absolute differences by 'modulus of d' or 'mod d'.

This is evident from the third column of the table below

| $X$   | $X - \bar{X} = d$ | $ d $ |
|-------|-------------------|-------|
| 4     | 0                 | 0     |
| 6     | 2                 | 2     |
| 2     | -2                | 2     |
| 0     | -4                | 4     |
| 3     | -1                | 1     |
| 5     | 1                 | 1     |
| 8     | 4                 | 4     |
| Total |                   | 14    |

By ignoring the sign of the deviations we have achieved a non-zero sum in our second column. Averaging these absolute differences, we obtain a measure of dispersion known as the mean deviation.

In other words, the mean deviation is given by the formula:

### **MEAN DEVIATION**

$$M.D. = \frac{\sum |d_i|}{n}$$

As we are averaging the absolute deviations of the observations from their mean, therefore the complete name of this measure is mean absolute deviation --- but generally we just say "mean deviation". Applying this formula in our example, we find that, the mean deviation of the number of fatalities is

$$M.D. = \frac{14}{7} = 2.$$

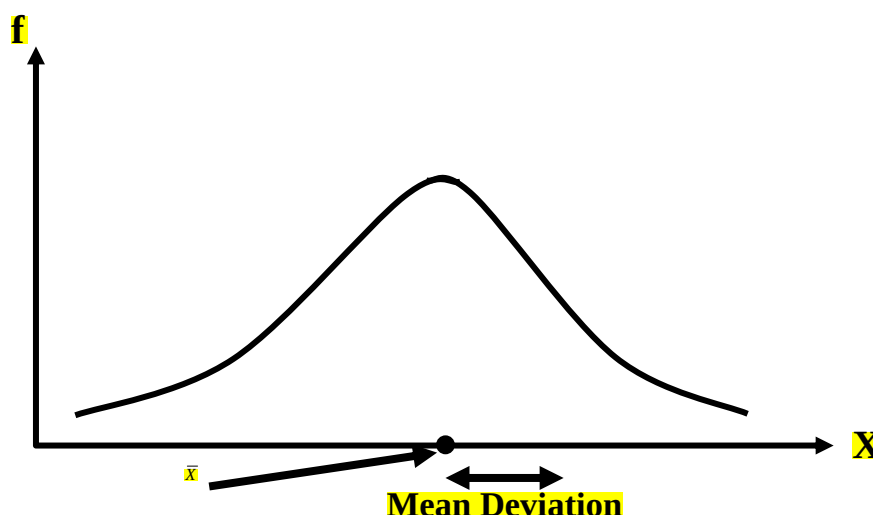
The formula that we have just considered is valid in the case of raw data. In case of grouped data i.e. a frequency distribution, the formula becomes



**MEAN DEVIATION FOR GROUPED DATA**

$$\text{M.D.} = \frac{\sum f_i |x_i - \bar{x}|}{n} = \frac{\sum f_i |d_i|}{n}$$

As far as the graphical representation of the mean deviation is concerned, it can be depicted by a horizontal line segment drawn below the X-axis on the graph of the frequency distribution, as shown below



The approach which we have adopted in the concept of the mean deviation is both quick and simple. But the problem is that we introduce a kind of artificiality in its calculation by ignoring the algebraic signs of the deviations.

In problems involving descriptions and comparisons alone, the mean deviation can validly be applied; but because the negative signs have been discarded, further theoretical development or application of the concept is impossible.

Mean deviation is an absolute measure of dispersion. Its relative measure, known as the co-efficient of mean deviation, is obtained by dividing the mean deviation by the average used in the calculation of deviations i.e. the arithmetic mean. Thus

**CO-EFFICIENT OF M.D**

Sometimes, the mean deviation is computed by averaging the absolute deviations of the data-values from the median i.e.

$$\text{Mean deviation} = \frac{\sum |x - \tilde{x}|}{n}$$

And when will we have a situation when we will be using the median instead of the mean? As discussed earlier, the median will be more appropriate than the mean in those cases where our data-set contains a few very high or very low values. In such a situation, the coefficient of mean deviation is given by:

**Co-efficient of M.D:**

$$= \frac{\text{M.D.}}{\text{Median}} = \frac{\text{M.D.}}{\text{Mean}}$$

Let us now consider the **standard deviation** --- that statistic which is the most important and the most widely used measure of dispersion.

The point that made earlier that from the mathematical point of view, it is not very preferable to take the absolute values of the deviations, *This problem is overcome by computing the standard deviation.*

In order to compute the standard deviation, rather than taking the absolute values of the deviations, we square the deviations.

Averaging these squared deviations, we obtain a statistic that is known as the variance.

VARIANCE

$$= \frac{\sum (x - \bar{x})^2}{n}$$

Let us compute this quantity for the data of the above example.  
Our X-values were:

| X |
|---|
| 4 |
| 6 |
| 2 |
| 0 |
| 3 |
| 5 |
| 8 |

Taking the deviations of the X-values from their mean, and then squaring these deviations, we obtain:

| X | $(x - \bar{x})$ | $(x - \bar{x})^2$ |
|---|-----------------|-------------------|
| 4 | 0               | 0                 |
| 6 | + 2             | 4                 |
| 2 | - 2             | 4                 |
| 0 | - 4             | 16                |
| 3 | - 1             | 1                 |
| 5 | + 1             | 1                 |
| 8 | + 4             | 16                |
|   |                 | 42                |

Obviously, both  $(-2)^2$  and  $(2)^2$  equal 4, both  $(-4)^2$  and  $(4)^2$  equal 16, and both  $(-1)^2$  and  $(1)^2 = 1$ .  
Hence  $\sum (x - \bar{x})^2 = 42$  is now positive, and this positive value has been achieved without ‘bending’ the rules of mathematics. Averaging these squared deviations, the variance is given by:

**Variance:**

$$= \frac{\sum (x - \bar{x})^2}{n}$$

$$= \frac{42}{7} = 6$$

The variance is frequently employed in statistical work, but it should be noted that the figure achieved is in ‘squared’ units of measurement.

In the example that we have just considered, the variance has come out to be “6 squared fatalities”, which does not seem to make much sense! In order to obtain an answer which is in the original unit of measurement, we take the positive square root of the variance. The result is known as the standard deviation.

**STANDARD DEVIATION**

$$S = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

Hence, in this example, our standard deviation has come out to be 2.45 fatalities.

In computing the standard deviation (or variance) it can be tedious to first ascertain the arithmetic mean of a series, then subtract it from each value of the variable in the series, and finally to square each deviation and then sum. It is very much more straight-forward to use the short cut formula given below:

**SHORT CUT FORMULA FOR THE STANDARD DEVIATION**

$$S = \sqrt{\left\{ \frac{\sum x^2}{n} - \left( \frac{\sum x}{n} \right)^2 \right\}}$$

In order to apply the short cut formula, we require only the aggregate of the series ( $\sum x$ ) and the aggregate of the squares of the individual values in the series ( $\sum x^2$ ).

In other words, only two columns of figures are called for. The number of individual calculations is also considerably reduced, as seen below:

| X     | X <sup>2</sup> |
|-------|----------------|
| 4     | 16             |
| 6     | 36             |
| 2     | 4              |
| 0     | 0              |
| 3     | 9              |
| 5     | 25             |
| 8     | 64             |
| Total | 28      154    |

Therefore

$$S = \sqrt{\left\{ \frac{154}{7} - \left( \frac{28}{7} \right)^2 \right\}} = \sqrt{(22 - 16)}$$

$$= \sqrt{6} = 2.45 \text{ fatalities}$$

The formulae that we have just discussed are valid in case of raw data. In case of grouped data i.e. a frequency distribution, each squared deviation round the mean must be multiplied by the appropriate frequency figure i.e.

**STANDARD DEVIATION IN CASE OF GROUPED DATA**

$$S = \sqrt{\frac{\sum f(x - \bar{x})^2}{n}}$$

And the short cut formula in case of a frequency distribution is:

**SHORT CUT FORMULA OF THE STANDARD DEVIATION IN CASE OF GROUPED DATA**

$$S = \sqrt{\left\{ \frac{\sum fx^2}{n} - \left( \frac{\sum fx}{n} \right)^2 \right\}}$$

Which is again preferred from the computational standpoint

For example, the standard deviation life of a batch of electric light bulbs would be calculated as follows:

**EXAMPLE**

| Life (in Hundreds of Hours) | No. of Bulbs f | Mid-point x | fx     | fx <sup>2</sup> |
|-----------------------------|----------------|-------------|--------|-----------------|
| 0 – 5                       | 4              | 2.5         | 10.0   | 25.0            |
| 5 – 10                      | 9              | 7.5         | 67.5   | 506.25          |
| 10 – 20                     | 38             | 15.0        | 570.0  | 8550.0          |
| 20 – 40                     | 33             | 30.0        | 990.0  | 29700.0         |
| 40 and over                 | 16             | 50.0        | 800.0  | 40000.0         |
|                             | 100            |             | 2437.5 | 78781.25        |

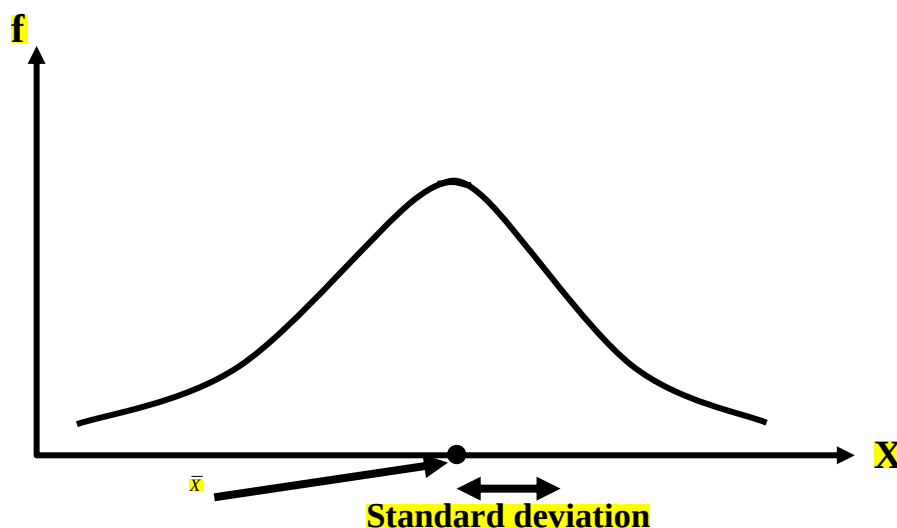
Therefore, standard deviation:

$$S = \sqrt{\left\{ \frac{78781.25}{100} - \left( \frac{2437.5}{100} \right)^2 \right\}}$$

$$= 13.9 \text{ hundred hours}$$

$$= 1390 \text{ hours}$$

As far as the graphical representation of the standard deviation is concerned, a horizontal line segment is drawn below the X-axis on the graph of the frequency distribution --- just as in the case of the mean deviation.



The standard deviation is an absolute measure of dispersion. Its relative measure called coefficient of standard deviation is defined as:

**COEFFICIENT OF S.D**

$$= \frac{\text{Standard Deviation}}{\text{Mean}}$$

And, multiplying this quantity by 100, we obtain a very important and well-known measure called the coefficient of variation.

**COEFFICIENT OF VARIATION**

$$C.V. = \frac{S}{\bar{X}} \times 100$$

As mentioned earlier, the standard deviation is expressed in absolute terms and is given in the same unit of measurement as the variable itself.

There are occasions, however, when this absolute measure of dispersion is inadequate and a relative form becomes preferable. For example, if a comparison between the variability of distributions with different variables is required, or when we need to compare the dispersion of distributions with the same variable but with very different arithmetic means. To illustrate the usefulness of the coefficient of variation, let us consider the following two examples.

**EXAMPLE-1**

Suppose that, in a particular year, the mean weekly earnings of skilled factory workers in one particular country were \$ 19.50 with a standard deviation of \$ 4, while for its neighboring country the figures were Rs. 75 and Rs. 28 respectively.

From these figures, it is not immediately apparent which country has the GREATER VARIABILITY in earnings. The coefficient of variation quickly provides the answer:

**For country No. 1:**

$$\frac{4}{19.5} \times 100 = 20.5 \text{ per cent,}$$

**And for country No. 2:**

$$\frac{28}{75} \times 100 = 37.3 \text{ per cent.}$$

From these calculations, it is immediately obvious that the spread of earnings in country No. 2 is greater than that in country No. 1, and the reasons for this could then be sought.

**EXAMPLE-2:**

The crop yield from 20 acre plots of wheat-land cultivated by ordinary methods averages 35 bushels with a standard deviation of 10 bushels. The yield from similar land treated with a new fertilizer averages 58 bushels, also with a standard deviation of 10 bushels. At first glance, the yield variability may seem to be the same, but in fact it has improved (i.e. decreased) in view of the higher average to which it relates.

Again, the coefficient of variation shows this very clearly:

**Untreated land:**

$$\frac{10}{35} \times 100 = 28.57 \text{ per cent}$$

**Treated land:**

$$\frac{10}{58} \times 100 = 17.24 \text{ per cent}$$

The coefficient of variation for the untreated land has come out to be 28.57 percent, whereas the coefficient of variation for the treated land is only 17.24 percent.

## LECTURE NO. 12

- Chebychev's Inequality
- The Empirical Rule
- The Five-Number Summary

In the last lecture, we discussed the concept of standard deviation in quite a lot of detail.

It is an extremely important concept, and it is very important that we appreciate and understand its role in statistical analysis. We've seen that if we are comparing the variability of two samples selected from a population, the sample with the larger standard deviation is the more variable of the two.

Thus, we know how to interpret the standard deviation on a relative or comparative basis, but we haven't considered how it provides a measure of variability for a single sample.

To understand how the standard deviation provides a measure of variability of a data set, consider a specific data set and answer the following questions:

**Question-1**

'How many measurements are within 1 standard deviation of the mean?'

**Question-2**

'How many measurements are within 2 standard deviations?'

and so on.

For any specific data set, we can answer these questions by counting the number of measurements in each of the intervals. However, if we are interested in obtaining a general answer to these questions the problem is a bit more difficult. We will discuss to you two sets of answers to the questions of how many measurements fall within 1, 2, and 3 standard deviations of the mean. General answer to these questions the problem is a bit more difficult. The first, which applies to any set of data, is derived from a theorem proved by Russian mathematician, P.L. Chebychev (1821-1894). The second, which applies to mound-shaped, symmetric distributions of data, is based upon empirical evidence that has accumulated over the years. And this set of answers is valid and applicable even if our distribution is slightly skewed. Let us begin with the Chebychev's theorem.

Chebychev's Rule applies to any data set, regardless of the shape of the frequency distribution of the data.

**CHEBYCHEV'S THEOREM**

For any number  $k$  greater than 1, at least  $1 - 1/k^2$  of the data-values fall within  $k$  standard deviations of the mean, i.e., within the interval  $(\bar{X} - kS, \bar{X} + kS)$

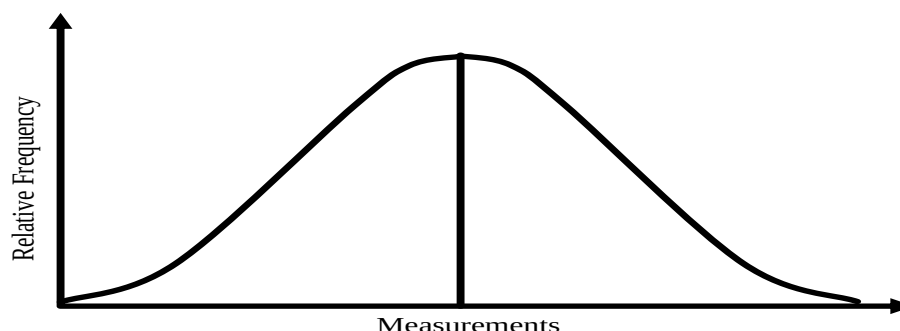
This means that:

- At least  $1 - 1/2^2 = 3/4$  will fall within 2 standard deviations of the mean, i.e. within the interval  $(\bar{X} - 2S, \bar{X} + 2S)$ .
- At least  $1 - 1/3^2 = 8/9$  of the data-values will fall within 3 standard deviations of the mean, i.e. within the interval  $(\bar{X} - 3S, \bar{X} + 3S)$

Because of the fact that Chebychev's theorem requires  $k$  to be greater than 1, therefore no useful information is provided by this theorem on the fraction of measurements that fall within 1 standard deviation of the mean, i.e. within the interval  $(\bar{X} - S, \bar{X} + S)$ .

Next, let us consider the Empirical Rule mentioned above.

This is a rule of thumb that applies to data sets with frequency distributions that are mound-shaped and symmetric, as follows:



According to this empirical rule:

- Approximately 68% of the measurements will fall within 1 standard deviation of the mean, i.e. within the interval  $(\bar{X} - S, \bar{X} + S)$
- Approximately 95% of the measurements will fall within 2 standard deviations of the mean, i.e. within the interval  $(\bar{X} - 2S, \bar{X} + 2S)$ .
- Approximately 100% (practically all) of the measurements will fall within 3 standard deviations of the mean, i.e. within the interval  $(\bar{X} - 3S, \bar{X} + 3S)$ .

Let us understand this point with the help of an example:

### EXAMPLE

The 50 companies' percentages of revenues spent on R&D (i.e. Research and Development) are:

|      |     |      |     |     |      |     |     |      |      |
|------|-----|------|-----|-----|------|-----|-----|------|------|
| 13.5 | 9.5 | 8.2  | 6.5 | 8.4 | 8.1  | 6.9 | 7.5 | 10.5 | 13.5 |
| 7.2  | 7.1 | 9.0  | 9.9 | 8.2 | 13.2 | 9.2 | 6.9 | 9.6  | 7.7  |
| 9.7  | 7.5 | 7.2  | 5.9 | 6.6 | 11.1 | 8.8 | 5.2 | 10.6 | 8.2  |
| 11.3 | 5.6 | 10.1 | 8.0 | 8.5 | 11.7 | 7.1 | 7.7 | 9.4  | 6.0  |
| 8.0  | 7.4 | 10.5 | 7.8 | 7.9 | 6.5  | 6.9 | 6.5 | 6.8  | 9.5  |

Calculate the proportions of these measurements that lie within the intervals  $\bar{X} \pm S$ ,  $\bar{X} \pm 2S$ , and  $\bar{X} \pm 3S$ , and compare the results with the theoretical values. The mean and standard deviation of these data come out to be 8.49 and 1.98, respectively.

Mean:

$$\bar{X} = 8.49$$

Standard deviation:

$$S = 1.98$$

Hence

$$\begin{aligned} &(\bar{X} - S, \bar{X} + S) \\ &= (8.49 - 1.98, 8.49 + 1.98) \\ &= (6.51, 10.47) \end{aligned}$$

A check of the measurement reveals that 34 of the 50 measurements, or 68%, fall between 6.51 and 10.47. Similarly, the interval

$$\begin{aligned} &(\bar{X} - 2S, \bar{X} + 2S) \\ &= (8.49 - 3.96, 8.49 + 3.96) \\ &= (4.53, 12.45) \end{aligned}$$

Contains 47 of the 50 measurements, i.e. 94% of the data-values

$$\begin{aligned} \text{Finally, the 3-standard deviation interval around } \bar{X}, \text{ i.e. } &(\bar{X} - 3S, \bar{X} + 3S) \\ &= (8.49 - 5.94, 8.49 + 5.94) \\ &= (2.55, 14.43) \text{ contains all, or 100\%, of the measurements.} \end{aligned}$$

In spite of the fact that the distribution of these data is skewed to the right, the percentages of data-values falling within 1, 2, and 3 standard deviations of the mean are remarkably close to the theoretical values (68%, 95%, and 100%) given by the Empirical Rule.

The fact of the matter is that, unless the distribution is extremely skewed, the mound-shaped approximations will be reasonably accurate. Of course, no matter what the shape of the distribution, Chebychev's Rule, assures that at least 75% and at least 89% (8/9) of the measurements will lie within 2 and 3 standard deviations of the mean, respectively.

In this example, 94% of the values are lying inside the interval  $\bar{X} \pm 2S$ , and this percentage is greater than 75%.

Similarly, 100% of the values are lying inside the interval  $\bar{X} \pm 3S$ , and this percentage is greater than 89%.

But, before we discuss all these new concepts, let us revise the concept of the Chebychev's Inequality. In the last lecture, we noted that when all the values in a set of data are located near their mean, they exhibit a small amount of variation or dispersion.

And those sets of data in which some values are located far from their mean have a large amount of dispersion. Expressing these relationships in terms of the standard deviation, which measures dispersion, we can say that when the values of a set of data are concentrated near their mean, the standard deviation is small. And when the values of a set of data are scattered widely about the mean, the standard deviation is large. In exactly the same way, if the standard deviation computed from a set of data is large, the values from which it is computed are dispersed widely about their mean. A useful rule that illustrates the relationship between dispersion and standard deviation is given by *Chebychev's theorem*, named after the Russian mathematician P.L. Chebychev (1821-1894). This theorem enables us to calculate for any set of data the minimum proportion of values that can be expected to lie within a specified number of standard deviations of the mean.

The theorem tells us that at least 75% of the values in a set of data can be expected to fall within two standard deviations of the mean, at least 89% (8/9) within three standard deviations of the mean, and at least 94% (15/16) within four standard deviations of the mean.

In general, Chebychev's theorem may be stated as follows:

### **CHEBYCHEV'S THEOREM**

Given a set of  $n$  observations  $x_1, x_2, x_3, \dots, x_n$  on the variable  $X$ , the probability is at least  $(1 - 1/k^2)$  that  $X$  will take on a value within  $k$  standard deviations of the mean of the set of observations (where  $k > 1$ ). Chebychev's theorem is applicable to any set of observations, so we can use it for either samples or populations. Let us now see how we can suppose that a set of data has a mean of 150 and a standard deviation of 25. Putting  $k = 2$  in the Chebychev's theorem, at least  $1 - 1/(2)^2 = 75\%$  of the data-values will take on a value within two standard deviations of the mean.

Apply it in practice.

Since the standard deviation is 25, hence  $2(25) = 50$ , and at least 75% of the data-values will take on a value between  $150 - 50 = 100$  and  $150 + 50 = 200$ . Consequently, we can say that we can expect at least 75% of the values to be between 100 and 200. By similar calculations we find that we can expect at least 89% to be between 75 and 225, and at least 96% to be between 25 and 275.

(The last statement has been made by putting  $k = 5$  in the formula  $1 - 1/k^2$ )

Suppose that another set of data has the same mean as before, i.e. 150, but a standard deviation of 10. Applying Chebychev's theorem, for this set of data we can expect at least 75% of the values to be between 130 and 170, at least 89% to be between 120 and 180, and at least 96% to be between 100 and 200.

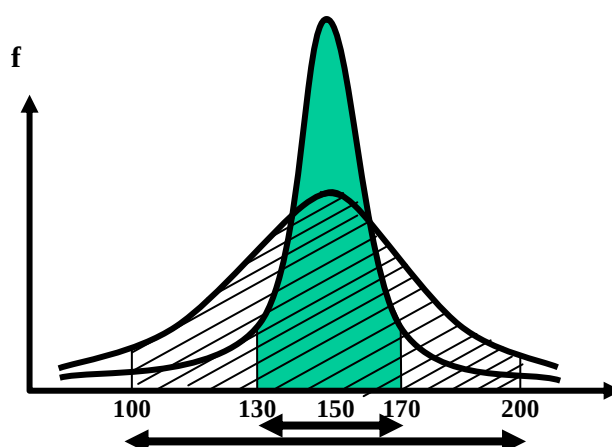
The above results are summarized in the following table:

| PERCENTAGE OF DATA | FOR DATA-SET NO. 1     | FOR DATA-SET NO. 2     |
|--------------------|------------------------|------------------------|
| At least 75 %      | Lies Between 100 & 200 | Lies Between 130 & 170 |
| At least 89 %      | Lies Between 75 & 225  | Lies Between 120 & 180 |
| At least 96 %      | Lies Between 25 & 275  | Lies Between 100 & 200 |

Thus the intervals computed for the latter set of data are all narrower than those for the former.

For two symmetric, hump-shaped distributions having the same mean, this point is depicted in the following diagram:

### **THE SYMMETRIC CURVE**



Therefore, we see that for a set of data with a small standard deviation, a larger proportion of the values will be concentrated near the mean than for a set of data with a large standard deviation.

A limitation of the Chebychev's theorem is that it gives no information at all about the probability of observing a value within one standard deviation of the mean, since  $1 - 1/k^2 = 0$  when  $k = 1$ . Also, it should be noted that the Chebychev's theorem provides weak information for our variable of interest. For many random variables, the probability of observing a value within 2 standard deviations of the mean is far greater than  $1 - 1/2^2 = 0.75$ .

In this way, the Chebychev's theorem and the Empirical Rule play an important role in understanding the nature and importance of the standard deviation as a measure of dispersion.



The next topic of today's lecture is the five-number summary. (Now that we have studied the three major properties of numerical data (i.e. **central tendency, variation, and shape**), it is important that we identify and describe the major features of the data in a summarized format.)

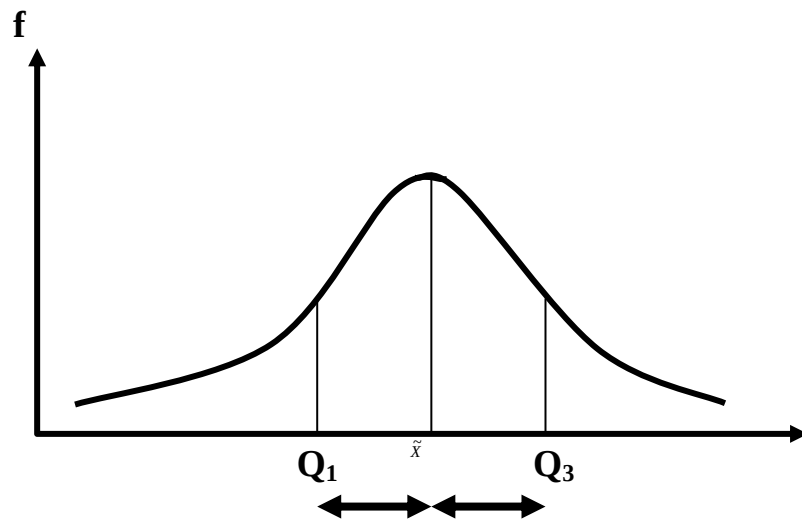
One approach to this "exploratory data analysis" is to develop a five-number summary.

### FIVE-NUMBER SUMMARY

A five-number summary consists of  $X_0, Q_1$ , Median,  $Q_3$ , and  $X_m$ ; It provides us quite a good idea about the shape of the distribution. If the data were perfectly symmetrical, the following would be true:

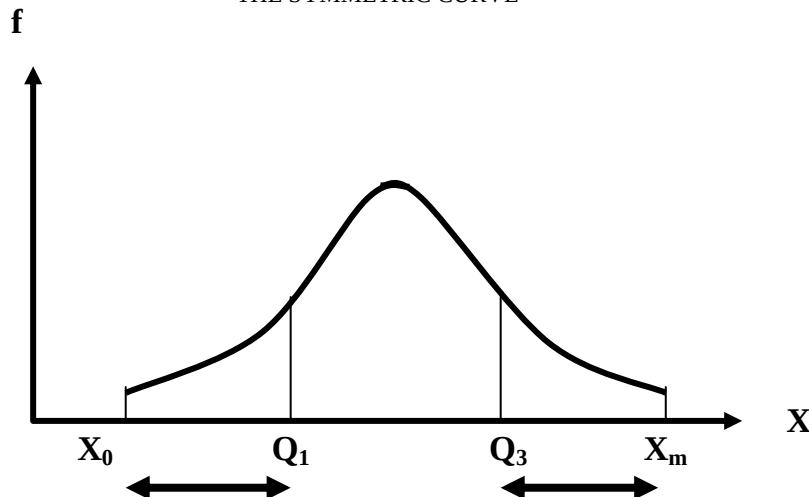
1. The distance from  $Q_1$  to the median would be equal to the distance from the median to  $Q_3$ :

THE SYMMETRIC CURVE



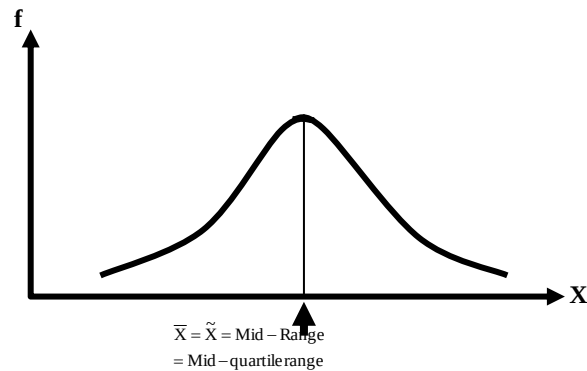
3. The distance from  $X_0$  to  $Q_1$  would be equal to the distance from  $Q_3$  to  $X_m$ .

THE SYMMETRIC CURVE



3. The median, the mid-quartile range, and the midrange would all be equal. All these measures would also be equal to the arithmetic mean of the data:

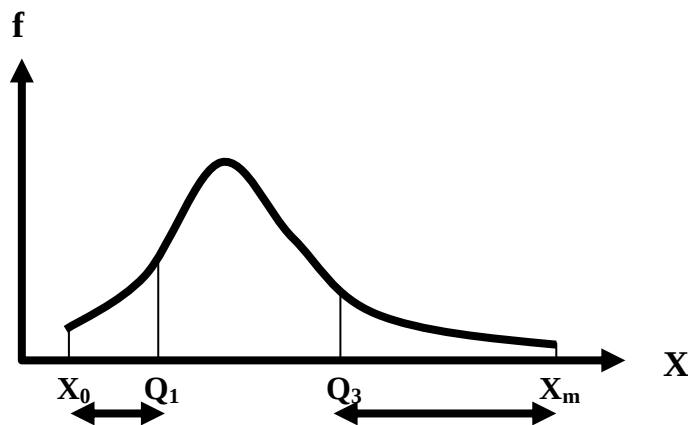
THE SYMMETRIC CURVE



On the other hand, for non-symmetrical distributions, the following would be true:

1. In right-skewed distributions the distance from  $Q_3$  to  $X_m$  greatly exceeds the distance from  $X_0$  to  $Q_1$ .

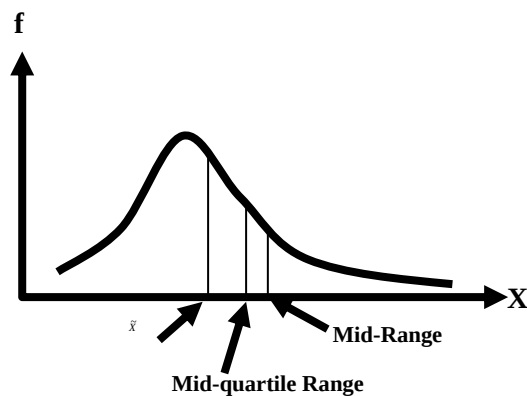
THE POSITIVELY SKEWED CURVE



2. In right-skewed distributions,

median < mid-quartile range < midrange:

THE POSITIVELY SKEWED CURVE



Similarly, in left-skewed distributions, the distance from  $X_0$  to  $Q_1$  greatly exceeds the distance from  $Q_3$  to  $X_m$ . Also, in left-skewed distributions, midrange < mid-quartile range < median. Let us try to understand this concept with the help of an example

### EXAMPLE

Suppose that a study is being conducted regarding the annual costs incurred by students attending public versus private colleges and universities in the United States of America. In particular, suppose, for exploratory purposes, our sample consists of 10 Universities whose athletic programs are members of the 'Big Ten' Conference. The annual costs incurred for tuition fees, room, and board at 10 schools belonging to Big Ten Conference are given as follows:

| Name of University            | Annual Costs<br>(in \$000) |
|-------------------------------|----------------------------|
| Indiana University            | 15.6                       |
| Michigan State University     | 17.0                       |
| Ohio State University         | 15.2                       |
| Pennsylvania State University | 16.4                       |
| Purdue University             | 15.2                       |
| University of Illinois        | 15.4                       |
| University of Iowa            | 13.0                       |
| University of Michigan        | 23.1                       |
| University of Minnesota       | 14.3                       |
| University of Wisconsin       | 14.9                       |

If we wish to state the five-number summary for these data, the first step will be to arrange our data-set in ascending order:

Ordered Array:

|              |      |      |      |      |      |      |      |      |              |
|--------------|------|------|------|------|------|------|------|------|--------------|
| $X_0 = 13.0$ | 14.3 | 14.9 | 15.2 | 15.2 | 15.4 | 15.6 | 16.4 | 17.0 | $X_m = 23.1$ |
|--------------|------|------|------|------|------|------|------|------|--------------|

And if we carry out the relevant computations, we find that:

- The median for this data comes out to be 15.30 thousand dollars.
- The first quartile comes out to be 14.90 thousand dollars, and
- The third quartile comes out to be 16.40 thousand dollars.

Therefore, the five-number summary for this data-set is:

**The Five-Number Summary:**

|       |       |             |       |       |
|-------|-------|-------------|-------|-------|
| $X_0$ | $Q_1$ | $\tilde{X}$ | $Q_3$ | $X_m$ |
| 13.0  | 14.9  | 15.3        | 16.4  | 23.1  |

If we apply the rules that I am conveyed to you a short while ago, it is clear that the annual cost data for our sample are right-skewed. We come to this conclusion because of two reasons:

- The distance from  $Q_3$  to  $X_m$  (i.e., 6.7) greatly exceeds the distance from  $X_0$  to  $Q_1$  (i.e., 1.9).
- If we compare the median (which is 15.3), the mid-quartile range (which is 15.65), and the midrange (which is 18.05), we observe that the median < the mid-quartile range < the midrange.

Both these points clearly indicate that our distribution is positively skewed.

The gist of the above discussion is that the five-number summary is a simple yet effective way of determining the shape of our frequency distribution --- without actually drawing the graph of the frequency distribution.

## LECTURE NO. 13

- Box and Whisker Plot
- Pearson's Coefficient of Skewness

Prior to discussing the THE BOX-AND-WHISKER PLOT, let us review the concept of THE FIVE-NUMBER SUMMARY. As indicated in the last lecture, once we have studied the three major properties of numerical data (i.e. central tendency, variation, and shape), it is important that we identify and describe the major features of the data in a SUMMARIZED format. One way of doing this is to develop a five-number summary.

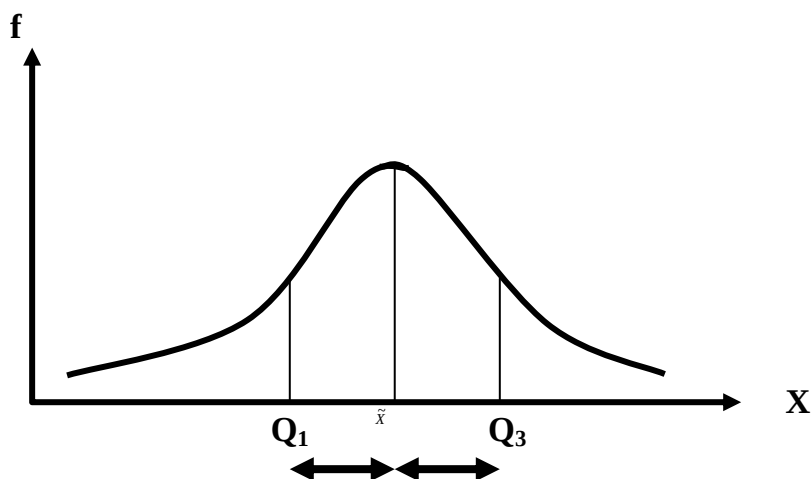
FIVE-NUMBER SUMMARY

A five-number summary consists of  $X_0, Q_1$ , Median,  $Q_3$ ;  $X_m$ . It provides us a better idea as to the SHAPE of the distribution, as explained below:

If the data were perfectly symmetrical, the following would be true:

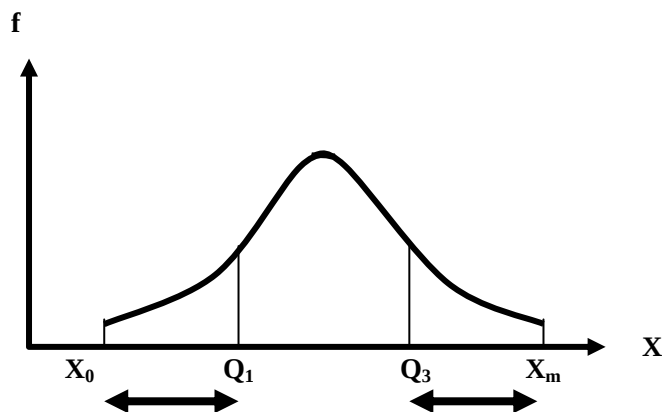
1. The distance from  $Q_1$  to the median would be equal to the distance from the median to  $Q_3$ , as shown below:

THE SYMMETRIC CURVE

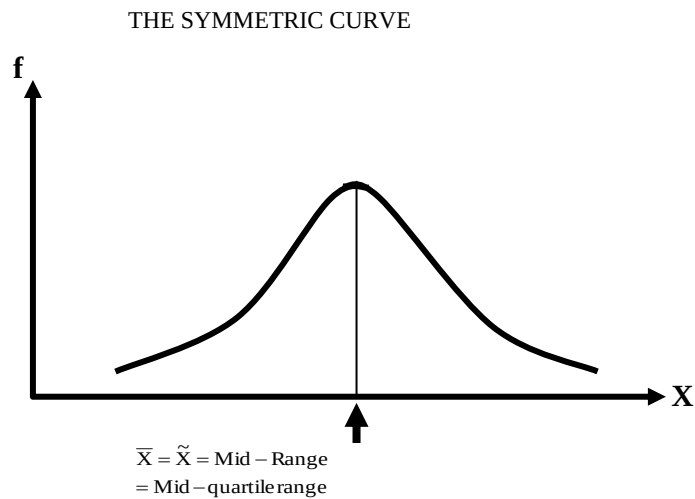


2. The distance from  $X_0$  to  $Q_1$  would be equal to the distance from  $Q_3$  to  $X_m$ , as shown below:

THE SYMMETRIC CURVE

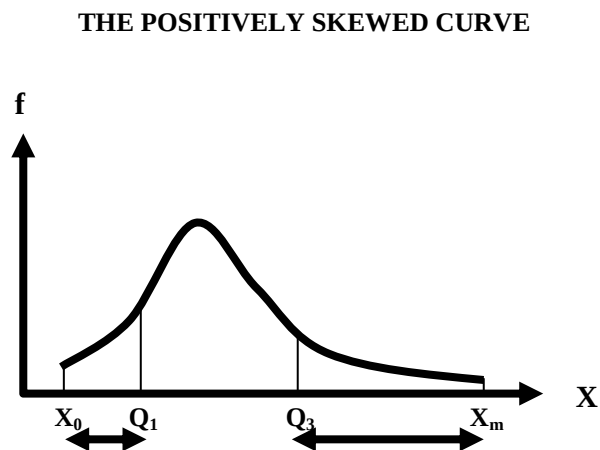


3. The median, the mid-quartile range, and the midrange would ALL be equal. These measures would also be equal to the arithmetic mean of the data, as shown below:



On the other hand, for non-symmetrical distributions, the following would be true:

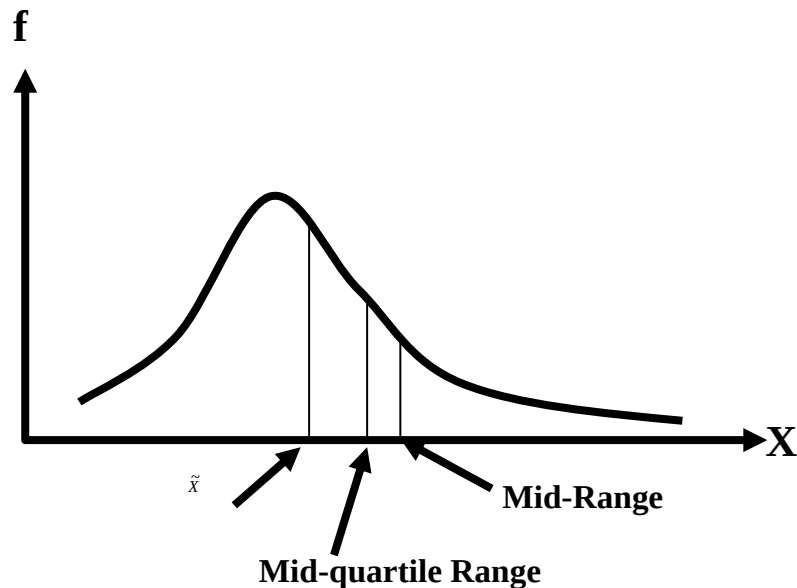
1. In right-skewed (positively-skewed) distributions the distance from  $Q_3$  to  $X_m$  greatly EXCEEDS the distance from  $X_0$  to  $Q_1$ , as shown below:



2. In right-skewed distributions,  
 median < mid-quartile range < midrange

This is indicated in the following figure:

### THE POSITIVELY SKEWED CURVE



Similarly, in left-skewed distributions, the distance from  $X_0$  to  $Q_1$  greatly exceeds the distance from  $Q_3$  to  $X_m$ . Also, in left-skewed distributions,  $\text{midrange} < \text{mid-quartile range} < \text{median}$ .

Let us try to understand this concept with the help of an example:

#### EXAMPLE

Suppose that a study is being conducted regarding the annual costs incurred by students attending public versus private colleges and universities in the United States of America. In particular, suppose, for exploratory purposes, our sample consists of 10 Universities whose athletic programs are members of the 'Big Ten' Conference? The annual costs incurred for tuition fees, room, and board at 10 schools belonging to Big Ten Conference are given in the following table; state the five-number summary for these data.

Annual Costs Incurred on Tuition Fees, etc.

| Name of University            | Annual Costs<br>(in \$000) |
|-------------------------------|----------------------------|
| Indiana University            | 15.6                       |
| Michigan State University     | 17.0                       |
| Ohio State University         | 15.2                       |
| Pennsylvania State University | 16.4                       |
| Purdue University             | 15.2                       |
| University of Illinois        | 15.4                       |
| University of Iowa            | 13.0                       |
| University of Michigan        | 23.1                       |
| University of Minnesota       | 14.3                       |
| University of Wisconsin       | 14.9                       |

#### SOLUTION:

For our sample, the ordered array is

|              |      |      |      |      |      |      |      |      |              |
|--------------|------|------|------|------|------|------|------|------|--------------|
| $X_0 = 13.0$ | 14.3 | 14.9 | 15.2 | 15.2 | 15.4 | 15.6 | 16.4 | 17.0 | $X_m = 23.1$ |
|--------------|------|------|------|------|------|------|------|------|--------------|

The median for this data comes out to be 15.30 thousand dollars. The first quartile comes out to be 14.90 thousand dollars, and the third quartile comes out to be 16.40 thousand dollars.

Therefore, the five-number summary is:

| $X_0$ | $Q_1$ | $\tilde{X}$ | $Q_3$ | $X_m$ |
|-------|-------|-------------|-------|-------|
| 13.0  | 14.9  | 15.3        | 16.4  | 23.1  |

We may now use the five-number summary to study the *shape* of this distribution:

We notice that

1. The distance from  $Q_3$  to  $X_m$  (i.e., 6.7) greatly exceeds the distance from  $X_0$  to  $Q_1$  (i.e., 1.9).

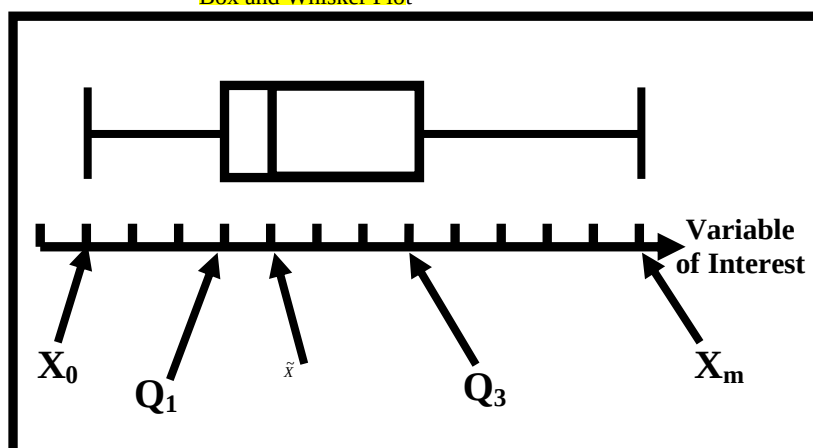
2. If we compare the median (which is 15.3), the mid-quartile range (which is 15.65), and the midrange (which is 18.05), we observe that the median < the mid-quartile range < the midrange.

Hence, from the preceding rules, it is clear that the annual cost data for our sample are *right-skewed*. The gist of the above discussion is that the five-number summary is a SIMPLE yet effective way of determining the shape of our frequency distribution --- WITHOUT actually drawing the graph of the frequency distribution. The concept of the five number summary is directly linked with the concept of the box and whisker plot:

**BOX AND WHISKER PLOT** It is used to display the data and the spread of the data

In its simplest form, a box-and-whisker plot provides a graphical representation of the data through its five-number summary.

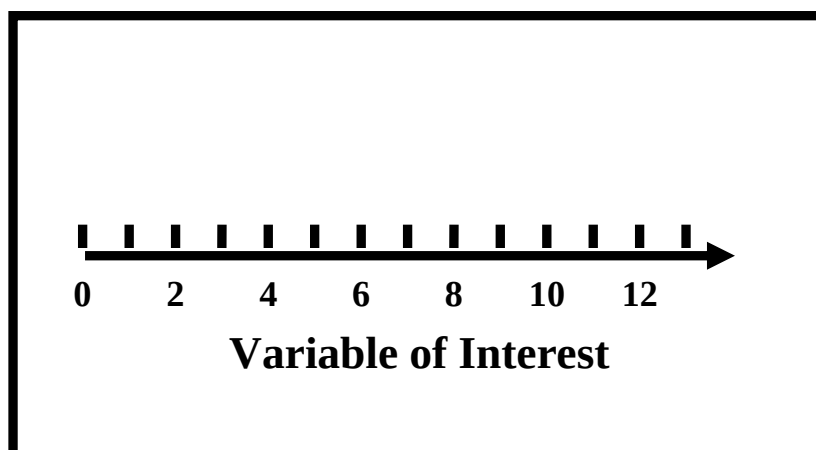
Box and Whisker Plot



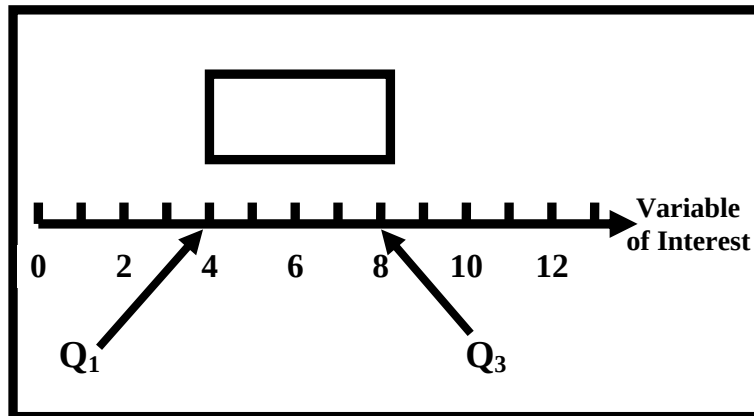
To construct a box-and-whisker plot, we proceed as follows:

Steps involved in the construction of the Box and Whisker Plot:

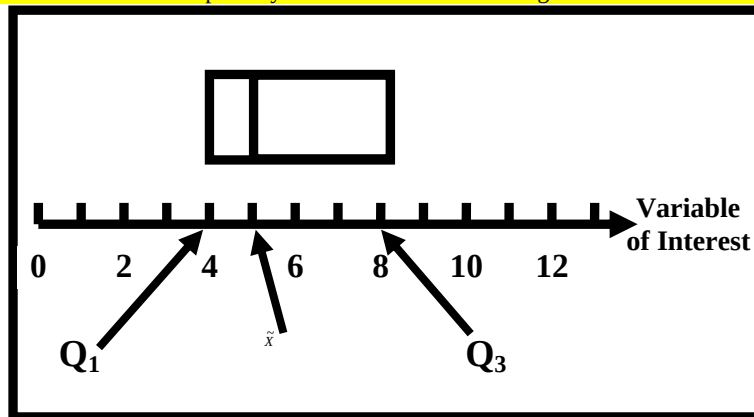
1. The variable of interest is represented on the horizontal axis.



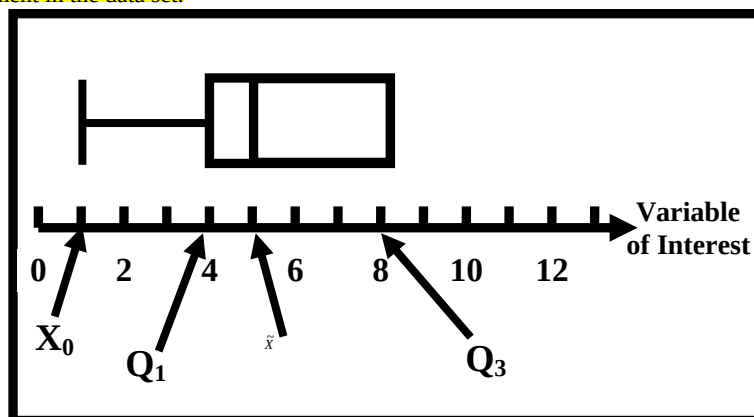
2. A BOX is drawn in the space above the horizontal axis in such a way that the left end of the box aligns with the first quartile  $Q_1$  and the right end of the box is aligned with the third quartile  $Q_3$ .



3. The box is divided into two parts by a VERTICAL line that aligns with the MEDIAN.

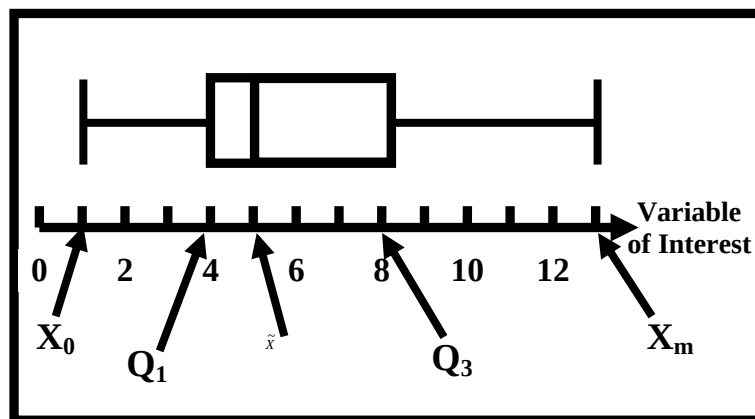


4. A line, called a whisker, is extended from the LEFT end of the box to a point that aligns with  $X_0$ , the smallest measurement in the data set.





5. Another line, or whisker, is extended from the RIGHT end of the box to a point that aligns with the LARGEST measurement in the data set.



Let us understand the construction of the box-and-whisker plot with reference to an example:

### EXAMPLE

The following table shows the downtime, in hours, recorded for 30 machines owned by a large manufacturing company. The period of time covered was the same for all machines.

DOWNTIME IN HOURS OF 30 MACHINES

|   |    |    |    |    |   |
|---|----|----|----|----|---|
| 4 | 4  | 1  | 4  | 1  | 4 |
| 6 | 10 | 5  | 5  | 8  | 2 |
| 1 | 6  | 10 | 1  | 13 | 5 |
| 8 | 4  | 3  | 9  | 4  | 9 |
| 1 | 4  | 4  | 11 | 8  | 9 |

In order to construct a box-and-whisker plot for these data, we proceed as follows:

First of all, we determine the two extreme values in our data-set:

The smallest and largest values are  $X_0 = 1$  and  $X_m = 13$ , respectively.

As far as the computation of the quartiles is concerned, we note that, in this example, we are dealing with raw data.

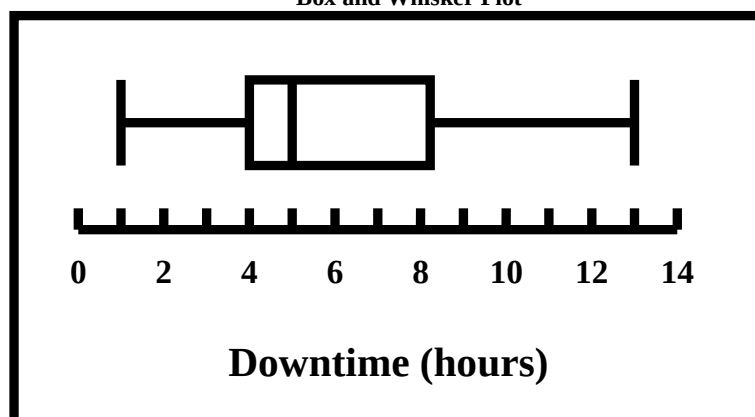
The first quartile is the  $(30 + 1)/4 = 7.75^{\text{th}}$  ordered measurement and is equal to 4.

The median is the  $(30 + 1)/2 = 15.5^{\text{th}}$  measurement, or 5, and

The third quartile is the  $3(30 + 1)/4 = 23.25^{\text{th}}$  ordered measurement, which is 8.25.

As a result, we obtain the following box and whisker plot:

Box and Whisker Plot



**INTERPRETATION OF THE BOX AND WHISKER PLOT**

With regard to the *interpretation* of the Box and Whisker Plot, it should be noted that, by looking at a box-and-whisker plot, one can quickly form an impression regarding the amount of SPREAD, location of CONCENTRATION, and SYMMETRY of our data set.

A glance at the box and whisker plot of the example that we just considered reveals that:

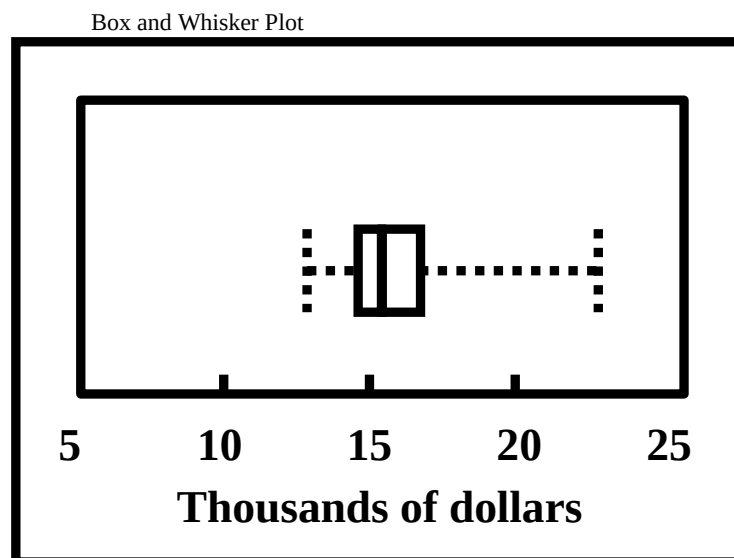
- 50% of the measurements are between 4 and 8.25.
- The median is 5, and the range is 12. and, most importantly:
- Since the median line is closer to the *left* end of the box, hence the data are SKEWED to the RIGHT. (The fundamental point is that in a perfectly symmetrical data set, the median line will be EXACTLY HALFWAY between the two ends of the box, and in a data set that is skewed to the LEFT, the median line will be CLOSER TO THE RIGHT END of the box.)

Let us consolidate all the above ideas by going back to the example of the Big Ten Universities in which the annual costs incurred for tuition fees, room, and board at 10 schools belonging to Big Ten Conference were given as follows:

| Name of University            | Annual Costs<br>(in \$000) |
|-------------------------------|----------------------------|
| Indiana University            | 15.6                       |
| Michigan State University     | 17.0                       |
| Ohio State University         | 15.2                       |
| Pennsylvania State University | 16.4                       |
| Purdue University             | 15.2                       |
| University of Illinois        | 15.4                       |
| University of Iowa            | 13.0                       |
| University of Michigan        | 23.1                       |
| University of Minnesota       | 14.3                       |
| University of Wisconsin       | 14.9                       |

As stated earlier, the Five-Number Summary of this data-set is :

For this data, the Box and Whisker Plot is of the form given below:



As indicated earlier, the vertical line drawn within the box represents the location of the median value in the data; the vertical line at the LEFT side of the box represents the location of Q1, and the vertical line at the RIGHT side of the box represents the location of Q3. Therefore, the BOX contains the middle 50% of the observations in the distribution. The lower 25% of the data are represented by the whisker that connects the *left* side of the box to the location of the *smallest* value,  $X_0$ , and the upper 25% of the data are represented by the whisker connecting the *right* side of the box to  $X_m$ .

**INTERPRETATION OF THE BOX AND WHISKER PLOT**

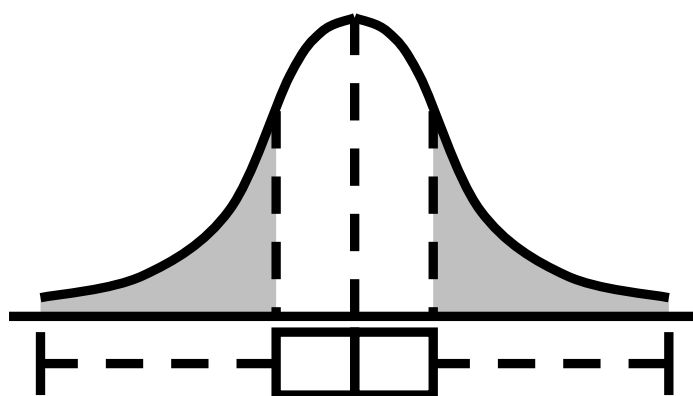
We note that (1) the vertical median line is CLOSER to the left side of the box, and (2) the left side whisker length is clearly SMALLER than the right side whisker length. Because of these observations, we The gist of the above discussion is that if the median line is at a greater distance from the left side of the box as compared with its distance from the right side of the box, our distribution will be skewed to the left.

In this situation, the whisker appearing on the left side of the box and whisker plot will be longer than the whisker of the right side. Conclude that the data-set of the annual costs is RIGHT-skewed.

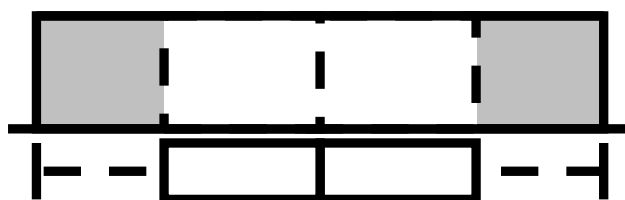
The gist of the above discussion is that if the median line is at a greater distance from the left side of the box as compared with its distance from the right side of the box, our distribution will be skewed to the left. In this situation, the whisker appearing on the left side of the box and whisker plot will be longer than the whisker of the right side. The Box and Whisker Plot comes under the realm of “exploratory data analysis” (EDA) which is a relatively new area of statistics. The following figures provide a comparison between the Box and Whisker Plot and the traditional procedures such as the frequency polygon and the frequency curve with reference to the SKEWNESS present in the data-set.

Four different types of hypothetical distributions are depicted through their box-and-whisker plots and corresponding frequency curves.

1) When a data set is perfectly symmetrical, as is the case in the following two figures, the mean, median, midrange, and mid-quartile range will be the SAME:



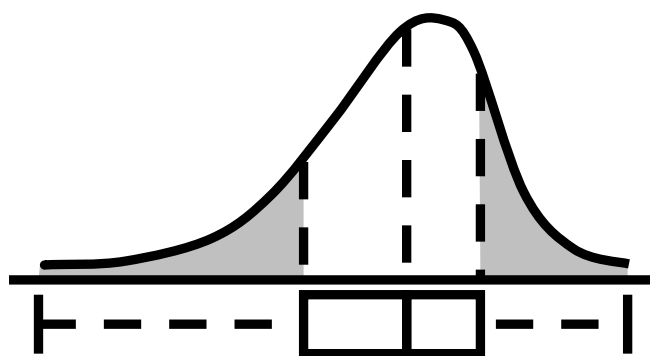
(a) Bell-shaped distribution



(b) Rectangular distribution

In ADDITION, the length of the left whisker will be equal to the length of the right whisker, and the median line will divide the box in HALF. (In *practice*, it is unlikely that we will observe a data set that is perfectly symmetrical. However, we should be able to state that our data set is *approximately* symmetrical *if* the lengths of the two whiskers are *almost* equal and the *median* line *almost* divides the box in HALF.)

2) When our data set is LEFT-skewed as in the following figure, the few small observations pull the midrange and mean toward the LEFT tail:

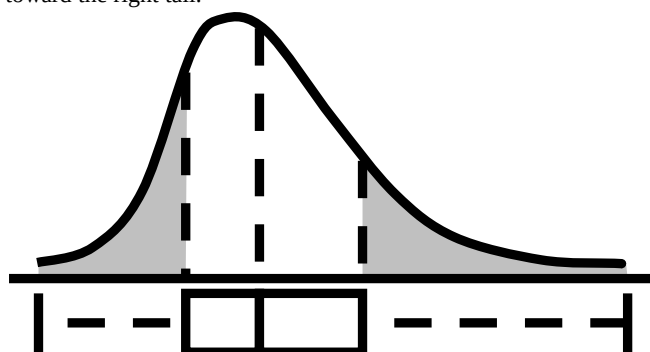


Left-skewed distribution

For this LEFT-skewed distribution, we observe that the skewed nature of the data set indicates that there is a HEAVY CLUSTERING of observations at the HIGH END of the scale (i.e., the RIGHT side).

75% of all data values are found between the left edge of the box (Q1) and the end of the right whisker ( $X_m$ ). Therefore, the LONG left whisker contains the distribution of only the smallest 25% of the observations, demonstrating the distortion from symmetry in this data set.

3) If the data set is RIGHT-skewed as shown in the following figure, the few large observations PULL the midrange and mean toward the right tail.



Right-skewed distribution

For the right-skewed data set, the concentration of data points is on the LOW end of the scale (i.e., the left side of the box-and-whisker plot). Here, 75% of all data values are found between the beginning of the left whisker ( $X_0$ ) and the RIGHT edge of the box (Q3), and the remaining 25% of the observations are DISPERSED ALONG the LONG right whisker at the upper end of the scale. This brings us to the end of the discussion of the five number summary and the box and whisker plot.

Next, we discuss *another* way of determining the skewness of the data-set and that is the **PEARSON'S**

#### **COEFFICIENT OF SKEWNESS**

In this connection, the first thing to note is that, by providing information about the location of a series and the dispersion within that series it might appear that we have achieved a PERFECTLY adequate overall description of the data. But, the fact of the matter is that, it is quite possible that two series are decidedly dissimilar and yet have exactly the same arithmetic mean AND standard deviation: Let us understand this point with the help of an example:

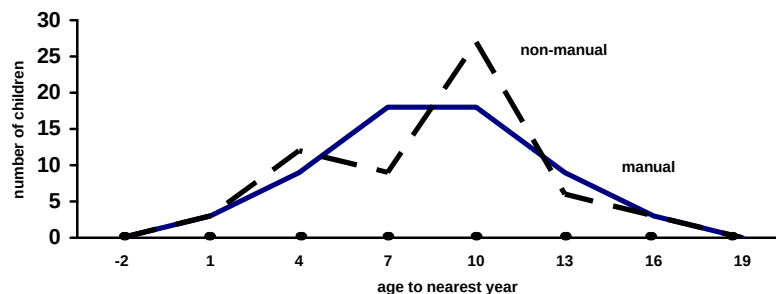
#### **EXAMPLE:**

| Age of Onset of Nervous Asthma in Children (to Nearest Year) | Children of Manual Workers | Children of Non-Manual Workers |
|--------------------------------------------------------------|----------------------------|--------------------------------|
| 0 – 2                                                        | 3                          | 3                              |
| 3 – 5                                                        | 9                          | 12                             |
| 6 – 8                                                        | 18                         | 9                              |
| 9 – 11                                                       | 18                         | 27                             |
| 12 – 14                                                      | 9                          | 6                              |
| 15 – 17                                                      | 3                          | 3                              |
|                                                              | 60                         | 60                             |

In order to compute the mean and standard deviation for each distribution, we carry out the following calculations:

| Age of Onset of Nervous Asthma in Children (to Nearest Year) |    | Children of Manual Workers |                  |                               | Children of Non-Manual Workers |                  |                               |
|--------------------------------------------------------------|----|----------------------------|------------------|-------------------------------|--------------------------------|------------------|-------------------------------|
| Age Group                                                    | X  | f <sub>1</sub>             | f <sub>1</sub> X | f <sub>1</sub> X <sup>2</sup> | f <sub>2</sub>                 | f <sub>2</sub> X | f <sub>2</sub> X <sup>2</sup> |
| 0 – 2                                                        | 1  | 3                          | 3                | 3                             | 3                              | 3                | 3                             |
| 3 – 5                                                        | 4  | 9                          | 36               | 144                           | 12                             | 48               | 192                           |
| 6 – 8                                                        | 7  | 18                         | 126              | 882                           | 9                              | 63               | 441                           |
| 9 – 11                                                       | 10 | 18                         | 180              | 1800                          | 27                             | 270              | 2700                          |
| 12 – 14                                                      | 13 | 9                          | 117              | 1521                          | 6                              | 78               | 1014                          |
| 15 – 17                                                      | 16 | 3                          | 48               | 768                           | 3                              | 48               | 768                           |
|                                                              | 51 | 60                         | 510              | 5118                          | 60                             | 510              | 5118                          |

We find that, for each of the two distributions, the mean is 8.5 years and the standard deviation is 3.61 years. The frequency polygons of the two distributions are as follows:



By inspecting these, it can be seen that one distribution is symmetrical while the other is quite different. The distinguishing feature here is the degree of asymmetry or *SKEWNESS* in the two polygons. In order to *measure* the skewness in our distribution, we compute the **PEARSON'S COEFFICIENT OF SKEWNESS** which is defined as: Pearson's Coefficient of Skewness:

$$\frac{\text{mean} - \text{mode}}{\text{standard deviation}}$$

Applying the *empirical relation* between the mean, median and the mode, the Pearson's Coefficient of Skewness is given by:

**Pearson's Coefficient of Skewness**

$$= \frac{3(\text{mean} - \text{median})}{\text{standard deviation}}$$

For a *symmetrical* distribution the coefficient will always be ZERO, for a distribution skewed to the RIGHT the answer will always be positive, and for one skewed to the LEFT the answer will always be negative.

Let us now calculate this coefficient for the example of the children of the manual and non-manual workers. Sample statistics pertaining to the ages of these children are as follows:

|                    | Children of Manual Workers | Children of Non-Manual Workers |
|--------------------|----------------------------|--------------------------------|
| Mean               | 8.50 years                 | 8.50 years                     |
| Standard deviation | 3.61 years                 | 3.61 years                     |
| Median             | 8.50 years                 | 9.16 years                     |
| Q <sub>1</sub>     | 6.00 years                 | 5.50 years                     |
| Q <sub>3</sub>     | 11.00 years                | 10.83 years                    |
| Quartile deviation | 2.50 years                 | 2.66 years                     |

The Pearson's Coefficient of Skewness is calculated for each of the two categories of children, as shown below:  
Pearson's Coefficient of Skewness (Modified):

$$\frac{\text{Ages of Children of Manual Workers}}{3(8.50 - 8.50)} = 0$$

$$\frac{\text{Ages of Children of Non-Manual Workers}}{3(8.50 - 9.16)} = -0.55$$

For the data pertaining to children of manual workers, the coefficient is zero, whereas, for the children of non-manual workers, the coefficient has turned out to be a negative number. This indicates that the distribution of the ages of the children of **the manual workers is symmetric** whereas the distribution of the ages of the children of the non-manual workers is **negatively skewed**.

The students are encouraged to draw the frequency polygon and the frequency curve for each of the two distributions, and to compare the results that have just been obtained with the *shapes* of the two distributions.

## LECTURE NO. 14

- Bowley's coefficient of skewness
- The Concept of Kurtosis
- Percentile Coefficient of Kurtosis
- Moments & Moment Ratios
- Sheppard's Corrections
- The Role of Moments in Describing Frequency Distributions

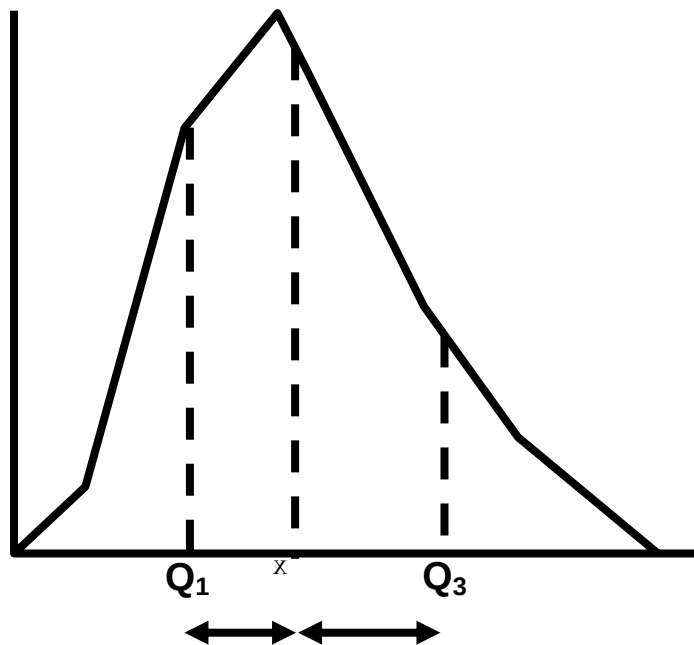
You will recall that the Pearson's coefficient of skewness is defined as (mean - mode)/standard deviation, and if we apply the empirical relation between the mean, median and the mode, then the coefficient is given by:

**PEARSON'S COEFFICIENT OF SKEWNESS:**

$$= \frac{3(\text{mean} - \text{median})}{\text{standard deviation}}$$

As you can see, this coefficient involves the calculation of the mean as well as the standard deviation. Actually, the numerator is divided by the standard deviation in order to obtain a pure number. If the analysis of a data-set is being undertaken using the median and quartiles alone, then we use a measure called Bowley's coefficient of skewness.

The advantage of this particular formula is that it requires NO KNOWLEDGE of the MEAN or STANDARD DEVIATION. In an asymmetrical distribution, the quartiles will NOT be equidistant from the median, and the AMOUNT by which each one deviates will give an indication of skewness. Where the distribution is positively skewed, Q1 will be closer to the median than Q3. In other words, the distance between Q3 and the median will be greater than the distance between the median and Q1.

**POSITIVE SKEWNESS**

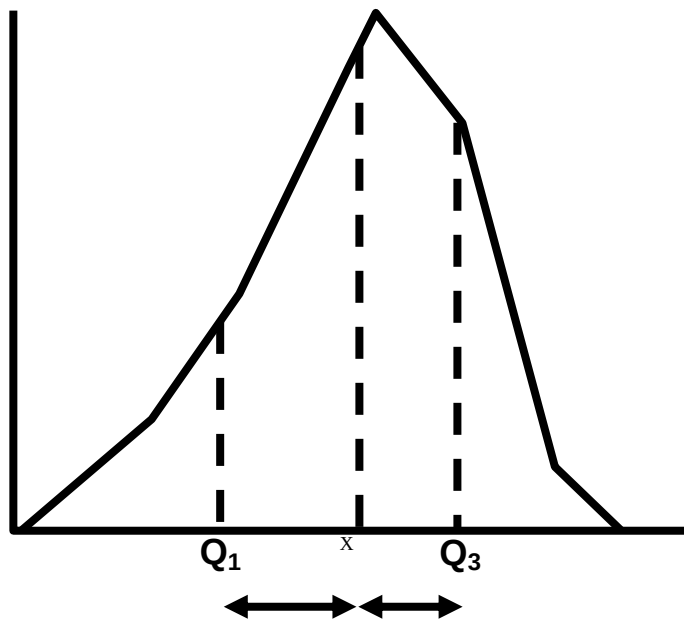
And hence, if we subtract the distance median - Q1 from the distance Q3 - median, we will obtain a positive answer.

In case of a positively skewed distribution:

$$(Q3 - \text{median}) - (\text{Median} - Q1) > 0$$

$$\text{i.e. } Q1 + Q3 - 2 \text{ median} > 0$$

The opposite is true for skewness to the left

**NEGATIVE SKEWNESS**

In this case:

$$(Q_3 - \text{median}) - (\text{Median} - Q_1) < 0 \quad \text{i.e.}$$

$$Q_1 + Q_3 - 2 \text{ median} < 0$$

The gist of the above discussion is that in case of a positively skewed distribution, the quantity

$$Q_1 + Q_3 - 2\tilde{X}$$

will be positive, whereas in case of a negatively distribution, this quantity will be negative.

**A RELATIVE** measure of skewness is obtained by dividing

$$Q_1 + Q_3 - 2\tilde{X}$$

by the inter-quartile range i.e.  $Q_3 - Q_1$ , so that **Bowley's coefficient of skewness is given by:**

Bowley's coefficient of Skewness

$$= \frac{(Q_1 + Q_3 - 2\tilde{X})}{Q_3 - Q_1}$$

**It is a pure (unit less) number, and its value lies between 0 and  $\pm 1$ .**

For a **positively skewed distribution, this coefficient will turn out to be positive**, and for a negatively skewed distribution this coefficient will come out **to be negative**. Let us apply this concept to the example regarding the ages of children of the manual and non-manual workers that we considered in the last lecture.

| Age of Onset of Nervous Asthma in Children (to Nearest Year) | Children of Manual Workers | Children of Non-Manual Workers |
|--------------------------------------------------------------|----------------------------|--------------------------------|
| 0 – 2                                                        | 3                          | 3                              |
| 3 – 5                                                        | 9                          | 12                             |
| 6 – 8                                                        | 18                         | 9                              |
| 9 – 11                                                       | 18                         | 27                             |
| 12 – 14                                                      | 9                          | 6                              |
| 15 – 17                                                      | 3                          | 3                              |
|                                                              | 60                         | 60                             |



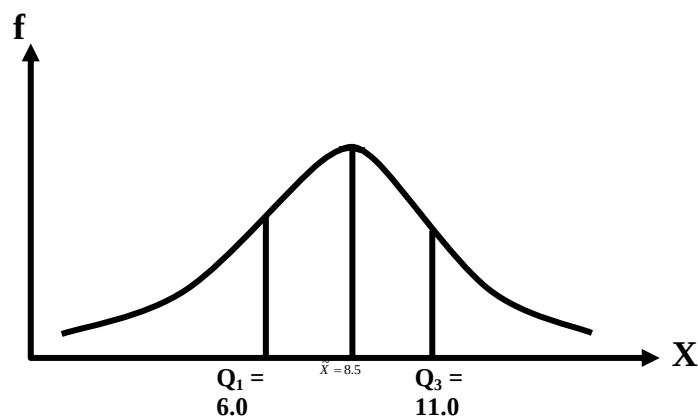
**EXAMPLE:**

Sample statistics pertaining to ages of children of manual and non-manual workers:

|                    | Children of Manual Workers | Children of Non-Manual Workers |
|--------------------|----------------------------|--------------------------------|
| Mean               | 8.50 years                 | 8.50 years                     |
| Standard deviation | 3.61 years                 | 3.61 years                     |
| Median             | 8.50 years                 | 9.16 years                     |
| $Q_1$              | 6.00 years                 | 5.50 years                     |
| $Q_3$              | 11.00 years                | 10.83 years                    |
| Quartile deviation | 2.50 years                 | 2.66 years                     |

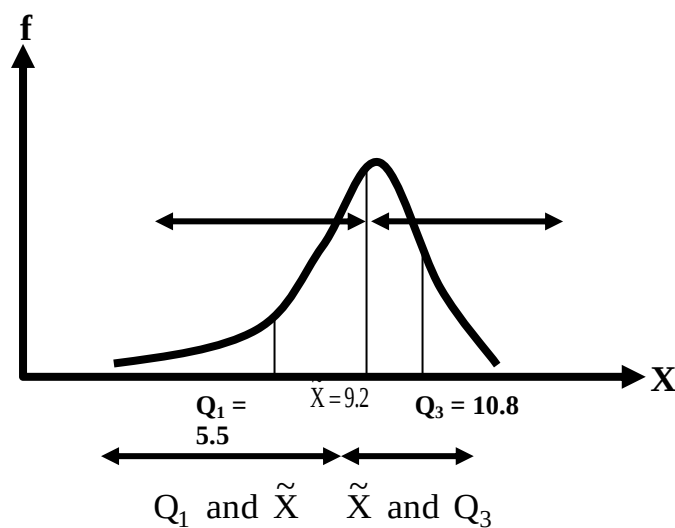
The statistics pertaining to children of manual workers yield the following PICTURE:

**Ages of Children of Manual Workers**



On the other hand, the statistics pertaining to children of non-manual workers yield the following PICTURE:

**Ages of Children of Non-Manual Workers**



The diagram pertaining to children of non-manual workers clearly shows that the distance between

is much greater than the distance between which happens whenever we are dealing with a negatively skewed distribution. If we compute the Bowley's coefficient of skewness for each of these two data-sets, we obtain:

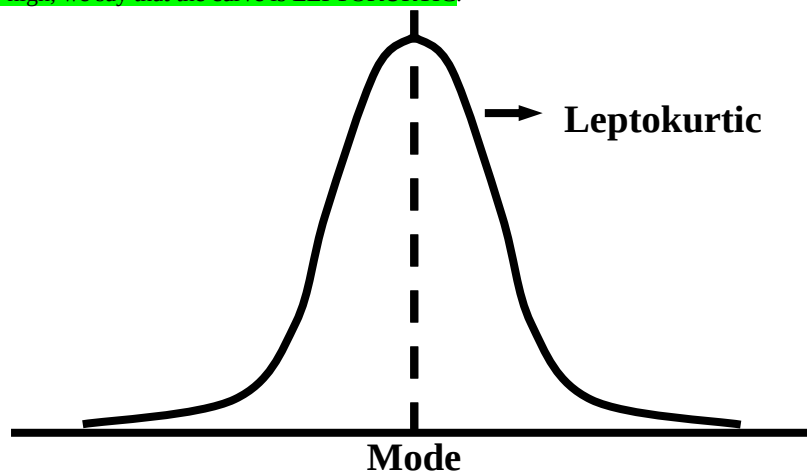
#### Bowley's Coefficient of Skewness

| <b>Ages of Children<br/>of Manual Workers</b> | <b>Ages of Children<br/>of Non-Manual Workers</b>   |
|-----------------------------------------------|-----------------------------------------------------|
| $= \frac{11.00 + 6.00 - 2 \times 8.50}{2.50}$ | $\frac{10.83 + 5.50 - 2 \times 9.16}{10.83 - 5.50}$ |
| $= 0$                                         | $= -0.37$                                           |

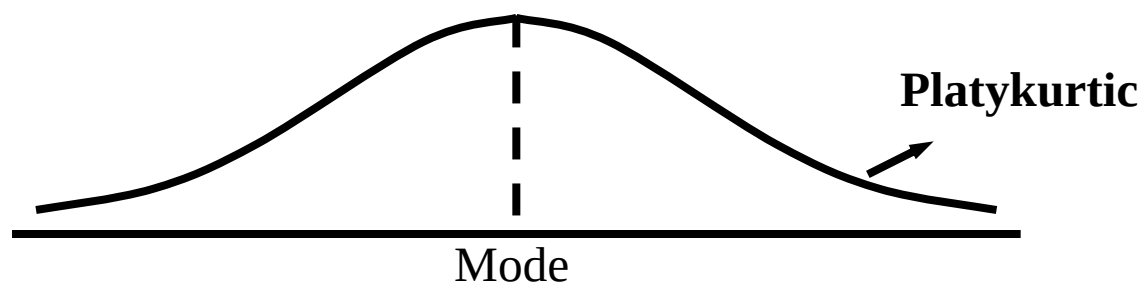
As you have noticed, for the children of the manual workers, the Bowley's coefficient has come out to be zero, whereas for the children of the non-manual workers, the coefficient has come out to be negative. This indicates that the distribution of the ages of the children of manual workers is symmetrical whereas the distribution of the ages of the children of the non-manual workers IS negatively skewed --- EXACTLY the same conclusion that we obtained when we computed the Pearson's coefficient of skewness.

#### KURTOSIS

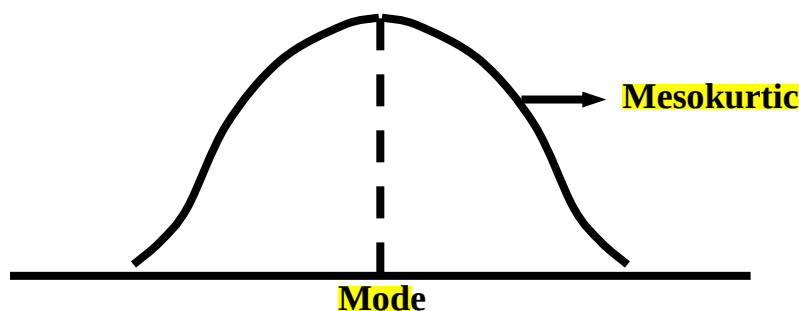
The term kurtosis was introduced by Karl Pearson. This word literally means 'the amount of hump', and is used to represent the degree of PEAKEDNESS or flatness of a unimodal frequency curve. When the values of a variable are closely BUNCHED round the mode in such a way that the peak of the curve becomes relatively high, we say that the curve is LEPTOKURTIC.



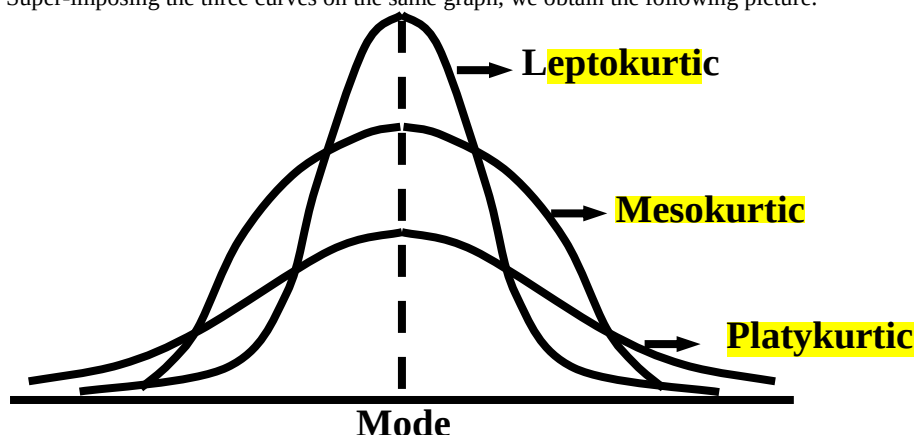
On the other hand, if the curve is flat-topped, we say that the curve is PLATYKURTIC:



The NORMAL curve is a curve which is neither very peaked nor very flat, and hence it is taken as A BASIS FOR COMPARISON. The normal curve itself is called MESOKURTIC. I will discuss with you the normal in detail when we discuss continuous probability distributions. At the moment, just think of the symmetric hump shaped curve shown below:



Super-imposing the three curves on the same graph, we obtain the following picture:



The tallest one is called leptokurtic, the intermediate one is called mesokurtic, and the flat one is called platykurtic. The question arises, “How will we MEASURE the degree of peakedness or kurtosis of a data-set?” A MEASURE of kurtosis based on quartiles and percentiles is

$$K = \frac{Q.D.}{P_{90} - P_{10}},$$

This is known as the **PERCENTILE COEFFICIENT OF KURTOSIS**.

It has been shown that K for a normal distribution is 0.263 and that it lies between 0 and 0.50.

In case of a leptokurtic distribution, the percentile coefficient of kurtosis comes out to be LESS THAN 0.263, and in the case of a platykurtic distribution, the percentile coefficient of kurtosis comes out to be GREATER THAN 0.263. The next concept that I am going to discuss with you is the concept of moments --- a MATHEMATICAL concept, and a very important concept in statistics.

### **MOMENTS**

A moment designates the power to which deviations are raised before averaging them. For example, the quantity

$$\frac{1}{n} \sum (x_i - \bar{x})^1 = \frac{1}{n} \sum (x_i - \bar{x}) \quad \frac{\sum (x_i - \bar{x})}{n}$$

is called the first sample moment about the mean, and is denoted by  $m_1$ .

Similarly, the quantity

$$\frac{1}{n} \sum (x_i - \bar{x})^2 \quad \frac{\sum (x_i - \bar{x})^2}{n}$$

is called the **second sample moment about the mean**, and is denoted by  $m_2$ . In general, the  $r$ th moment about the mean is: **the arithmetic mean of the  $r$ th power of the deviations of the observations from the mean**. In symbols, this means that

$$m_r = \frac{1}{n} \sum (x_i - \bar{x})^r \quad \text{for sample data.}$$

**Moments about the mean are also called the central moments or the mean moments.**

In a similar way, moments about an **arbitrary origin**, say  $\alpha$ , are defined by the relation

$$m'_r = \frac{1}{n} \sum (x_i - \alpha)^r \quad \text{for sample data}$$

For  $r = 1$ , we have

$$m_1 = \frac{1}{n} \sum (x_i - \bar{x}) = \frac{\sum x_i}{n} - \bar{x} = \bar{x} - \bar{x} = 0,$$

and

$$m'_1 = \frac{1}{n} \sum (x_i - \alpha) = \frac{\sum x_i}{n} - \alpha = \bar{x} - \alpha.$$

Putting  $r = 2$  in the relation for mean moments, we see that

$$m_2 = \frac{1}{n} \sum (x_i - \bar{x})^2$$

which is exactly the same as the sample variance.

If we take the positive square root of this quantity, we obtain the standard deviation.

In the formula,

$$m'_r = \frac{1}{n} \sum (x_i - \alpha)^r$$

if we put  **$\alpha = 0$ , we obtain**

$$m'_r = \frac{1}{n} \sum x_i^r$$

**and this is called the  $r$ th moment about zero, or the  $r$ th moment about the origin.**

Let us now consolidate the idea of moments by considering an example.

### EXAMPLE

Calculate the first four moments about the mean for the following set of examination marks: 45, 32, 37, 46, 39, 36, 41, 48 & 36.

For convenience, the observed values are written in an increasing sequence. The necessary calculations appear in the table below:

| $x_i$ | $x_i - \bar{x}$ | $(x_i - \bar{x})^2$ | $(x_i - \bar{x})^3$ | $(x_i - \bar{x})^4$ |
|-------|-----------------|---------------------|---------------------|---------------------|
| 32    | -8              | 64                  | -512                | 4096                |
| 36    | -4              | 16                  | -64                 | 256                 |
| 36    | -4              | 16                  | -64                 | 256                 |
| 37    | -3              | 9                   | -27                 | 81                  |
| 39    | -1              | 1                   | -1                  | 1                   |
| 41    | 1               | 1                   | 1                   | 1                   |
| 45    | 5               | 25                  | 125                 | 625                 |
| 46    | 6               | 36                  | 216                 | 1296                |
| 48    | 8               | 64                  | 512                 | 4096                |
| 360   | 0               | 232                 | 186                 | 10708               |

Now

$$\bar{x} = \frac{\sum x_i}{n} = \frac{360}{9} = 40 \quad \text{marks.}$$

Therefore

$$m_1 = \frac{\sum (x_i - \bar{x})}{n} = 0$$

$$m_2 = \frac{\sum (x_i - \bar{x})^2}{n} = \frac{232}{9} = 25.78 \text{ (marks)}^2$$

$$m_3 = \frac{\sum (x_i - \bar{x})^3}{n} = \frac{186}{9} = 20.67 \text{ (marks)}^3$$

$$m_4 = \frac{\sum (x_i - \bar{x})^4}{n} = \frac{10708}{9} = 1189.78 \text{ (marks)}^4$$

All the formulae that I have discussed until now pertain to the case of raw data. How will we compute the various moments in the case of grouped data?

### **MOMENTS IN THE CASE OF GROUPED DATA**

When the sample data are grouped into a frequency distribution having  $k$  classes with midpoints  $x_1, x_2, \dots, x_k$  and the corresponding frequencies  $f_1, f_2, \dots, f_k$ , ( $\sum f_i = n$ ), the  $r$ th sample moments are given by

$$m_r = \frac{1}{n} \sum f_i (x_i - \bar{x})^r, \text{ and}$$

$$m'_r = \frac{1}{n} \sum f_i (x_i - \alpha)^r.$$

In the calculation of moments from a grouped frequency distribution, an error is introduced by the assumption that the frequencies associated with a class are located at the MIDPOINT of the class interval. You remember the concept of grouping error that I discussed with you in an earlier lecture? Our moments therefore need corrections.

These corrections were introduced by W.F. Sheppard, and hence they are known as **SHEPPARD'S CORRECTIONS**: Sheppard's Corrections for Grouping Error:

It has been shown by W.F. Sheppard that, if the frequency distribution (i) is continuous and (ii) tails off to zero at each end, the corrected moments are as given below:

$$m_2 \text{ (corrected)} = m_2 \text{ (uncorrected)} - \frac{h^2}{12};$$

$$m_3 \text{ (corrected)} = m_3 \text{ (uncorrected)};$$

$$m_4 \text{ (corrected)} = m_4 \text{ (uncorrected)} - \frac{h^2}{2} \cdot m_2 \text{ (uncorrected)} + \frac{7}{240} \cdot h^4;$$

where  $h$  denotes the uniform class-interval

The important point to note here is that these corrections are NOT applicable to highly skewed distributions and distributions having unequal class-intervals. I am now going to discuss with you certain mathematical **RELATIONSHIPS that exist between the moments about the mean and the moments about an arbitrary origin.**

The reason for doing so is that, in many situations, it is easier to calculate the moments in the first instance, about an arbitrary origin. They are then transformed to the mean-moments using the relationships that I am now going to convey to you.

The equations are:

$$m_1 = 0$$

$$m_2 = m'_2 - (m'_1)^2;$$

$$m_3 = m'_3 - 3 m'_2 m'_1 + 2 (m'_1)^3, \text{ and}$$

$$m_4 = m'_4 - 4 m'_3 m'_1 + 6 m'_2 (m'_1)^2 - 3 (m'_1)^4$$

In this course, I will not be discussing the mathematical derivation of these relationships. You are welcome to study the mathematics behind these formulae if you are interested. (The derivation is available in your own text book.) But I would like to give you two tips for remembering these formulae:

- In each of these relations, the sum of the coefficients of various terms on the right hand side equals zero and
- Each term on the right is of the same dimension as the term on the left.

Let us now apply these concepts to an example:

### EXAMPLE

Compute the first four moments for the following distribution of marks after applying Sheppard's corrections:

|                 |   |   |   |    |    |    |    |    |    |    |    |
|-----------------|---|---|---|----|----|----|----|----|----|----|----|
| Marks out of 20 | 5 | 6 | 7 | 8  | 9  | 10 | 11 | 12 | 13 | 14 | 15 |
| No. of Students | 1 | 2 | 5 | 10 | 20 | 51 | 22 | 11 | 5  | 3  | 1  |

If we wish to compute the first four moments about the mean by the direct method, first of all, we will have to compute mean itself. The mean of this particular data-set comes out to be 10.06.

But, 10.06 is not a very convenient number to work with!

This is so because when we construct the columns of  $X - \bar{X}$ ,  $(X - \bar{X})^2$  etc.,

we will have a lot many decimals. An alternative way of computing the moments is to take a convenient number as the arbitrary origin and to compute the moments about this number. Later, we utilize the relationships between the moments about the mean and the moments about the arbitrary origin in order to find the moments about the mean.

In this example, we may select 10 as the arbitrary origin, which is the X-value corresponding to the highest frequency 51, and construct the column of D which is the same as  $X - 10$ . Next, we compute the columns of  $fD$ ,  $fD^2$ ,  $fD^3$ , and so on.

| Earnings in Rs. ( $x_i$ ) | No. of Men $f_i$ | $D_i$<br>( $x_i - 10$ ) | $f_i D_i$        | $f_i D_i^2$      | $f_i D_i^3$      | $f_i D_i^4$       |
|---------------------------|------------------|-------------------------|------------------|------------------|------------------|-------------------|
| 5                         | 1                | -5                      | -5               | 25               | -125             | 625               |
| 6                         | 2                | -4                      | -8               | 32               | -128             | 512               |
| 7                         | 5                | -3                      | -15              | 45               | -135             | 405               |
| 8                         | 10               | -2                      | -20              | 40               | -80              | 160               |
| 9                         | 20               | -1                      | -20              | 20               | -20              | 20                |
| 10                        | 51               | 0                       | 0                | 0                | 0                | 0                 |
| 11                        | 22               | 1                       | 22               | 22               | 22               | 22                |
| 12                        | 11               | 2                       | 22               | 44               | 88               | 176               |
| 13                        | 5                | 3                       | 15               | 45               | 135              | 405               |
| 14                        | 3                | 4                       | 12               | 48               | 192              | 768               |
| 15                        | 1                | 5                       | 5                | 25               | 125              | 625               |
| Sum                       | 131              | ..                      | 8                | 346              | 74               | 3718              |
| Sum ÷ n                   | 1                | ..                      | 0.06<br>= $m'_1$ | 2.64<br>= $m'_2$ | 0.56<br>= $m'_3$ | 28.38<br>= $m'_4$ |

Moments about the mean are:

$$m_1 = 0$$

$$m_2 = m'_2 - (m'_1)^2 = 2.64 - (0.06)^2 = 2.64$$

$$\begin{aligned} m_3 &= m'_3 - 3m'_2 m'_1 + 2(m'_1)^3 \\ &= 0.56 - 3(2.64)(0.06) + 2(0.06)^3 \\ &= 0.08 \end{aligned}$$

$$\begin{aligned} m_4 &= m'_4 - 4m'_3 m'_1 + 6m'_2 (m'_1)^2 - 3(m'_1)^4 \\ &= 28.38 - 4(0.56)(0.06) + 6(2.64)(0.06)^2 - 3(0.06)^4 \end{aligned}$$

= 28.30

Applying Sheppard's corrections, we have

$$m_2 (\text{corrected}) = m_2 (\text{uncorrected}) - = 2.64 - 0.08 = 2.56,$$

$$m_3 (\text{corrected}) = m_3 (\text{uncorrected}) = 0.08,$$

$$m_4 (\text{corrected}) = m_4 (\text{uncorrected}) - \frac{3}{2} m_2 (\text{uncorrected}) +$$

$$= 28.30 - 1.32 + 0.03 = 27.01$$

I have discussed with you in quite a lot of detail the concept of moments.

The question arises, "Why is it that we are going through all these lengthy calculations? What is the significance of computing moments?" "You will obtain the answer to this question when I discuss with you the concept of moment ratios. There are certain ratios in which both the numerators and the denominators are moments. The most common of these moment-ratios are denoted by  $b_1$  and  $b_2$ , and defined by the relations:

### **MOMENT RATIOS:**

$$b_1 = \frac{(m_3)^2}{(m_2)^3} \text{ and } b_2 = \frac{m_4}{(m_2)^2}$$

(in the case of sample data)

They are independent of origin and units of measurement, i.e. they are pure numbers.

$b_1$  is used to measure the skewness of our distribution, and  $b_2$  is used to measure the kurtosis of the distribution.

### **INTERPRETATION OF $b_1$**

For symmetrical distributions,  $b_1$  is equal to zero. Hence, for any data-set,  $b_1$  comes out to be zero, we can conclude that our distribution is symmetric. It should be noted that the measure which will indicate the direction of skewness is the third moment round the mean.

If our distribution is positively skewed,  $m_3$  will be positive, and if our distribution is negatively skewed,  $m_3$  will be negative.  $b_1$  will turn out to be positive in both situations because it is given by

$$b_1 = \frac{(m_3)^2}{(m_2)^3}$$

(Since  $m_3$  is being squared,  $b_1$  will be positive regardless of the sign of  $m_3$ .)

### **INTERPRETATION OF $b_2$**

For the normal distribution,  $b_2 = 3$ .

For a leptokurtic distribution,  $b_2 > 3$ , and for a platykurtic distribution,  $b_2 < 3$

You have noted that the third and fourth moments about the mean provide information about the skewness and the kurtosis of our data-set. This is so because  $m_3$  occurs in the numerator of  $b_1$  and  $m_4$  occurs in the numerator of  $b_2$ .

What about the dispersion and the centre of our data-set? Do you not remember that the second moment about the mean is exactly the same thing as the variance, the positive square root of which is the standard deviation --- the most important measure of dispersion? What about the centre of the distribution? You will be interested to note that the first moment about zero is NONE OTHER than the arithmetic mean!

This is so because  $\frac{1}{n} \sum (x_i - 0)^1$  is equal to  $\frac{1}{n} \sum x_i$

--- none other than the arithmetic mean! In this way, the first four moments play a KEY role in describing frequency distributions.

## LECTURE NO. 15

On numerous occasions, our interest lies not in just one single variable but in two, three, four or more variables. For example, if we talk about the yield of a crop, we realize that the yield of any crop depends on a variety of factors --- the fertility of the soil, the type of fertilizer used, the amount of rainfall, and so on.

- Simple Linear Regression
- Standard Error of Estimate
- Correlation

Let me begin the discussion of the bivariate situation by picking up an example.

**EXAMPLE:**

An important concern for any pharmaceutical company producing drugs is to determine how a particular drug will affect one's perception or general awareness. Suppose one such company wants to establish a relationship between the PERCENTAGE of a drug in the blood-stream and the LENGTH OF TIME it takes to respond to a stimulus.

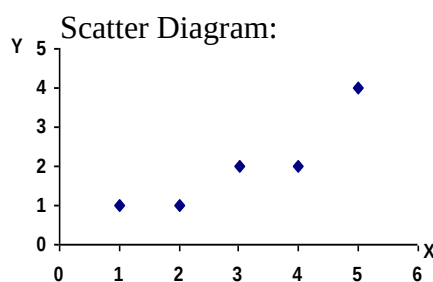
Suppose the company administers this drug on 5 subjects and obtains the following information:

| Subject | Percentage of drug | Reaction Time (milli-seconds) |
|---------|--------------------|-------------------------------|
|         | X                  | Y                             |
| A       | 1                  | 1                             |
| B       | 2                  | 1                             |
| C       | 3                  | 2                             |
| D       | 4                  | 2                             |
| E       | 5                  | 4                             |

In this example, the reaction time to the stimulus will **DEPEND** on the amount of drug in the blood-stream. As you must know, the dependent variable is denoted by Y, and the independent variable is denoted by X. In this example, the reaction time will be denoted by Y, and the percentage of drug in the blood stream by X. Going back to the example that we were just considering, it is obvious that we are interested in determining the nature of the relationship between the amount of drug in the blood stream and the time it takes to react to a stimulus.

In order to ascertain the nature of the relationship between these two variables, the first step is to draw a **SCATTER DIAGRAM** --- which is a simple graph of the X-values against the Y-values depicted on the graph paper in the form of points.

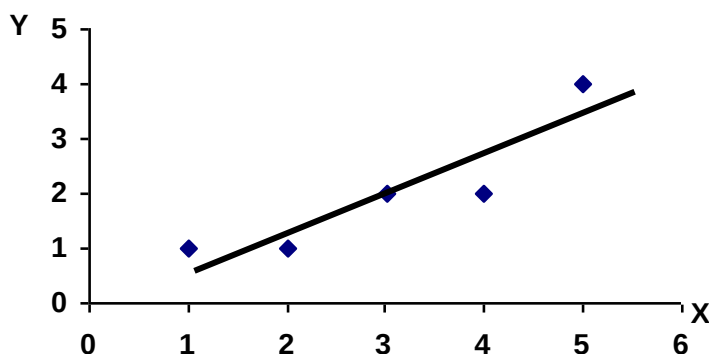
In this example, the scatter diagram is as follows:



As you can see, there is an upward trend in the scatter diagram i.e. it is clear that as X increases, Y also increases. Of course, the points are not all falling on a straight line, but if we look carefully, we find an overall linear pattern as shown below:



## Scatter Diagram:



It will be very RARE in the field of behavioral or social sciences to find two sets of data which are related perfectly by a straight line: it is more likely that only a general linear pattern or tendency will be apparent.

WHY is it that we will not get an exact linear relationship?

Let me explain this to you with the help of an example:

Suppose one is studying the relationship between the research and development expenditure and the profit margin on products of a number of firms. While it may be generally true to state that the two will increase together, it is INEVITABLE that some firms' profit margin will be higher than others with the SAME R and D expenditure, and vice versa. The reasons for this may be that the conditions under which the various firms are operating may be very different. The goods being produced, the firm's share of the market, the efficiency of the firm etc. will ALL play a part in determining the individual results.

A linear relationship between two variables is a SURPRISINGLY common occurrence, and even where a refined non-linear curve might prove slightly superior, the SIMPLER form will often be quite adequate in the context of the problem under consideration. Having plotted the n pairs of values in the form of a scatter diagram, IF an overall linear pattern emerges, then the object of regression is to superimpose on this pattern the general relationship between y and x in the linear form which will REMOVE the effect of outside factors. I am sure that you are aware of the equation of a straight line.

Do you not remember the equation  $Y = mX + c$ , where Y represents the slope of the line, and c represent the Y-intercept?

This equation can also be stated as  $Y = c + mX$ , and if we rename c and m as 'a' and 'b', the equation becomes  $Y = a + bX$ .

### EQUATION OF A STRAIGHT LINE

$$Y = a + bX$$

Where

- Y represents the dependent variable
- X represents the independent variable
- a represents the Y-intercept

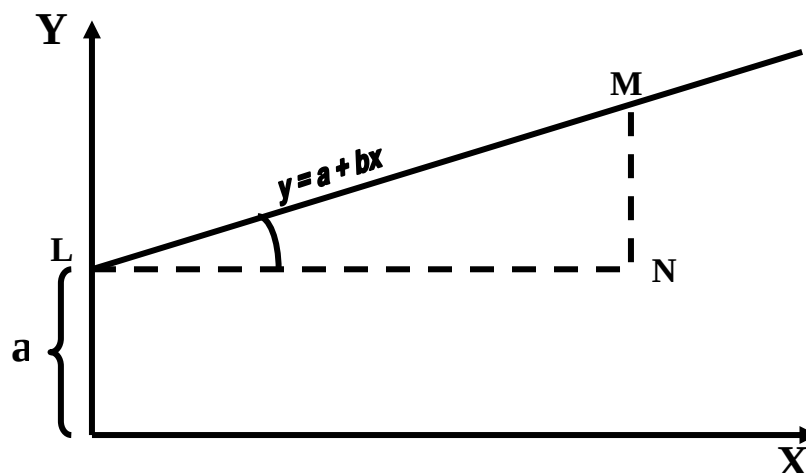
(i.e. the value of Y when X is equal to zero)

- b represents the slope of the line

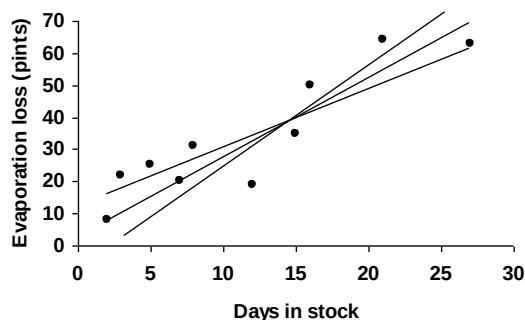
(i.e. the value of the  $\tan \theta$ , where  $\theta$  represents the angle between the line and the horizontal axis)

**Interpretation of ‘a’ and ‘b’:**

$$b = \tan \theta = \frac{MN}{NL}$$



A very important point to note is that MANY lines can be drawn through the same scatter diagram.

**THE LINEAR PATTERN:**

Even with the greatest care and skill, a line drawn between the points with a ruler will be highly SUBJECTIVE, and different individuals will arrive at different lines.

The *real* objective is to find the line of BEST fit. For this, we use a method known as THE METHOD OF LEAST SQUARES. The line of best fit obtained by the method of least squares is called the REGRESSION LINE of Y on X.

And, this whole process is known as simple linear regression. A very important point to note here is that, from the MATHEMATICAL standpoint, simple linear regression requires that X is a NON-RANDOM variable, whereas Y is a RANDOM variable. For example, consider the case of agricultural experiments. If we conduct an experiment to determine the optimal amount of a particular fertilizer to obtain the maximum yield of a certain crop, then the amount of fertilizer is a non-random variable whereas the yield is a random variable. This is so because the amount of fertilizer is in our OWN control. But, the yield is a random variable because it is NOT in our control. In connection with determining the line of BEST fit, the first point is that.

If we use the ‘FREE-hand’ method of curve-fitting in order to represent the relationship between X and Y as portrayed by the scatter diagram, one tends, consciously or subconsciously, to draw the straight line such that there is EQUAL numbers of points located on either side of the line.

What is more, the eye will automatically try to judge and EQUATE the total distances between the points above and below the line.

This is a recognition of the fact that the line of best fit must be an ‘AVERAGE’ line in the true sense.

You will recall that the sum of the deviations round the ARITHMETIC MEAN of a data-set is always equal to zero i.e. the positive and negative deviations CANCEL each other.

Similarly, POSITIVE and NEGATIVE deviations round a line of BEST fit must CANCEL out.

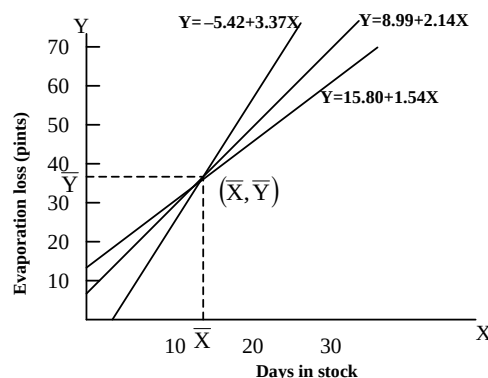
This is the first of the conditions or requirements for an optimal line.

But, the point to understand is that there are an INFINITELY large number of straight lines which will satisfy this condition.

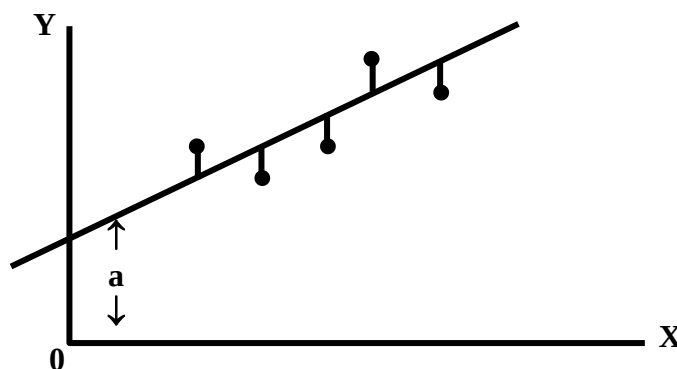
Any line that passes through the point  $(\bar{X}, \bar{Y})$  will satisfy this condition, and as shown in the following figure, numerous lines can pass through the point  $(\bar{X}, \bar{Y})$

### Three equations where

$$\sum(Y - \hat{Y}) = 0:$$



For each of the three lines that you see, the SUM of the VERTICAL deviations between the data-points and the line is ZERO. These deviations are depicted by the following diagram:



How will we calculate these vertical deviations?

The values of Y obtained from the line are denoted by  $\hat{Y}$ .

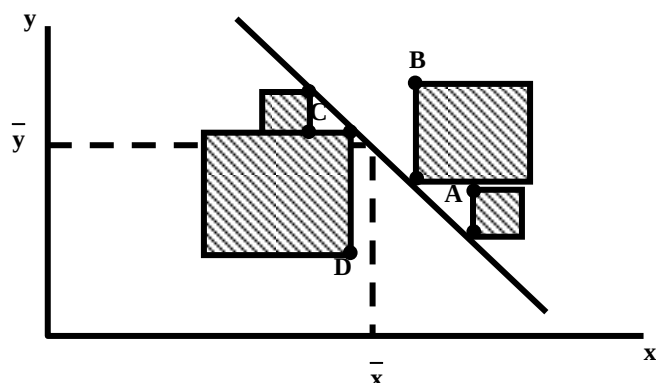
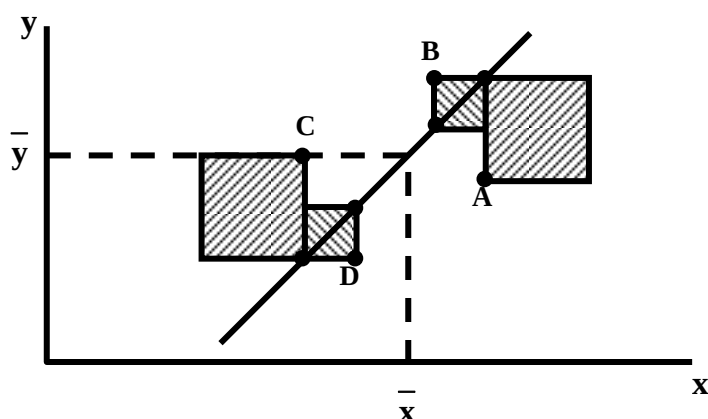
And the deviations of the actual Y-values from the corresponding Y-values obtained from the line are obtained by subtracting  $\hat{Y}$  from Y.

In all the cases --- as long as our line passes through the point  $(\bar{X}, \bar{Y})$  ---, we find that the sum of the deviations of the *actual* Y-values from the corresponding Y-values obtained from the *line* is zero.

Hence, it appears that we need some SECOND criterion for establishing a *unique* position for the BEST-fitting line.

Our interest is NOT simply in achieving a non-zero sum: it is the MAGNITUDE of the sum which is our main concern.

In the two figures that follow, the DIFFERENCES in the sums of squared deviations for two different lines passing through the SAME scatter diagram are clearly portrayed. In position 1, the shaded areas are relatively large, but AS the line is rotated around the point  $(\bar{X}, \bar{Y})$  in a clockwise direction to position 2, the areas become smaller.

**Position 1****Position 2**

There would seem to be some UNIQUE position at which the sum of the square deviations is at a MINIMUM. This is the position of *least squares*.

If we can ascertain the location of THIS particular straight line in terms of the constants  $a$  and  $b$  of the linear equation  $Y = a + bX$ , then we have found the line of BEST fit.

The rationale is that the SMALLER the sum of the squared deviations round the mean, the LESS dispersed are the data points around the fitted line.

**THE PRINCIPAL OF LEAST SQUARES**

According to the principal of least squares, the best-fitting line to a set of points is the one for which the sum of the squares of the vertical distances between the points and the line is minimum.

The line  $Y = a + bX$  is the one that best fits the given set of points according to the principal of least squares.

And, this best fitting line is obtained by solving simultaneously two equations which are known as the normal equations.

**NORMAL EQUATIONS**

$$\left. \begin{aligned} \sum Y &= na + b \sum X \\ \sum XY &= a \sum X + b \sum X^2 \end{aligned} \right\}$$

In connection with these two equations, two points should be noted:

- 1) I will not be discussing the mathematical derivation of these equations.
- 2) The word “normal” here has nothing to do with the well-known normal distribution.

For any bivariate data-set, obviously we will have available to us two columns, a column of  $X$  and a column of  $Y$ . Hence, obviously, we will be in a position to compute sums like  $\sum X$ ,  $\sum Y$ ,  $\sum XY$ , and so on.

Hence, the only unknown quantities in the two normal equations are a and b, as shown below:

#### **NORMAL EQUATIONS**

$$\left. \begin{aligned} \sum Y &= na + b \sum X \\ \sum XY &= a \sum X + b \sum X^2 \end{aligned} \right\}$$

Hence, when we solve the two normal equations simultaneously, we will obtain the values of a and b, and these are EXACTLY the two quantities that we need in order to obtain the BEST-fitting line.

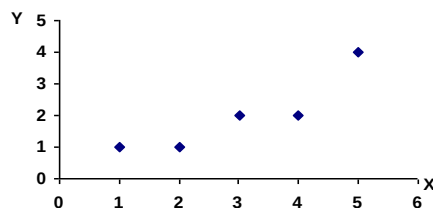
Let me explain this whole concept to you with the help of the same example that I picked up in the beginning of today's lecture:

#### **EXAMPLE**

An important concern for any pharmaceutical company producing drugs is to determine how a particular drug will affect one's perception or general awareness. Suppose one such company wants to establish a relationship between the PERCENTAGE of a drug in the blood-stream and the LENGTH OF TIME it takes to respond to a stimulus. Suppose the company administers this drug on 5 subjects and obtains the following information:

| Subject | Percentage of drug | Reaction Time (milli-seconds) |
|---------|--------------------|-------------------------------|
|         | X                  | Y                             |
| A       | 1                  | 1                             |
| B       | 2                  | 1                             |
| C       | 3                  | 2                             |
| D       | 4                  | 2                             |
| E       | 5                  | 4                             |

#### **Scatter Diagram:**



In order to find a and b, we need to solve the two normal equations, and for this purpose, we will carry out computations as shown below:

| X  | Y  | X <sup>2</sup> | XY |
|----|----|----------------|----|
| 1  | 1  | 1              | 1  |
| 2  | 1  | 4              | 2  |
| 3  | 2  | 9              | 6  |
| 4  | 2  | 16             | 8  |
| 5  | 4  | 25             | 20 |
| 15 | 10 | 55             | 37 |

$$\begin{aligned}
 10 &= 5a + 15b & \text{--- 1} \\
 37 &= 15a + 55b & \text{--- 2} \\
 -7 &= -10b \\
 b &= \frac{-7}{-10} = 0.7 \\
 10 &= 5a + 15b \\
 \therefore 10 - 15b &= 5a \\
 \therefore 10 - 15(0.7) &= 5a \\
 \therefore 10 - 10.5 &= 5a \\
 \therefore -0.5 &= 5a \quad \therefore a = \frac{-0.5}{5} = -0.1
 \end{aligned}$$

Hence our straight line is given by  
 $Y = -0.1 + 0.7 X$

A hat is placed on top of the Y so as to differentiate the Y values obtained from the line from the ones that pertain to the actual data-points.

The question is, "What is the advantage of fitting this line?"

The answer to this question is that this line can be used to ESTIMATE the value of the dependent variable corresponding to some particular value of the independent variable. In this example, suppose that we are interested in finding out what will be the reaction time of a person who has 4.33% of the drug in his blood stream? The answer will be obtained by putting  $X=4.33$  in the equation that we just obtained.

Our regression line is

$$= -0.1 + 0.7 X$$

Putting  $X = 4.33$ , we obtain

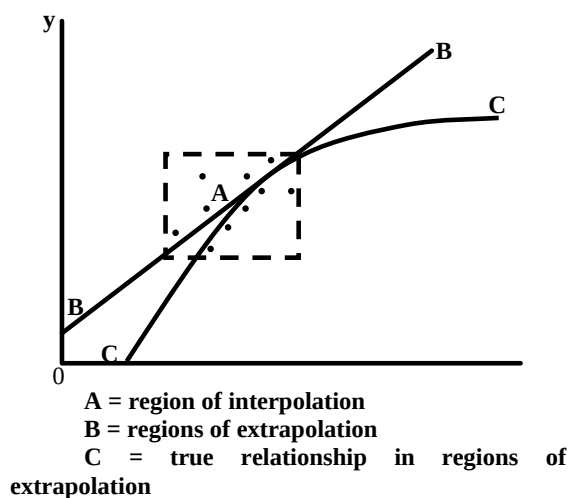
$$= -0.1 + 0.7 (4.33)$$

$$= -0.1 + 3.031 = 2.931$$

Hence we conclude that it can be expected that a person having 4.33% of the drug in his blood stream will take 2.9 milli-seconds to react to the stimulus. A point to be noted here is that this procedure of estimating the value of the dependent variable should not be used for extrapolation. Extrapolation means the making of estimates or predictions outside the range of the experimental data, and in some situations, this can be very unwise.

Let me explain this point with the help of the following diagram:

### The extrapolation trap



While a set of observations may show a good linear relationship between the variables, there is NEVER any *guarantee* that the SAME linear form is present over THOSE ranges of the variable NOT under consideration. I would now like to convey to you another point: All the discussion that I have done until now assumes that Y is the dependent variable and X is the independent variable, and therefore, we are regressing Y on X. But, in some situations, we may be interested in just the OPPOSITE --- i.e. we may wish to regress X on Y. In this situation, all we have to do is to interchange the roles of X and Y. I would like to encourage you to work on this on your own, and to establish the normal equations that will

be required in this situation. You may be thinking, “Why should we go through this hassle? Can’t we use the equation that we have just fitted i.e.  $Y = a + bX$  to estimate  $X$  from  $Y$ ?”

It is important to note that this is not the case. If we are confronted with a situation where we require to predict  $Y$  from  $X$  AND  $X$  from  $Y$ , then *two* DISTINCT equations need to be found. The regressions of  $Y$  on  $X$  and  $X$  on  $Y$  for the same bivariate data are NOT identical.

The next concept that I am going to discuss with you is the STANDARD ERROR OF ESTIMATE.

### **STANDARD ERROR OF ESTIMATE**

The observed values of  $(X, Y)$  do not all fall on the regression line but they scatter away from it. The degree of scatter of the observed values about the regression line is measured by what is called the standard deviation of regression or the standard error of estimate of  $Y$  on  $X$ .

For sample data, the standard error of estimate is obtained from the formula

$$S_{y.x} = \sqrt{\frac{\sum (Y - \hat{Y})^2}{n - 2}}$$

where  $Y$  denotes an observed values, denotes the corresponding values obtained from the least-squares line and  $n$  denotes the sample size.

The formula that I just conveyed to you is a bit cumbersome to apply because, in order to apply it, we first need to compute corresponding to all our  $X$ -values.

Alternative formula for  $S_{y.x}$

The standard error of estimate can be more conveniently computed from the alternative formula

$$S_{y.x} = \sqrt{\frac{\sum Y^2 - a \sum Y - b \sum XY}{n - 2}}$$

### **INTERPRETATION OF $S_{y.x}$**

The range within which  $sy.x$  lies is given by  $0 < sy.x < sy$  (where  $sy$  denotes the standard deviation of the  $y$  values).  $sy.x$  will be zero when all the observed points fall on the regression line (denoting perfect relationship between the two variables).  $sy.x$  will be equal to  $sy$  when there is no relationship between the two variables. Hence the closer  $sy.x$  is to zero (the further away it is from  $sy$ ), the closer the points are to the line, and the more RELIABLE is our line for purposes of prediction.

Let us apply this concept to the example of the amount of drug in the blood stream and the time taken to react to a stimulus:

| X  | Y  | $X^2$ | $Y^2$ | XY |
|----|----|-------|-------|----|
| 1  | 1  | 1     | 1     | 1  |
| 2  | 1  | 4     | 1     | 2  |
| 3  | 2  | 9     | 4     | 6  |
| 4  | 2  | 16    | 4     | 8  |
| 5  | 4  | 25    | 16    | 20 |
| 15 | 10 | 55    | 26    | 37 |

$$\begin{aligned} a &= -0.1 \\ b &= 0.7 \end{aligned}$$

$$\begin{aligned} S_{y.x} &= \sqrt{\frac{\sum Y^2 - a \sum Y - b \sum XY}{n - 2}} \\ &= \sqrt{\frac{26 - (-0.1)(10) - (0.7)(37)}{5 - 2}} \\ &= \sqrt{\frac{26 + 1 - 25.9}{3}} \\ &= \sqrt{\frac{1.1}{3}} = \sqrt{0.3667} = 0.61 \end{aligned}$$

$$\begin{aligned}
 \text{Also, } s_y &= \sqrt{\frac{\sum Y^2}{n} - \left(\frac{\sum Y}{n}\right)^2} \\
 &= \sqrt{\frac{26}{5} - \left(\frac{10}{5}\right)^2} = \sqrt{5.2 - 4} \\
 &= \sqrt{1.2} = 1.10
 \end{aligned}$$

**INTERPRETATION**

$s_{y.x}$  is NOT very small compared with  $s_y$ , hence our least-squares line.

$$\hat{Y} = -0.1 + 0.7 X$$

is probably NOT very reliable for purposes of prediction. As it explained a short while ago, the smaller our Standard Error of Estimate, the closer the data-points will be to the line --- i.e. the more REPRESENTATIVE our line will be of the data-points --- and, hence, the more RELIABLE our line will be for estimation purposes.

The next concept that I am going to discuss with you is the concept of CORRELATION.

It is a concept that is very closely linked with the concept of linear regression.

**CORRELATION**

is a measure of the strength or the degree of relationship between two RANDOM variables.

A numerical measure of the strength of the linear relationship between two random variables  $X$  and  $Y$  is known as Pearson's Product-Moment Coefficient of Correlation.

**PEARSON'S COEFFICIENT OF CORRELATION**

where, covariance of  $X$  and  $Y$  is defined as  $r = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}$

$$\text{Cov}(X, Y) = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{n}$$

This formula is a bit cumbersome to apply. Therefore, we may use the following short cut formula:

**SHORT CUT FORMULA FOR THE PEARSON'S COEFFICIENT OF CORRELATION**

$$r = \frac{\sum XY - (\sum X)(\sum Y)/n}{\sqrt{[\sum X^2 - (\sum X)^2/n][\sum Y^2 - (\sum Y)^2/n]}}$$

It should be noted that

$r$  is a pure number that lies between  $-1$  and  $1$  i.e.

$$-1 < r < 1$$

Actually, the mathematical expressions that you have just seen is a combination of three different mathematical expressions:

**Case 1:**

Positive correlation:  $0 < r < 1$

**Case 2:**

No correlation:  $r = 0$

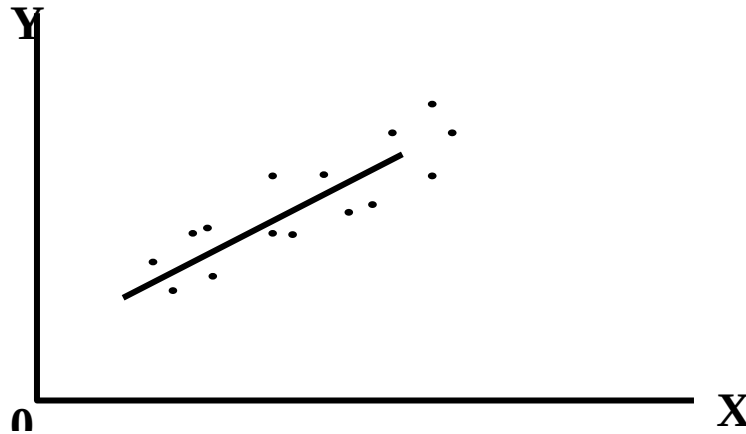
**Case 3:**

Negative correlation:  $-1 < r < 0$



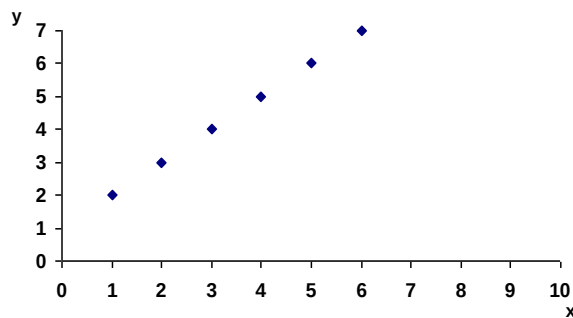
**Case 1:**

In case of a positive linear relationship,  $r$  lies between 0 and 1.



In this case, the closer the points are to the UPWARD-going line, the STRONGER is the positive linear relationship, and the closer  $r$  is to 1.

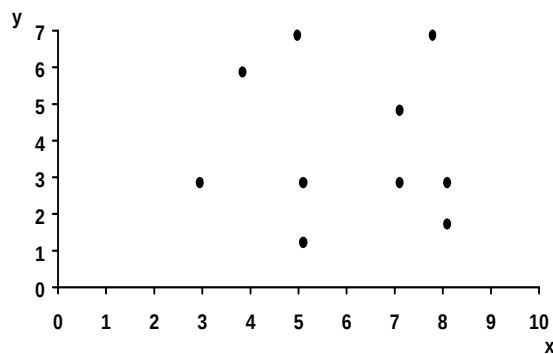
**Perfect Positive Linear Correlation ( $r = 1$ ):**



In this case, the closer the points are to the DOWNWARD-going line, the stronger is the linear relationship, and the closer  $r$  is to  $-1$ .

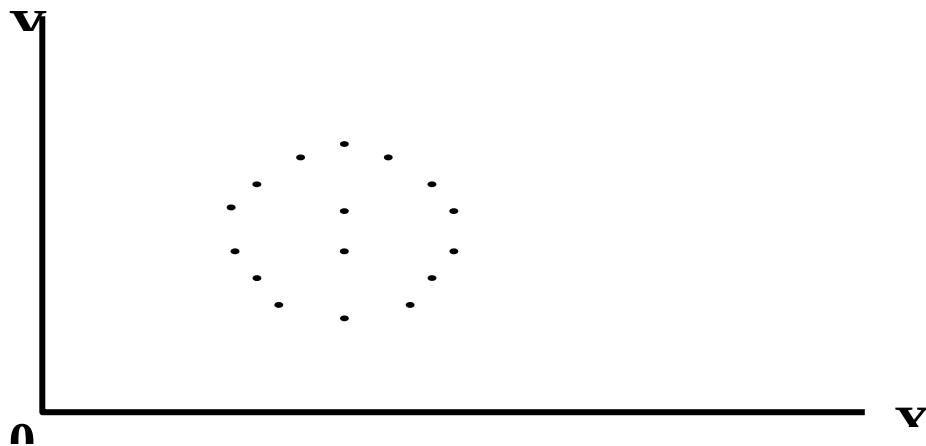
**Case 3:**

In a situation where neither an upward linear trend nor a downward linear trend can be visualized,  $r \simeq 0$



Here, the bivariate data seem to be completely random.

**The extreme of dissociation (zero correlation ( $r = 0$ )):**

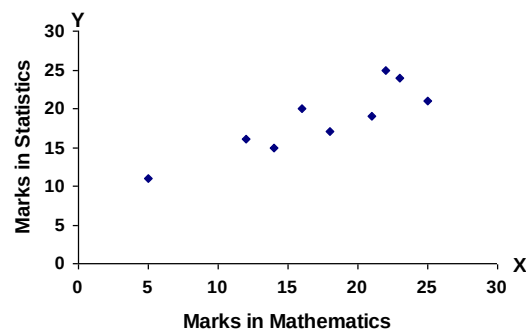


In such a situation, X and Y are said to be uncorrelated.

**EXAMPLE**

Suppose that the principal of a college wants to know if there exists any correlation between grades in Mathematics and grades in Statistics. Suppose that he selects a random sample of 9 students out of all those who take this combination of subjects. The following information is obtained:

| Student | Marks in Mathematics<br>(Total Marks: 25) | Marks in Statistics<br>(Total Marks: 25) |
|---------|-------------------------------------------|------------------------------------------|
|         | X                                         | Y                                        |
| A       | 5                                         | 11                                       |
| B       | 12                                        | 16                                       |
| C       | 14                                        | 15                                       |
| D       | 16                                        | 20                                       |
| E       | 18                                        | 17                                       |
| F       | 21                                        | 19                                       |
| G       | 22                                        | 25                                       |
| H       | 23                                        | 24                                       |
| I       | 25                                        | 21                                       |

**SCATTER DIAGRAM**

In order to compute the correlation coefficient, we carry out the following computations,

| X   | Y   | $X^2$ | $Y^2$ | XY   |
|-----|-----|-------|-------|------|
| 5   | 11  | 25    | 121   | 55   |
| 12  | 16  | 144   | 256   | 192  |
| 14  | 15  | 196   | 225   | 210  |
| 16  | 20  | 256   | 400   | 320  |
| 18  | 17  | 324   | 289   | 306  |
| 21  | 19  | 441   | 361   | 399  |
| 22  | 25  | 484   | 625   | 550  |
| 23  | 24  | 529   | 576   | 552  |
| 25  | 21  | 625   | 441   | 525  |
| 156 | 168 | 3024  | 3294  | 3109 |

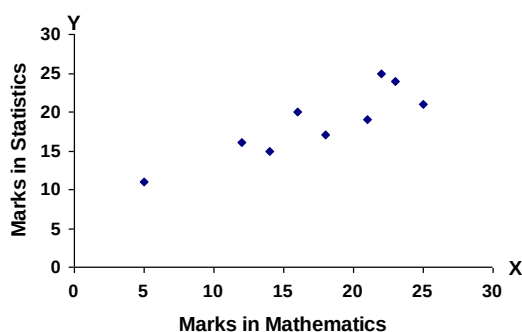
$$\begin{aligned}
 r &= \frac{\sum XY - (\sum X)(\sum Y)/n}{\sqrt{[\sum X^2 - (\sum X)^2/n][\sum Y^2 - (\sum Y)^2/n]}} \\
 &= \frac{3109 - (156)(168)/9}{\sqrt{[3024 - (156)^2/9][3294 - (168)^2/9]}} \\
 &= \frac{3109 - 2912}{\sqrt{[3024 - 2704][3294 - 3136]}} \\
 &= \frac{197}{\sqrt{320 \times 158}} = \frac{197}{224.86} = 0.88
 \end{aligned}$$

### INTERPRETATION

There exists a strong *positive* linear correlation between marks in Mathematics and marks in Statistics for these 9 students who have been taken into consideration.

The conclusion that we have just drawn i.e. strong positive linear correlation --- this conclusion is supported by the scatter diagram.

### SCATTER DIAGRAM



As you can see in the scatter diagram, the data-points appear to follow a linear pattern quite strongly.

In today's lecture, I have discussed with you the concept of regression and correlation. Although I have conveyed to you a number of interesting concepts, believe me, this is only the BEGINNING of a very vast and important area of Statistics. You can study this concept further, and, if possible, to study a little bit about MULTIPLE regression and correlation as well --- the situation when we try to study the relationship between three or more variables.

This brings us to the end of the FIRST part of this course i.e. Descriptive Statistics.

This brings us to the end of the FIRST part of this course i.e. Descriptive Statistics.

## LECTURE NO. 16

- Set Theory
- Counting Rules:
- The Rule of Multiplication

**“SET”**

A set is any well-defined collection or list of distinct objects, e.g. a group of students, the books in a library, the integers between 1 and 100, all human beings on the earth, etc. The term well-defined here means that any object must be classified as either belonging or not belonging to the set under consideration, and the term distinct implies that each object must appear only once. The objects that are in a set, are called members or elements of that set. Sets are usually denoted by capital letters such as A, B, C, Y, etc., while their elements are represented by small letters such as, a, b, c, y, etc.

Elements are enclosed by parentheses to represent a set.

For example:

**EXAMPLES OF SETS:**

$$A = \{a, b, c, d\} \text{ or}$$

$$B = \{1, 2, 3, 7\}$$

The Number of a set A, written as  $n(A)$ , is defined as the number of elements in A.

If x is an element of a set A, we write  $x \in A$  which is read as “x belongs to A” or x is in A. If x does not belong to A, i.e. x is not an element of A, we write  $x \notin A$ .

A set that has no elements is called an empty or a null set and is denoted by the symbol  $\phi$ . (It must be noted that  $\{0\}$  is not an empty set as it contains an element 0.)

If a set contains only one element, it is called a unit set or a singleton set.

It is also important to note the difference between an element “x” and a unit set  $\{x\}$ .

A set may be specified in two ways:

1. We may give a list of all the elements of a set (the “Roster” method),  
e.g.

$$A = \{1, 3, 5, 7, 9, 11\};$$

$$B = \{\text{a book, a city, a clock, a teacher}\};$$

2. We may state a rule that enables us to determine whether or not a given object is a member of the set (the “Rule” method or the “Set Builder” method),  
e.g.

$A = \{x \mid x \text{ is an odd number and } x < 12\}$  meaning that A is a set of all elements x such that x is an odd number and x is less than 12. (The vertical line is read as “such that”). *An important point to note is that:* The repetition or the order in which the elements of a set occur, does not change the nature of the set. The size of a set is given by the number of elements present in it. This number may be finite or infinite. Thus a set is finite when it contains a finite number of elements; otherwise it is an infinite set.

The Empty set is regarded as a Finite set.

**EXAMPLES OF FINITE SETS**

i)  $A = \{1, 2, 3, \dots, 99, 100\};$

ii)  $B = \{x \mid x \text{ is a month of the year}\};$

iii)  $C = \{x \mid x \text{ is a printing mistake in a book}\};$

iv)  $D = \{x \mid x \text{ is a living citizen of Pakistan}\};$

Examples of infinite sets:

i)  $A = \{x \mid x \text{ is an even integer}\};$

ii)  $B = \{x \mid x \text{ is a real number between 0 and 1 inclusive}\},$   
i.e.  $B = \{x \mid 0 \leq x \leq 1\}$

iii)  $C = \{x \mid x \text{ is a point on a line}\};$

iv)  $D = \{x \mid x \text{ is a sentence in a English language}\}; \text{ etc}$

**SUBSETS**

A set that consists of some elements of another set, is called a subset of that set.  
For example, if B is a subset of A, then every member of set B is also a member of set A.

If B is a subset of A, we write:

$B \subset A$  or equivalently:

$A \supset B$

'B is a subset of A' is also read as 'B is contained in A',  
or 'A contains B'.

**EXAMPLE**

If  $A = \{1, 2, 3, 4, 5, 10\}$

and  $B = \{1, 3, 5\}$

then  $B \subset A$ ,

i.e. B is contained in A.

It should be noted that any set is always regarded a subset of itself.

and an empty set  $\phi$  is considered to be a subset of every set.

Two sets A and B are Equal or Identical, if and only if they contain exactly the same elements.

In other words,  $A = B$  if and only if  $A \subset B$  and  $B \subset A$ .

**PROPER SUBSET**

If a set B contains some but not all of the elements of another set A, while A contains each element of B, i.e. if

$B \subset A$  and  $B \neq A$

then the set B is defined to be a proper subset of A.

**Universal Set:**

The original set of which all the sets we talk about, are subsets, is called the universal set (or the space) and is generally denoted by S or  $\Omega$ .

The universal set thus contains all possible elements under consideration.

A set S with n elements will produce  $2^n$  subsets, including S and  $\phi$ .

**EXAMPLE;**

Consider the set  $A = \{1, 2, 3\}$ .

All possible subsets of this set are:

$\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}$  and  $\{1, 2, 3\}$

Hence, there are  $2^3 = 8$  subsets of the set A.

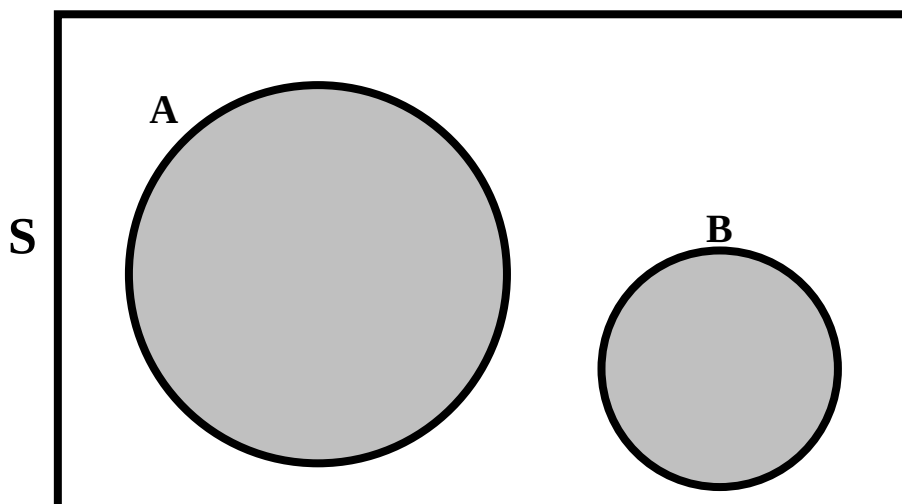
**VENN DIAGRAM**

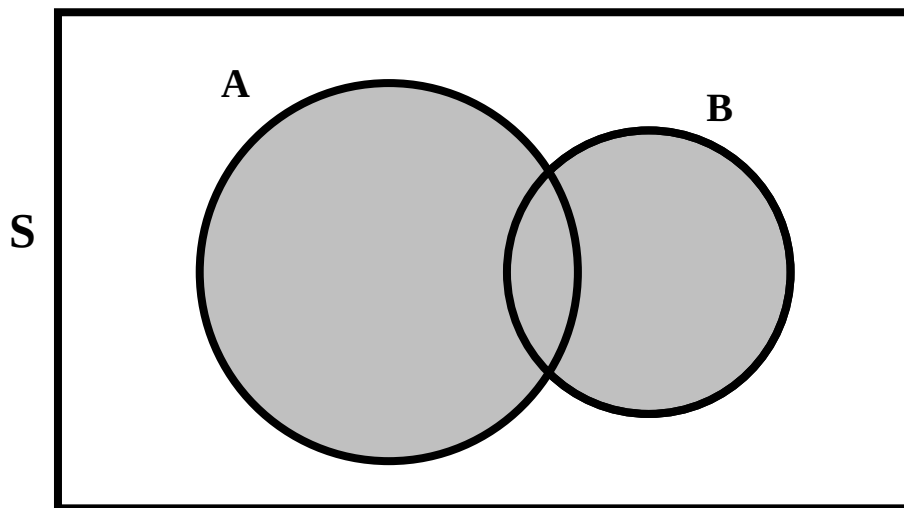
A diagram that is understood to represent sets by circular regions, parts of circular regions or their complements with respect to a rectangle representing the space S is called a Venn diagram, named after the English logician John Venn (1834-1923).

The Venn diagrams are used to represent sets and subsets in a pictorial way and to verify the relationship among sets and subsets.

**A Simple Venn diagram:**

Disjoint Sets





### OPERATIONS ON SETS

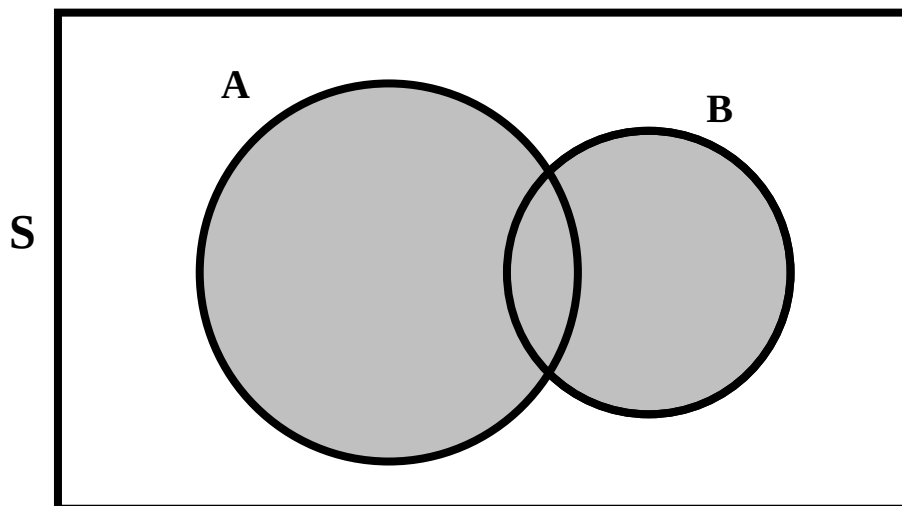
Let the sets A and B be the subsets of some universal set S. Then these sets may be combined and operated on in various ways to form new sets which are also subsets of S. The basic operations are union, intersection, difference and complementation.

### UNION OF SETS

The union or sum of two sets A and B, denoted by  $A \cup B$ , and read as “A union B”, means the set of all elements that belong to at least one of the sets A and B, that is

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

By means of a Venn Diagram,  $A \cup B$  is shown by the shaded area as below:



$A \cup B$  is shaded

### EXAMPLE

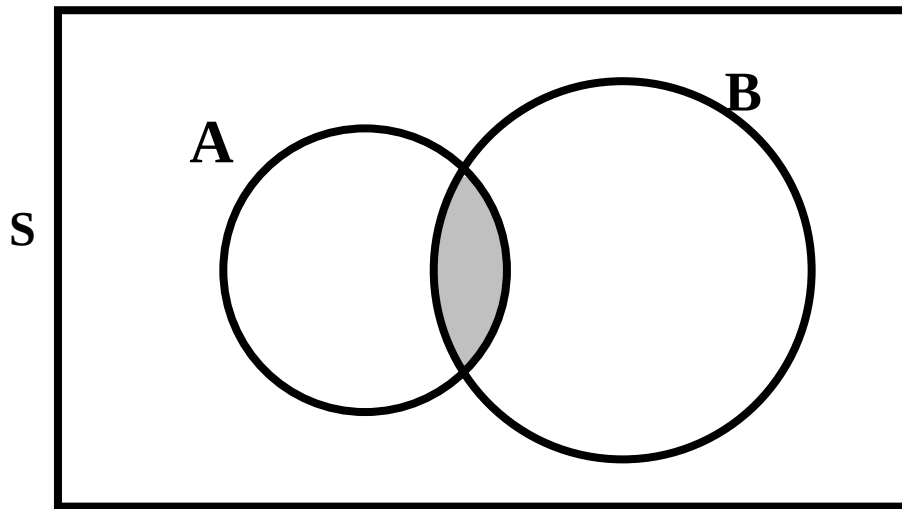
Let  $A = \{1, 2, 3, 4\}$  and  $B = \{3, 4, 5, 6\}$   
Then  $A \cup B = \{1, 2, 3, 4, 5, 6\}$

**INTERSECTION OF SETS**

The intersection of two sets A and B, denoted by  $A \cap B$ , and read as “A intersection B”, means that the set of all elements that belong to both A and B; that is

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

Diagrammatically,  $A \cap B$  is shown by the shaded area as below:



$A \cap B$  is shaded

**EXAMPLE**

Let  $A = \{1, 2, 3, 4\}$  and  $B = \{3, 4, 5, 6\}$

Then  $A \cap B = \{3, 4\}$

The operations of union and intersection that have been defined for two sets may conveniently be extended to any finite number of sets.

**DISJOINT SETS**

Two sets A and B are defined to be disjoint or mutually exclusive or non-overlapping when they have no elements in common, i.e. when their intersection is an empty set

$$\text{i.e. } A \cap B = \phi.$$

On the other hand, two sets A and B are said to be conjoint when they have at least one element in common.

**SET DIFFERENCE**

The difference of two sets A and B, denoted by  $A - B$  or by  $A - (A \cap B)$ , is the set of all elements of A which do not belong to B.

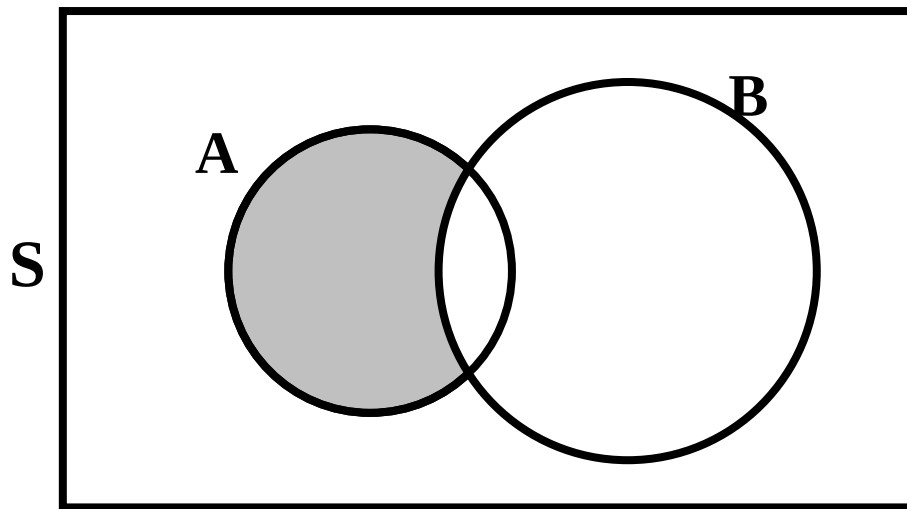
Symbolically,

$$A - B = \{x \mid x \in A \text{ and } x \notin B\}$$

It is to be pointed out that in general  $A - B \neq B - A$ .

The shaded area of the following Venn diagram shows the difference  $A - B$ :





**Difference  $A - B$  is shaded**

It is to be noted that  $A - B$  and  $B$  are disjoint sets. If  $A$  and  $B$  are disjoint, then the difference  $A - B$  coincides with the set  $A$ .

#### **COMPLEMENTATION**

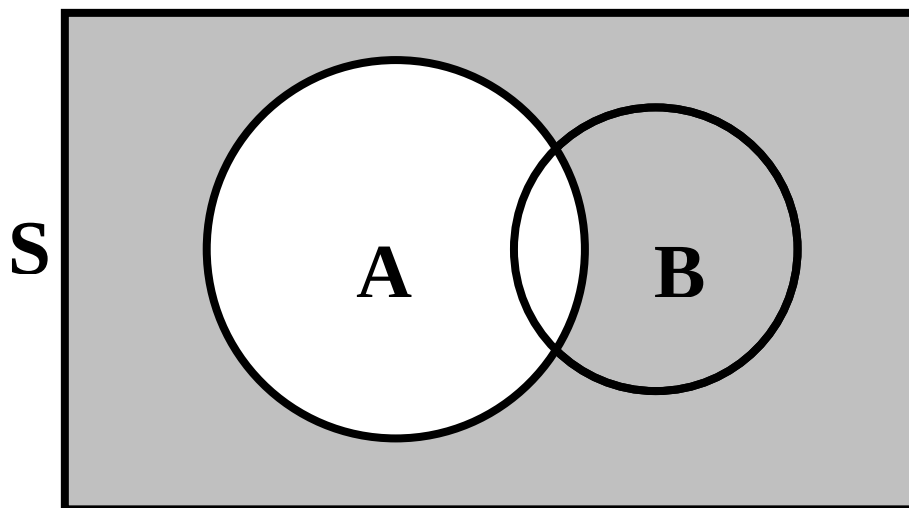
The particular difference  $S - A$ , that is, the set of all those elements of  $S$  which do not belong to  $A$ , is called the complement of  $A$  and is denoted by  $\bar{A}$  or by  $A^c$ .

In symbols:

$$\bar{A} = \{x \mid x \in S \text{ and } x \notin A\}$$

The complement of  $S$  is the empty set  $\phi$ .

The complement of  $A$  is shown by the shaded portion in the following Venn diagram.



**$\bar{A}$  is shaded**

It should be noted that  $A - B$  and  $A \cap \bar{B}$ , where  $\bar{B}$  is the complement of set  $B$ , are the same set. Next, we consider the Algebra of Sets. The algebra of sets provides us with laws which can be used to solve many problems in probability calculations.

Let  $A$ ,  $B$  and  $C$  be any subsets of the universal set  $S$ . Then, we have:

#### **1. Commutative laws:**

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

#### **2. Associative laws:**

$$(A \cup B) \cup C = A \cup (B \cup C)$$

$$(A \cap B) \cap C = A \cap (B \cap C)$$

**3. Distributive laws**

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

**4. Idempotent laws**

$$A \cup A = A$$

$$A \cap A = A$$

**5. Identity laws**

$$A \cup S = S,$$

$$A \cap S = A,$$

$$A \cup \phi = A, \text{ and}$$

$$A \cap \phi = \phi.$$

**6. Complementation laws**

$$A \cup \bar{A} = S,$$

$$A \cap \bar{A} = \phi,$$

$$(\bar{\bar{A}}) = A,$$

$$\bar{\bar{S}} = \phi, \text{ and}$$

$$\bar{\phi} = S$$

**7. De Morgan's laws:**

$$\overline{(A \cup B)} = \bar{A} \cap \bar{B},$$

$$\text{and } \overline{(A \cap B)} = \bar{A} \cup \bar{B}$$

**PARTITION OF SETS**

A partition of a set S is a sub-division of the set into non-empty subsets that are disjoint and exhaustive, i.e. their union is the set S itself.

This implies that:

- i)  $A_i \cap A_j = \phi$ , where  $i \neq j$ ;
- ii)  $A_1 \cap A_2 \cup \dots \cup A_n = S.$

The subsets in a partition are called cells.

**EXAMPLE**

Let us consider a set  $S = \{a, b, c, d, e\}$ .

Then  $\{a, b\}$ , and  $\{c, d, e\}$  is a partition of S as each element of S belongs to exactly one cell.

**CLASS OF SETS**

A set of sets is called a class. For example, in a set of lines, each line is a set of points.

**POWER SET**

The class of ALL subsets of a set A is called the Power Set of A and is denoted by  $P(A)$ .

For example, if  $A = \{H, T\}$ , then  $P(A) = \{\phi, \{H\}, \{T\}, \{H, T\}\}$ .

**CARTESIAN PRODUCT OF SETS**

The Cartesian product of sets A and B, denoted by  $A \times B$ , (read as “A cross B”), is a set that contains all ordered pairs  $(x, y)$ , where x belongs to A and y belongs to B.

Symbolically, we write

$$A \times B = \{(x, y) \mid x \in A \text{ and } y \in B\}$$

This set is also called the Cartesian set of A and B set of A and B, named after the French mathematician Rene' Descartes (1596-1605).

The product of a set A by itself is denoted by  $A^2$ .

This concept of product may be extended to any finite number of sets.

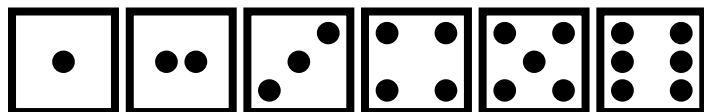
**EXAMPLE**

Let  $A = \{H, T\}$  and  $B = \{1, 2, 3, 4, 5, 6\}$ . Then the Cartesian product set is the collection of the following twelve ( $2 \times 6$ ) ordered pairs:

$$A \times B = \{(H, 1); (H, 2); (H, 3); (H, 4); (H, 5); (H, 6); (T, 1); (T, 2); (T, 3); (T, 4); (T, 5); (T, 6)\}$$

Clearly, these twelve elements together make up the universal set  $S$  when a COIN and a DIE are tossed together.

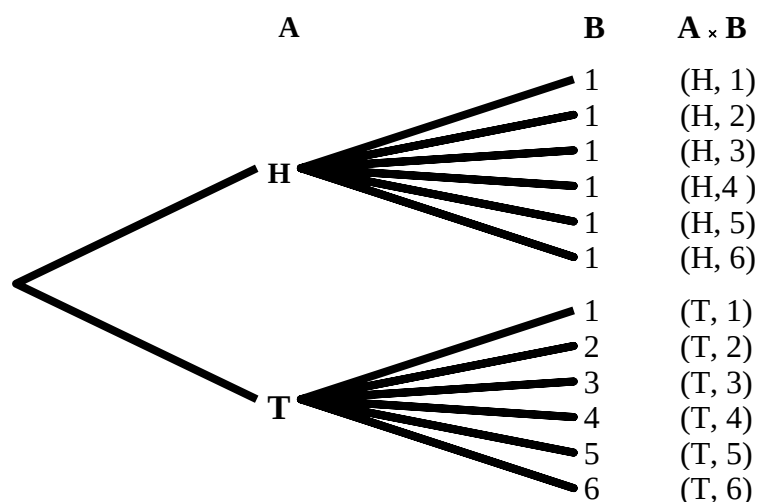
A die is a cube of wood or ivory whose six faces are marked with dots are shown below:



The plural of the word 'die' is 'dice'.

The product  $A \times B$  may conveniently be found by means of the so-called tree diagram shown below:

Tree Diagram

**TREE DIAGRAM**

The "tree" is constructed from the left to the right. A "tree diagram" is a useful device for enumerating all the possible outcomes of two or more sequential events.

The possible outcomes are represented by the individual paths or branches of the tree.

It is relevant to note that, in general

$$A \times B \neq B \times A$$

Having reviewed the basics of set theory, let us now review the COUNTING RULES that facilitate the computation of probabilities in a number of problems.

**RULE OF MULTIPLICATION**

If a compound experiment consists of two experiments which that the first experiment has exactly  $m$  distinct outcomes and, if corresponding to each outcome of the first experiment there can be  $n$  distinct outcomes of the second experiment, then the compound experiment has exactly  $mn$  outcomes.

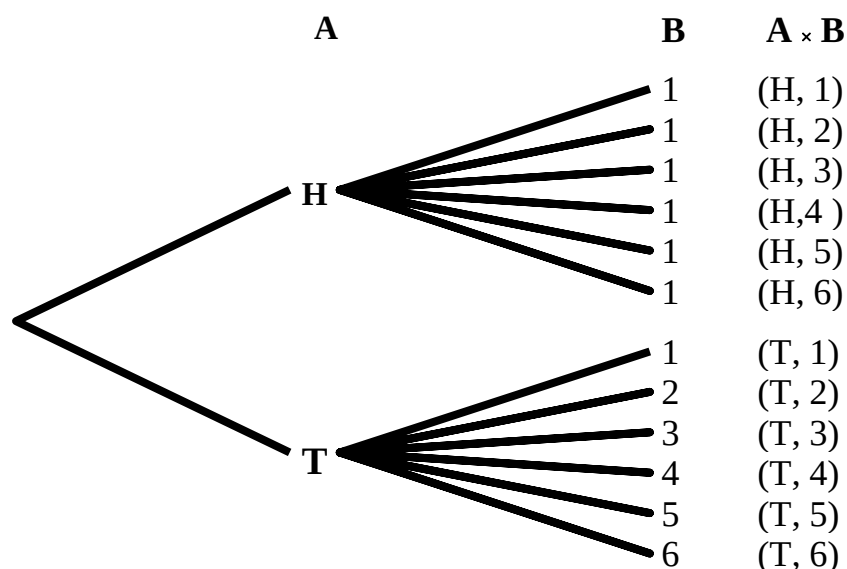
**EXAMPLE**

The compound experiment of tossing a coin and throwing a die together consists of two experiments. The coin-tossing experiment consists of two distinct outcomes (H, T), and the die-throwing experiment consists of six distinct outcomes (1, 2, 3, 4, 5, 6).

The total number of possible distinct outcomes of the compound experiment is therefore  $2 \times 6 = 12$  as each of the two outcomes of the coin-tossing experiment can occur with each of the six outcomes of die-throwing experiment. As stated earlier, if  $A = \{H, T\}$  and  $B = \{1, 2, 3, 4, 5, 6\}$ , then the Cartesian product set is the collection of the following twelve  $(2 \times 6)$  ordered pairs:

$$A \times B = \{ (H, 1); (H, 2); (H, 3); (H, 4); (H, 5); (H, 6); (T, 1); (T, 2); (T, 3); (T, 4); (T, 5); (T, 6) \}$$

### Tree Diagram



The rule of multiplication can be readily extended to compound experiments consisting of any number of experiments performed in a given sequence.

This rule can also be called the Multiple Choice Rule, as illustrated by the following example:

#### EXAMPLE

Suppose that a restaurant offers three types of soups, four types of sandwiches, and two types of desserts. Then, a customer can order any one out of  $3 \times 4 \times 2 = 24$  different meals.

#### EXAMPLE

Suppose that we have a combination lock on which there are eight rings. In how many ways can the lock be adjusted?

#### **Solution:**

The logical way to look at this problem is to see that there are eight rings on the lock, each of which can have any of the 10 figures 0 to 9:

$$\overline{A} \quad \overline{B} \quad \overline{C} \quad \overline{D} \quad \overline{E} \quad \overline{F} \quad \overline{G} \quad \overline{H}$$

ring A can have any of the digits 0 to 9 and ring B can have any of the digits 0 to 9 and ring C can have any of the digits 0 to 9 and

·  
·  
·

ring H can have any of the digits 0 to 9

Hence the total No. of ways in which these 8 rings can be filled is

$$10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10$$

$$= 10^8$$

i.e. 100,000,000 — one hundred million.

**LECTURE NO. 17**

- Permutations
- Combinations
- Random Experiment
- Sample Space
- Events
  - Mutually Exclusive Events
  - Exhaustive Events
  - Equally Likely Events

**COUNTING RULES**

As discussed in the last lecture, there are certain rules that facilitate the calculations of probabilities in certain situations. They are known as counting rules and include concepts of:

- Multiple Choice
- Permutations
- Combinations

We have already discussed the rule of multiplication in the last lecture. Let us now consider the rule of permutations.

**RULE OF PERMUTATION**

A permutation is any ordered subset from a set of  $n$  distinct objects.

For example, if we have the set  $\{a, b\}$ , then one permutation is  $ab$ , and the other permutation is  $ba$ . The number of permutations of  $r$  objects, selected in a definite order from  $n$  distinct objects is denoted by the symbol  ${}^n P_r$  and is given by

$${}^n P_r = n(n-1)(n-2) \dots (n-r+1)$$

$$= \frac{n!}{(n-r)!}$$

**FACTORIALS**

$$7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$$

$$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1$$

.

.

.

$$1! = 1$$

Also, we define  $0! = 1$ .

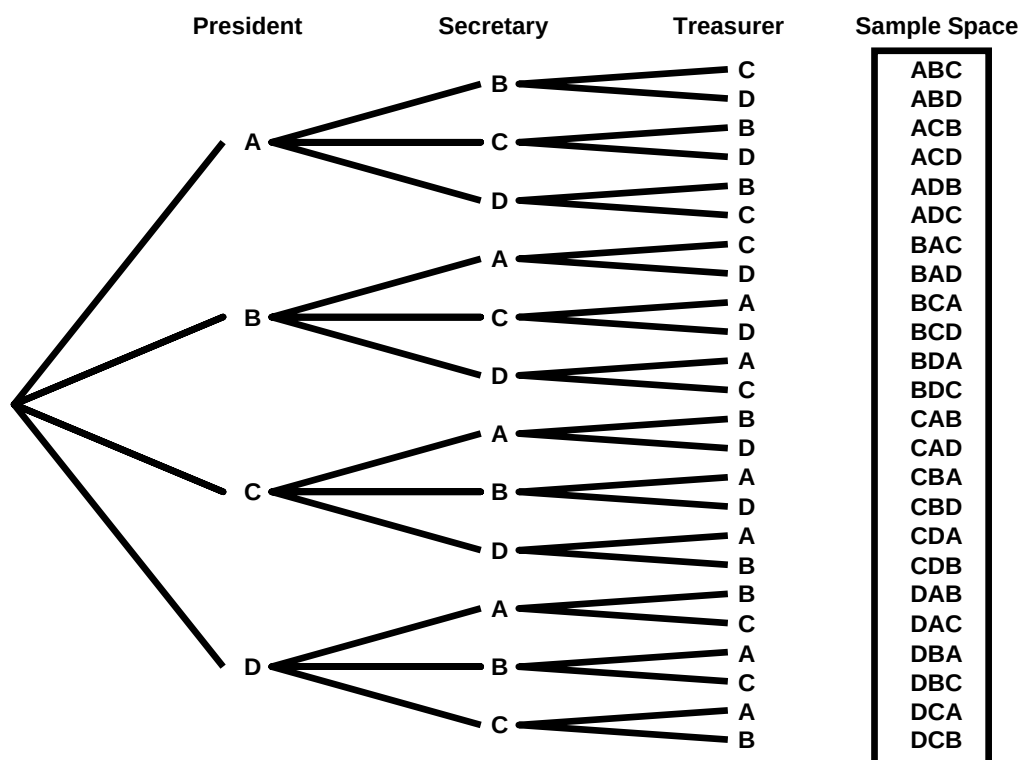
**EXAMPLE**

A club consists of four members. How many ways are there of selecting three officers: president, secretary and treasurer? It is evident that the order, in which 3 officers are to be chosen, is of significance. Thus there are 4 choices for the first office, 3 choices for the second office, and 2 choices for the third office. Hence the total number of ways in which the three offices can be filled is  $4 \times 3 \times 2 = 24$ .

The same result is obtained by applying the rule of permutations:

$$\begin{aligned} {}^4 P_3 &= \frac{4!}{(4-3)!} \\ &= 4 \times 3 \times 2 \\ &= 24 \end{aligned}$$

Let the four members be, A, B, C and D. Then a tree diagram which provides an organized way of listing the possible arrangements, for this example, is given below:



### PERMUTATIONS

In the formula of  ${}^nP_r$ , if we put  $r = n$ , we obtain:

$${}^nP_n = n(n-1)(n-2) \dots 3 \times 2 \times 1$$

$$= n!$$

I.e. the total number of permutations of  $n$  distinct objects, taking all  $n$  at a time, is equal to  $n!$

### EXAMPLE

Suppose that there are three persons A, B & D, and that they wish to have a photograph taken.

The total number of ways in which they can be seated on three chairs (placed side by side) is

$${}_3P_3 = 3! = 6$$

These are:

ABD,  
ADB,  
BAD,  
BDA,  
DAB,  
DBA

The above discussion pertained to the case when all the objects under consideration are distinct objects. If some of the objects are not distinct, the formula of permutations modifies as given below:

The number of permutations of  $n$  objects, selected all at a time, when  $n$  objects consist of  $n_1$  of one kind,  $n_2$  of a second kind, ...,  $n_k$  of a  $k$ th kind,

$$\text{is } P = \frac{n!}{n_1! n_2! \dots n_k!}$$

(where  $\sum n_i = n$ )

**EXAMPLE**

How many different (meaningless) words can be formed from the word ‘committee’?

In this example:

$n = 9$  (because the total number of letters in this word is 9)

$n_1 = 1$  (because there is one c)

$n_2 = 1$  (because there is one o)

$n_3 = 2$  (because there are two m's)

$n_4 = 1$  (because there is one i)

$n_5 = 2$  (because there are two t's)

and

$n_6 = 2$  (because there are two e's)

Hence, the total number of (meaningless) words (permutations) is:

$$\begin{aligned} P &= \frac{n!}{n_1! n_2! \dots n_k!} \\ &= \frac{9!}{1! 1! 2! 1! 2! 2!} \\ &= \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{1 \times 1 \times 2 \times 1 \times 1 \times 2 \times 1 \times 2 \times 1} \\ &= 45360 \end{aligned}$$

Next, let us consider the rule of combinations.

**RULE OF COMBINATION**

A combination is any subset of  $r$  objects, selected without regard to their order, from a set of  $n$  distinct objects. The total number of such combinations is denoted by the symbol

and is given by

$${}^n C_r \text{ or } \binom{n}{r},$$

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

where  $r < n$ .

It should be noted that

$${}^n P_r = r! \binom{n}{r}$$

In other words, every combination of  $r$  objects (out of  $n$  objects) generates  $r!$  Permutations

**EXAMPLE**

Suppose we have a group of three persons, A, B, & C. If we wish to select a group of two persons out of these three, the three possible groups are {A, B}, {A, C} and {B, C}. In other words, the total number of combinations of size two out of this set of size three is 3.

Now, suppose that our interest lies in forming a committee of two persons, one of whom is to be the president and the other the secretary of a club.

The six possible committees are:

(A, B), (B, A),  
(A, C), (C, A),  
(B, C) & (C, B)

In other words, the total number of permutations of two persons out of three is 6.

And the point to note is that each of three combinations mentioned earlier generates  $2 = 2!$  Permutations, I.e. the combination {A, B} generates the permutations (A, B) and (B, A) and the combination {A, C} generates the permutations (A, C) and (C, A); and the combination {B, C} generates the permutations (B, C) and (C, B).

The quantity

$$\binom{n}{r}$$

or  ${}^nC_r$  is also called a binomial co-efficient because of its appearance in the binomial expansion of

$$(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r.$$

The binomial co-efficient has two important properties.

$$\text{i) } \binom{n}{r} = \binom{n}{n-r}, \text{ and}$$

$$\text{ii) } \binom{n}{n-r} + \binom{n}{r} = \binom{n+1}{r}$$

Also, it should be noted that

$$\binom{n}{0} = 1 = \binom{n}{n}$$

and

$$\binom{n}{1} = n = \binom{n}{n-1}$$

### EXAMPLE

A three-person committee is to be formed out of a group of ten persons. In how many ways can this be done?

Since the order in which the three persons of the committee are chosen, is unimportant, it is therefore an example of a problem involving combinations. Thus the desired number of combinations is

$$\begin{aligned} \binom{n}{r} &= \binom{10}{3} = \frac{10!}{3!(10-3)!} = \frac{10!}{3!7!} \\ &= \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1} \\ &= 120 \end{aligned}$$

In other words, there are one hundred and twenty different ways of forming a three-person committee out of a group of only ten persons!

### EXAMPLE

In how many ways can a person draw a hand of 5 cards from a well-shuffled ordinary deck of 52 cards?

The total number of ways of doing so is given by

$$\binom{n}{r} = \binom{52}{5} = \frac{52 \times 51 \times 50 \times 49 \times 48}{5 \times 4 \times 3 \times 2 \times 1} = 2,598,960$$

Having reviewed the counting rules that facilitate calculations of probabilities in a number of problems, let us now begin the discussion of concepts that lead to the formal definitions of probability. The first concept in this regard is the concept of Random Experiment. The term experiment means a planned activity or process whose results yield a set of data. A single performance of an experiment is called a trial. The result obtained from an experiment or a trial is called an outcome.

### RANDOM EXPERIMENT

An experiment which produces different results even though it is repeated a large number of times under essentially similar conditions is called a Random Experiment.



The tossing of a fair coin, the throwing of a balanced die, drawing of a card from a well-shuffled deck of 52 playing cards, selecting a sample, etc. are examples of random experiments.

### **PROPERTIES OF A RANDOM EXPERIMENT**

A random experiment has three properties:

- The experiment can be repeated, practically or theoretically, any number of times.
- The experiment always has two or more possible outcomes. An experiment that has only one possible outcome is not a random experiment.
- The outcome of each repetition is unpredictable, i.e. it has some degree of uncertainty.

Considering a more realistic example, interviewing a person to find out whether or not he or she is a smoker is an example of a random experiment. This is so because this example fulfils all the three properties that have just been discussed:

- This process of interviewing can be repeated a large number of times.
- To each interview, there are at least two possible replies: 'I am a smoker' and 'I am not a smoker'.
- For any interview, the answer is not known in advance i.e. there is an element of uncertainty regarding the person's reply.

A concept that is closely related with the concept of a random experiment is the concept of the Sample Space.

### **SAMPLE SPACE**

A set consisting of all possible outcomes that can result from a random experiment (real or conceptual), can be defined as the sample space for the experiment and is denoted by the letter S. Each possible outcome is a member of the sample space, and is called a sample point in that space. Let us consider a few examples:

#### **EXAMPLE-1**

The experiment of tossing a coin results in either of the two possible outcomes: a head (H) or a tail (T). (We assume that it is not possible for the coin to land on its edge or to roll away). The sample space for this experiment may be expressed in set notation as  $S = \{H, T\}$ . 'H' and 'T' are the two sample points.

#### **EXAMPLE-2**

The sample space for tossing two coins once (or tossing a coin twice) will contain four possible outcomes denoted by  $S = \{HH, HT, TH, TT\}$ .

In this example, clearly, S is the Cartesian product  $A \times A$ , where  $A = \{H, T\}$ .

#### **EXAMPLE-3**

The sample space S for the random experiment of throwing two six-sided dice can be described by the Cartesian product  $A \times A$ , where  $A = \{1, 2, 3, 4, 5, 6\}$ . In other words,  $S = A \times A = \{(x, y) \mid x \in A \text{ and } y \in A\}$ , Where x denotes the number of dots on the upper face of the first die, and y denotes the number of dots on the upper face of the second die. Hence, S contains 36 outcomes or sample points, as shown below:

$S =$        $\{(1, 1), (1, 2), (1, 3), (1, 5), (1, 6),$   
               $(2, 1), (2, 2), (2, 3), (2, 5), (2, 6),$   
               $(3, 1), (3, 2), (3, 3), (3, 5), (3, 6),$   
               $(4, 1), (4, 2), (4, 3), (4, 5), (4, 6),$   
               $(5, 1), (5, 2), (5, 3), (5, 5), (5, 6),$   
               $(6, 1), (6, 2), (6, 3), (6, 5), (6, 6)\}$

The next concept is that of events.

### **EVENTS**

Any subset of a sample space S of a random experiment, is called an event. In other words, an event is an individual outcome or any number of outcomes (sample points) of a random experiment.

### **SIMPLE & COMPOUND EVENTS**

An event that contains exactly one sample point is defined as a simple event. A compound event contains more than one sample point, and is produced by the union of simple events.

**EXAMPLE**

The occurrence of a 6 when a die is thrown, is a simple event, while the occurrence of a sum of 10 with a pair of dice, is a compound event, as it can be decomposed into three simple events (4, 6), (5, 5) and (6, 4).

**OCCURRENCE OF AN EVENT**

An event  $A$  is said to occur if and only if the outcome of the experiment corresponds to some element of  $A$ .

**EXAMPLE**

Suppose we toss a die, and we are interested in the occurrence of an even number.

If ANY of the three numbers '2', '4' or '6' occurs, we say that the event of our interest has occurred.

In this example, the event  $A$  is represented by the set  $\{2, 4, 6\}$ , and if the outcome '2' occurs, then, since this outcome is corresponding to the first element of the set  $A$ , therefore, we say that  $A$  has occurred.

**COMPLEMENTARY EVENT**

The event "not- $A$ " is denoted by  $\bar{A}$  or  $A^c$  and called the negation (or complementary event) of  $A$ .

**EXAMPLE**

If we toss a coin once, then the complement of "heads" is "tails". If we toss a coin four times, then the complement of "at least one head" is "no heads". A sample space consisting of  $n$  sample points can produce  $2^n$  different subsets (or simple and compound events).

**EXAMPLE**

Consider a sample space  $S$  containing 3 sample points, i.e.  $S = \{a, b, c\}$ .

Then the  $2^3 = 8$  possible subsets are

$$\phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \\ \{a, c\}, \{b, c\}, \{a, b, c\}$$

Each of these subsets is an event. The subset  $\{a, b, c\}$  is the sample space itself and is also an event. It always occurs and is known as the certain or sure event. The empty set  $\phi$  is also an event, sometimes known as impossible event, because it can never occur.

**MUTUALLY EXCLUSIVE EVENTS**

Two events  $A$  and  $B$  of a single experiment are said to be mutually exclusive or disjoint if and only if they cannot both occur at the same time i.e. they have no points in common.

**EXAMPLE-1**

When we toss a coin, we get *either* a head *or* a tail, but *not* both at the same time. The two events head and tail are therefore mutually exclusive.

**EXAMPLE-2**

When a die is rolled, the events 'even number' and 'odd number' are mutually exclusive as we can get either an even number or an odd number in one throw, not both at the same time. Similarly, a student *either* qualifies or fails, a single birth must be *either* a boy or a girl, it cannot be both, etc., etc. Three or more events originating from the same experiment are mutually exclusive if pair wise they are mutually exclusive. If the two events *can* occur at the same time, they are not mutually exclusive, e.g., if we draw a card from an ordinary deck of 52 playing cards, it *can* be both a king and a diamond.

Therefore, kings and diamonds are not mutually exclusive. Similarly, inflation and recession are not mutually exclusive events. Speaking of playing cards, it is to be remembered that an ordinary deck of playing cards contains 52 cards arranged in 4 suits of 13 each. The four suits are called diamonds, hearts, clubs and spades; the first two are red and the last two are black. The face values called denominations, of the 13 cards in each suit are ace, 2, 3, ..., 10, jack, queen and king. The *face cards* are king, queen and jack. These cards are used for various games such as whist, bridge, poker, etc. We have discussed the concepts of mutually exclusive events. Another important concept is that of exhaustive events.

**EXHAUSTIVE EVENTS**

Events are said to be collectively exhaustive, when the union of mutually exclusive events is equal to the entire sample space  $S$ .

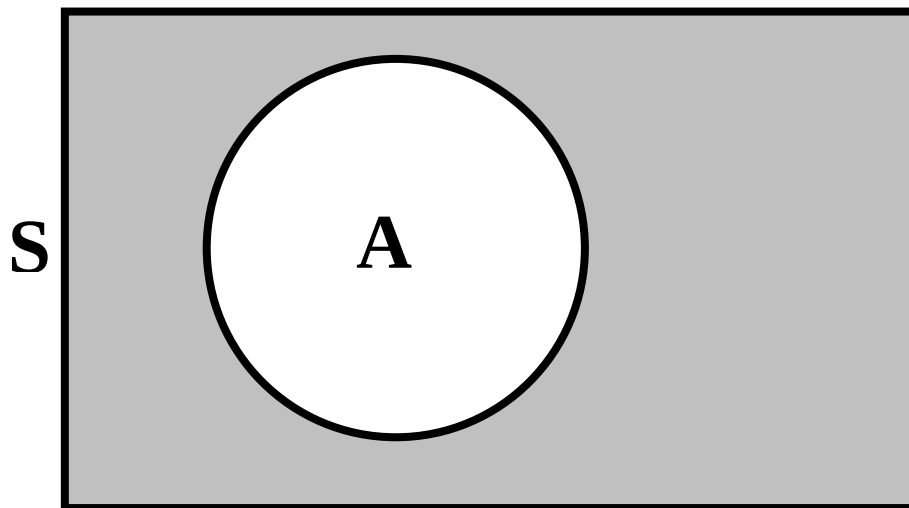
**EXAMPLES**

- In the coin-tossing experiment, 'head' and 'tail' are collectively exhaustive events.
- In the die-tossing experiment, 'even number' and 'odd number' are collectively exhaustive events.

In conformity with what was discussed in the last lecture:

**PARTITION OF THE SAMPLE SPACE**

A group of mutually exclusive and exhaustive events belonging to a sample space is called a partition of the sample space. With reference to any sample space  $S$ , events  $A$  and  $\bar{A}$  form a partition as they are mutually exclusive and their union is the entire sample space. The Venn Diagram below clearly indicates this point.



$\bar{A}$  is shaded

Next, we consider the concept of equally likely events:

**EQUALLY LIKELY EVENTS**

Two events  $A$  and  $B$  are said to be equally likely, when one event is as likely to occur as the other. In other words, each event should occur in equal number in repeated trials.

**EXAMPLE**

When a fair coin is tossed, the head is as likely to appear as the tail, and the proportion of times each side is expected to appear is  $1/2$ .

**EXAMPLE**

If a card is drawn out of a deck of well-shuffled cards, each card is equally likely to be drawn, and the probability that any card will be drawn is  $1/52$ .

## LECTURE NO. 18

### **DEFINITIONS OF PROBABILITY**

- Subjective Approach to Probability
- Objective Approach:
- Classical Definition of Probability

### **RELATIVE FREQUENCY DEFINITION OF PROBABILITY**

Before we begin the various definitions of probability, let us revise the concepts of:

- Mutually Exclusive Events
- Exhaustive Events
- Equally Likely Events

### **MUTUALLY EXCLUSIVE EVENTS**

Two events A and B of a single experiment are said to be mutually exclusive or disjoint if and only if they cannot both occur at the same time i.e. they have no points in common.

#### **EXAMPLE-1**

When we toss a coin, we get *either* a head *or* a tail, but *not* both at the same time. The two events head and tail are therefore mutually exclusive.

#### **EXAMPLE-2**

When a die is rolled, the events ‘even number’ and ‘odd number’ are mutually exclusive as we can get either an even number or an odd number in one throw, not both at the same time. Similarly, a student *either* qualifies *or* fails, a person is either a teenager or not a teenager, etc., etc.

Three or more events originating from the same experiment are mutually exclusive if pair wise they are mutually exclusive. If the two events *can* occur at the same time, they are not mutually exclusive, e.g., if we draw a card from an ordinary deck of 52 playing cards, it *can* be both a king and a diamond.

Therefore, kings and diamonds are not mutually exclusive. Speaking of playing cards, it is to be remembered that an ordinary deck of playing cards contains 52 cards arranged in 4 suits of 13 each. The four suits are called diamonds, hearts, clubs and spades; the first two are red and the last two are black. The face values called denominations, of the 13 cards in each suit are ace, 2, 3, ..., 10, jack, queen and king. The face values called denominations, of the 13 cards in each suit are ace, 2, 3, ..., 10, jack, queen and king. We have discussed the concepts of mutually exclusive events.

Another important concept is that of exhaustive events.

### **EXHAUSTIVE EVENTS**

Events are said to be collectively exhaustive, when the union of mutually exclusive events is equal to the entire sample space S.

#### **EXAMPLES**

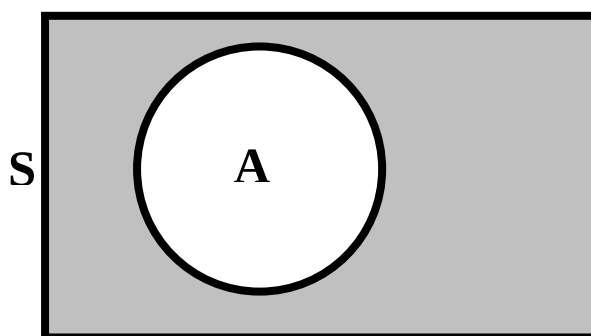
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In conformity with what was discussed in the last lecture:

### **PARTITION OF THE SAMPLE SPACE**

A group of mutually exclusive and exhaustive events belonging to a sample space is called a partition of the sample space. With reference to any sample space S, events A and  $\bar{A}$  form a partition as they are mutually exclusive and their union is the entire sample space. The Venn Diagram below clearly indicates this point.

## Venn Diagram



$\bar{A}$  is shaded

### EQUALLY LIKELY EVENTS

Two events A and B are said to be equally likely, when one event is as likely to occur as the other. In other words, each event should occur in equal number in repeated trials.

#### EXAMPLE

When a fair coin is tossed, the head is as likely to appear as the tail, and the proportion of times each side is expected to appear is  $1/2$ .

#### EXAMPLE

If a card is drawn out of a deck of well-shuffled cards, each card is equally likely to be drawn, and the proportion of times each card can be expected to be drawn in a very large number of draws is  $1/52$ . Having discussed basic concepts related to probability theory, we now begin the discussion of THE CONCEPT AND DEFINITIONS OF PROBABILITY. Probability can be discussed from two points of view: the subjective approach, and the objective approach.

### SUBJECTIVE OR PERSONALISTIC PROBABILITY

As its name suggests, the subjective or personality probability is a measure of the strength of a person's belief regarding the occurrence of an event A. Probability in this sense is purely subjective, and is based on whatever evidence is available to the individual. It has a disadvantage that two or more persons faced with the same evidence may arrive at different probabilities.

For example, suppose that a panel of three judges is hearing a trial. It is possible that, based on the evidence that is presented, two of them arrive at the conclusion that the accused is guilty while one of them decides that the evidence is NOT strong enough to draw this conclusion. On the other hand, objective probability relates to those situations where everyone will arrive at the same conclusion.

It can be classified into two broad categories, each of which is briefly described as follows:

#### 1. THE CLASSICAL OR 'A PRIORI' DEFINITION OF PROBABILITY

If a random experiment can produce  $n$  mutually exclusive and equally likely outcomes, and if  $m$  out of these outcomes are considered favorable to the occurrence of a certain event A, then the probability of the event A, denoted by  $P(A)$ , is defined as the ratio  $m/n$ .

Symbolically, we write

$$P(A) = \frac{m}{n}$$

$$= \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

This definition was formulated by the French mathematician P.S. Laplace (1749-1827) and can be very conveniently used in experiments where the total number of possible outcomes and the number of outcomes favorable to an event can be DETERMINED.

Let us now consider a few examples to illustrate the classical definition of probability:

### EXAMPLE-1

If a card is drawn from an ordinary deck of 52 playing cards, find the probability that i) the card is a red card, ii) the card is a 10.

#### SOLUTION:

The total number of possible outcomes is  $13+13+13+13 = 52$ , and we assume that all possible outcomes are equally likely. (It is well-known that an ordinary deck of cards contains 13 cards of diamonds, 13 cards of hearts, 13 cards of clubs, and 13 cards of spades.)

(i) Let A represent the event that the card drawn is a red card.

Then the number of outcomes favorable to the event A is 26 (since the 13 cards of diamonds and the 13 cards of hearts are red).

Hence

$$P(A) = \frac{m}{n} = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} = \frac{26}{52} = \frac{1}{2}$$

$$\text{Thus } P(B) = \frac{4}{52} = \frac{1}{13}.$$

### EXAMPLE-2

A fair coin is tossed three times. What is the probability that at least one head appears?

IMP

#### SOLUTION

The sample space for this experiment is

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

and thus the total number of sample points is 8 i.e.  $n(S) = 8$ . Let A denote the event that at least one head appears. Then

$$A = \{HHH, HHT, HTH, THH, HTT, THT, TTH\}$$

Therefore  $n(A) = 7$ .

Hence

$$P(A) = \frac{n(A)}{n(S)} = \frac{7}{8}.$$

### EXAMPLE-3

Four items are taken at random from a box of 12 items and inspected. The box is *rejected* if more than 1 item is found to be faulty. If there are 3 faulty items in the box, find the probability that the box is *accepted*.

#### SOLUTION

The sample space S contains  $\binom{12}{4} = 495$  Sample points

(Because there are  $\binom{12}{4}$  ways of selecting four items out of twelve)

The box contains 3 faulty and 9 good items. The box is accepted if there is (i) no faulty items, or (ii) one faulty item in the sample of 4 items *selected*.

Let A denote the event the number of faulty items chosen is 0 or 1.

Then

$$\begin{aligned} n(A) &= \binom{3}{0} \binom{9}{4} + \binom{3}{1} \binom{9}{3} \\ &= 126 + 252 = 378 \text{ sample points.} \end{aligned}$$

$$\therefore P(A) = \frac{m}{n} = \frac{378}{495} = 0.76$$

Hence the probability that the box is accepted is 76% , (in spite of the fact that the box *contains* 3 faulty items).

### THE CLASSICAL DEFINITION HAS THE FOLLOWING SHORTCOMINGS

- This definition is said to involve circular reasoning as the term equally likely really means equally probable.
- Thus probability is defined by introducing concepts that presume a *prior* knowledge of the *meaning* of probability.
- This definition becomes vague when the possible outcomes are INFINITE in number, or uncountable.
- This definition is NOT applicable when the assumption of equally likely does *not* hold. And the fact of the matter is that there are NUMEROUS situations where the assumption of equally likely cannot hold.

And these are the situations where we have to look for another definition of probability!

### THE RELATIVE FREQUENCY DEFINITION OF PROBABILITY

The essence of this definition is that if an experiment is repeated a large number of times under (more or less) identical conditions, and if the event of our interest occurs a certain number of times, then the *proportion* in which this event occurs is regarded as the probability of that event.

For example, we know that a large number of students sit for the matric examination every year. Also, we know that a certain proportion of these students will obtain the first division, a certain proportion will obtain the second division, --- and a certain proportion of the students will fail.

Since the total number of students appearing for the matric exam is very large, hence:

- The proportion of students who obtain the first division --- this proportion can be regarded as the *probability* of obtaining the first division,
- The proportion of students who obtain the second division --- this proportion can be regarded as the *probability* of obtaining the second division, and so on.